Stochastic Geometry and Compatible Histories in Three-Dimensional Navier–Stokes: Formal Results, Analytic Mechanisms and the Open Budget

The MeTTapedia formalization effort

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Abstract

Ben Goertzel's Navier–Stokes programme connects stochastic viscosity, geometric depletion of vorticity stretching, and control of complete histories. This note records the current mathematical implementation of that programme. The original SG–Cole–Hopf adjoint hypothesis fails in the audited Fourier shear model. The repaired approach retains stochastic-Lagrangian transport and geometric feedback, while making the remaining payment estimate an obligation about actual unforced solutions.

Local solutions, uniqueness, pressure reconstruction and conditional continuation are checked in Lean in the stated periodic and whole-space classes. Compatible whole-space solution segments have additive signed costs; one nonnegative reserve would bound all their continuations. Analytic calculations provide finite payments for original heat parents and several controlled interaction classes. A coupled periodic family generates positive self-stretching, has a paid leading return, and has a strictly depleting next amplitude coefficient. Its complete higher return remains uncontrolled. Distinction observers connect a uniform margin below saturation to continuation, and expose directional information lost by coarse summaries. The arbitrary-data finite-horizon history estimate remains open. Earlier formal derivations are retained as appendices, with their original hypotheses and model boundaries.

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1 The mechanism and the present mathematical target

Consider unforced incompressible Navier–Stokes,

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0, \qquad \nu > 0,$$

with admissible smooth initial velocity u_0. The intended conclusion is global regularity in both the periodic and whole-space settings. The exact datum classes and the maps to the official statements are specified in `FourierOfficialTheorems` and `MettapediaNS.WholeSpaceOfficialStatement`. A periodic construction does not, by itself, establish a whole-space assertion.

The mechanism is a question about renewed geometry. A thin or coherent configuration may stretch, rotate and transfer energy. What needs a bound is the accumulated effect when the surrounding fluid repeatedly recreates that configuration. Stochastic transport describes viscosity without deleting the deformation; a geometric estimate measures the dangerous stretching; a history reserve must pay for its renewal from the full original datum.

original fluid $\longrightarrow$ compatible evolving geometry $\longrightarrow$ signed history payment $\longrightarrow$ continuation.

The last implication is available. The uniform physical payment is the open step. A formal ledger, an initial-time sign, or a bounded comparison score alone does not supply it.

1.1 A sufficient estimate with the quantifiers visible

Write $\omega = \nabla \times u$ and

$$E_1(t) = \|\nabla u(t)\|_2^2, \qquad E_2(t) = \|\nabla \omega(t)\|_2^2, \qquad g(t) = \int \omega(t)^{\mathsf{T}} \nabla u(t) \omega(t).$$

For smooth solenoidal whole-space fields, these are the order-one and order-two energies and the physical stretching mass in the formal development. We also write $g(v)$ for the same spatial cubic contraction applied to a solenoidal field v. The exact balance is

$$E_1(t) + 2\nu \int_0^t E_2(s)\, ds = E_1(0) + 2 \int_0^t g(s)\, ds.$$

One sufficient target is

$$2 \int_0^t [g(s) - g(0)]\, ds \le \nu \int_0^t E_2(s)\, ds + K(u_0, \nu, H). \tag{1}$$

The requirement is

$$\forall u_0, \nu > 0, H > 0 \quad \exists K < \infty \quad \forall \text{ available smooth prefixes from } u_0, \quad 0 \le t \le T \le H.$$

The allowance must be controlled from the original datum, viscosity and prescribed horizon before the prefixes are considered. It cannot assume a bound on their unknown future. Under this condition the energy balance gives

$$E_1(t) \le E_1(0) + 2|g(0)|H + K.$$

The whole-space implication is Lean-certified in `JointCubicRenewal.hasNSEvolution_of_signed_renewal`. The periodic development separately has order-one and vorticity-integral continuation theorems. The estimate in (1) is not proved for arbitrary data.

2 Actual histories, stochastic transport and boundary summaries

2.1 Composition keeps the fluid that pays

`WholeSpaceHistoryBridge` uses actual unforced solution segments. Their endpoints contain the full velocity, and their clocks stay inside one prescribed horizon. At order n, the signed cost is the integrated physical energy production, proved equal to

$$\operatorname{cost}(u_s \rightsquigarrow u_t) = E_n(u_t) - E_n(u_s).$$

Segments with the same complete seam velocity concatenate into an actual solution. Their costs add. A nonnegative reserve satisfying

$$\operatorname{cost}(\gamma) + R(t, u_t) \le R(s, u_s)$$

would therefore give $E_n(u_t) \le E_n(u_0) + R(0, u_0)$ for every prefix. Intermediate reserves cancel; a newly generated parent receives no fresh initial allowance. These implications and the actual segment composition are Lean-certified. Constructing a reserve for order one is still open.

An eliminated region or set of modes may look like a force on the observed component. Its boundary history must nevertheless be generated by the remaining fluid's own equation. GSLT and Kron-style reduction are useful when they preserve compatible boundary histories and reconstruct their genuine interiors. Matching a pattern label or scalar energy is insufficient to justify physical composition.

2.2 The stochastic and control roles

The Constantin–Iyer repair transports momentum one-forms through a noisy, volume-preserving flow and recovers velocity by averaging and projection. The covector pushdown has a checked conditional NS equation with an explicit pressure. The periodic lane also constructs actual solution trajectories, inverse maps, Jacobians and measure preservation path by path for continuous additive noise. The Brownian path measure and the stochastic-calculus identification are not fully formalized by these pathwise constructions.

Bellman/HJB reasoning can seek an upper bound on the costs attainable along actual fluid histories. A relaxed control system is useful only after its domination of those physical costs is proved. It need not make every auxiliary optimizer a realizable fluid, and an auxiliary sampling control is not an external force in the NS equation. Schrödinger-bridge terminology does not remove the identification or domination obligation.

Joint stochastic information must be retained: an expectation of a product is not generally a product of expectations. Source deformation, covariance, pressure and the evolution of hidden parents remain part of the physical defect. The earlier scale-local pancake budgets are possible sufficient routes to the payment, not already discharged hypotheses.

3 A coupled-parent benchmark: growth, payment and depletion

All norms and integrals in this section use the normalized period-2π torus. Take real A, B, C and

$$u_0 = (A\cos y + B\cos z, 0, C\cos x), \qquad h(t) = e^{-\nu t}u_0, \qquad b = u - h.$$

Here h is the heat evolution of the *full original datum*, throughout the same history. For $N(v) = -\mathbb{P}[(v\cdot\nabla)v]$, its source is

$$n(t) = N(h(t)) = ACe^{-2\nu t}\sin x\cos y\, e_z.$$

Every $n(t)$ has $N(n(t)) = 0$ and $g(n(t)) = 0$. That source space is not invariant under interaction with the other parents. Actual pressure generates the initially absent transverse component:

$$\partial_t^2 u_y(0) = -\tfrac{2}{3} ABC \cos x \sin y \sin z.$$

The generated remainder then has a genuine positive local interval,

$$g(b(t)) = \frac{8393}{2903040} A^4 B^2 C^4 t^7 + O(t^8) \qquad (ABC \neq 0). \tag{2}$$

This is an analytic statement about actual local unforced solutions for any fixed amplitudes and positive viscosity, supported by exact PDE derivatives. It is not a blowup or an indefinite-renewal construction.

3.1 The entire leading coefficient has a finite price

For unit viscosity, the complete degree-ten amplitude coefficient is $A^4 B^2 C^4 G_{10}(t)$. Every convolution output and the canonical pressure are retained. An exact scalar differential cascade proves $G_{10}(t) > 0$ for $t > 0$, and termwise integration gives

$$C_{10} = \int_0^\infty G_{10}(s)\, ds = \frac{415522849111}{1911826022400000000}.$$

At general viscosity define

$$G_{\mathrm{lead}}(t) = A^4 B^2 C^4 \nu^{-7} G_{10}(\nu t), \qquad T_{10}(s) = \int_s^\infty G_{10}(r)\, dr.$$

The leading coefficient has a complete viscous lifetime; this is not a claim that a finite-amplitude nonlinear solution equals that coefficient.

There is an explicit original-parent payment. Set $S = |A| + |B| + |C|$, $E_0 = \|u_0\|_2^2$ and

$$\begin{aligned}
\chi(t) &= \exp[(5S/\nu) e^{-\nu t}], \qquad \chi_0 = \chi(0), \\
\mathcal{B}(t) &= \frac{A^2 C^2}{16\nu^2} e^{-4\nu t}, \\
E_* &= \min\{4E_0, e^{2S/\nu} A^2 C^2/(16\nu^2)\}, \\
F(t) &= E_* \frac{S}{\nu} e^{-\nu t} + 2\mathcal{B}(t), \\
J_{\mathrm{parent}}(t) &= \chi(t)[\|\nabla b(t)\|_2^2 + F(t)].
\end{aligned}$$

The analytic original-heat estimate pays the source and mixed-parent terms on every available prefix and gives

$$J'_{\mathrm{parent}} + \tfrac{3}{2}\nu\chi D_b \leq 2\chi g(b), \qquad D_b = \|\nabla \operatorname{curl} b\|_2^2.$$

Add the nonnegative remaining allowance

$$R_{10}(t) = 2\chi_0 A^4 B^2 C^4 \nu^{-8} T_{10}(\nu t).$$

Its derivative pays $2\chi G_{\mathrm{lead}}$ over the full lifetime. With the *exact* defect $\eta = g(b) - G_{\mathrm{lead}}$ and $J_{\mathrm{total}} = J_{\mathrm{parent}} + R_{10}$, one obtains

$$J'_{\mathrm{total}} + \tfrac{3}{2}\nu\chi D_b \leq 2\chi\eta. \tag{3}$$

This inequality does not assume that η is small. The sufficient remaining estimate for this benchmark is

$$2\int_0^t \chi\eta\,ds \le \nu\int_0^t \chi D_b\,ds + K(u_0,\nu,H), \qquad t \le T \le H, \tag{4}$$

with one original-data allowance for all same-datum prefixes. It would bound $\|\nabla b\|_2^2$ and hence the complete gradient energy. This estimate is open; its arbitrary-data analogue remains the larger objective.

3.2 The next complete correction is depleting

The complete degree-twelve coefficient at unit viscosity has three channels,

$$G_{12}(A,B,C;t) = A^4B^2C^4[A^2F_A(t) + B^2F_B(t) + C^2F_C(t)].$$

Exact differential cascades prove

$$F_A(t) < 0, \qquad F_B(t) < 0, \qquad F_C(t) < 0 \qquad \text{for every } t > 0.$$

Thus this correction depletes the positive leading return whenever $ABC \neq 0$. These are all-time signs of complete amplitude coefficients, proved by rational arithmetic identities and scalar Duhamel comparison, rather than by a sampled time grid.

For fixed A, B, C, $\nu > 0$, and $0 < t_0 \le t \le t_1 < \infty$, the actual solution with datum λu_0 satisfies, for sufficiently small λ,

$$g(b_\lambda(t)) = \lambda^{10}G_{\mathrm{lead}}(t) + \lambda^{12}\nu^{-9}G_{12}(A,B,C;\nu t) + O_{A,B,C,\nu,t_1}(\lambda^{14}),$$

uniformly on that interval. Analytic dependence and strict negativity give an actual comparison below the leading forecast for $0 < |\lambda| < \lambda_*(A,B,C,\nu,t_0,t_1)$ when $ABC \neq 0$. No numerical value of λ_* is supplied. There is no conclusion uniform as $t_0 \downarrow 0$, $t_1 \uparrow \infty$, or amplitudes become large. The complete higher remainder in (3) is still unpaid.

4 Distinction observers: what the interface must remember

The d-calculus application connects an observer to a continuation quantity. For $M > 0$ define

$$d_M(x,y) = \min\{1, |x-y|/M\}, \qquad \mathrm{score}_M(u) = d_M(E_1(u), 0).$$

Lean proves the metric law and, for $q < 1$, the equivalence

$$\mathrm{score}_M(u) \le q \quad \Longleftrightarrow \quad E_1(u) \le qM.$$

At energy at least M, the score is exactly one. A bound by one therefore does not control energy. The continuation theorem requires M and $q < 1$ fixed before every same-datum prefix on the prescribed horizon. Given a nonnegative signed reserve, $Z = E_1(u_0) + R(0,u_0)$ supplies $M = Z+1$, $q = Z/(Z+1) < 1$, uniformly over every time of a prefix. These implications are certified in `MettapediaNS.DistinctionHistory`; the arbitrary-flow reserve is not.

The physical benchmark separates a predictive summary from an upper bound. The parent triples $(2,1,1)$ and $(1,1,2)$ have the same normalized energy, all scalar isotropic integer derivative energies, complete original heat source and leading return. Nevertheless their next signed costs differ:

$$G_{12}(2,1,1;t) - G_{12}(1,1,2;t) = 48[F_A(t) - F_C(t)] < 0 \quad (t > 0).$$

For their actual local solutions at fixed positive viscosity,

$$g(b_{(2,1,1)}(t)) - g(b_{(1,1,2)}(t)) = -\frac{75692097}{11000\,9!}t^9 + O_\nu(t^{10}).$$

The local PDE identification is analytic with exact independent checks. Lean separately certifies the rational polynomial obstruction: the ninth-jet readout does not factor through the energy/source/leading-work summary. Retaining the ordered parent squares separates this particular pair; it is not asserted to give a complete predictive state for arbitrary flows.

Exact prediction from the coarse summary fails, while a common upper comparison at degree twelve survives because every channel is depleting. A cost-dominating summary can therefore be sufficient for control. It still needs an original-data price and an estimate valid as geometry, scales and parents change.

5 What the tests establish and what remains open

The calculations give positive mechanisms and precise restrictions:

- Decaying shear and triangular systems have actual stretching histories with finite original-data payments. These do not cover mutually amplifying parents of arbitrary size.
- Kinetic exchange between components has an exact signed balance. It does not decide the sign of derivative-weighted amplification; complete spherical-cutoff examples retain amplifying memory.
- Concentrated actual unforced bursts refute energy-only allowances uniform over the tested data and prescribed cutoffs. An allowance depending on the full datum, or a different datum-chosen cutoff, remains possible.
- A fixed smooth reserve depending only on the current local gradient cannot be universally nonincreasing in the tested whole-space class, unless it is constant. Nonlocal pressure and hidden parents must be accounted for. This does not rule out a history reserve.
- The imported OpenAI construction supplies forced singular solutions under its stated hypotheses. The full-field obstruction is Lean-certified; the frequency-tail analysis is analytic. Finite direct forcing work does not make the force irrelevant to the organization of the flow. No unforced blowup or general impossibility of internal realization has been established.

The next physical estimate concerns the complete defect in (3), not another isolated favorable coefficient. It must retain directional parent information, pressure and joint memory, derive their evolution along actual solutions, and compare their signed cost with the full diffusion. A successful estimate must survive concentration, changing geometry and repeated parent renewal outside a controlled neighborhood of a known regular flow. Extending the benchmark payment to arbitrary data and both domains is unproved.

6 Evidence and source map

Result	Evidence	Remaining boundary
Local actual evolution; pressure; uniqueness	Lean, separately on $\mathbb{T}^3$ and $\mathbb{R}^3$	No arbitrary-data global bound
Continuation and official statements	Lean implications from named finite-horizon bounds	Their arbitrary-data hypotheses
Noisy trajectories and Jacobians	Periodic pathwise Lean construction; conditional covector pushdown	Brownian measure and stochastic identification
Compatible signed histories	Actual whole-space segments and reserve consumers in Lean	Nonnegative order-one reserve
Original heat and coupled returns	Analytic PDE estimates and exact coefficient/sign checks	Complete higher nonlinear history
Distinction observer and summary obstruction	Lean observer/continuation and rational non-factorization; analytic PDE identification	Uniform nonsaturation and sufficient evolving summary

The current source entry points are:

- `lean/navier-stokes/README.md`: actual solution classes, continuation criteria, imported libraries and current mechanisms.
- `lean/navier-stokes/MettapediaNS/JointCubicRenewal.lean`, `WholeSpaceHistoryBridge.lean` and `DistinctionHistory.lean`: the formal contracts used above.
- `lean/navier-stokes/labs/README.md`: complete analytic derivations, especially the original-heat payment (254)–(257), coupled return CCR1–CCR10 and distinction application DCH1–DCH10.
- `coupled_caloric_return.py` and `distinction_coupled_history.py`, with their result archives in the same lab directory: exact recurrence and sign certificates.
- The companion tail-budget paper: forced obstructions, unforced concentration and separated-sector payments, with their distinct hypotheses.

At the 5 October 2026 checkpoint, the combined NS and live-source gate completed 11,927 jobs. Its 1,458 axiom reports, including the seven new distinction application reports, use only `propext`, `Classical.choice` and `Quot.sound`. The exact coupled-return and distinction checkers passed. These checks certify the stated formal implications and computational identities; they do not convert the analytic PDE arguments or the open payment into Lean theorems.

Earlier formal derivations

The appendices preserve the June–September SG audit, the cotangent repair and the microlocal pancake development. They establish useful identities, model obstructions and conditional routes. Their listings record that stage of the formalization; current Lean source and its audit are authoritative. The present trajectory and open estimate are stated in Sections 1–6.

A The problem and the role of hypothesis (H1)

A.1 The SG–Cole–Hopf route

We work on the flat three-torus $\mathbb{T}^3$ with smooth divergence-free initial data u_0. Ben Goertzel's manuscript *Global Regularity of 3D Navier–Stokes via SG–Cole–Hopf Program*, version v7 (hereafter "v7"), lifts the incompressible Navier–Stokes equation to a diffusion on the volume-preserving diffeomorphism group

$$\mathrm{SG} := \mathrm{SDiff}(\mathbb{T}^3),$$

equipped with a right-invariant Fourier frame $\{D_i\}_{i\geq 1}$ induced by a divergence-free frame $\{E_i\}_{i\geq 1}$ on $\mathbb{T}^3$. The generator and carré du champ are

$$L := \sum_{i\geq 1} D_i^2, \qquad \Gamma(F) := \sum_{i\geq 1} (D_i F)^2.$$

One solves the SG heat equation $\partial_t \Phi = \nu L \Phi$ with a strictly positive datum $\varphi = \exp(-W_0/(2\nu))$, applies the Cole–Hopf transform $W = -2\nu \log \Phi$ to obtain an SG Hamilton–Jacobi–Bellman potential, and pushes the Γ-gradient down at the identity to a projected Navier–Stokes solution $u(t)$ on $\mathbb{T}^3$.

The analytic content is concentrated entirely in closing the Beale–Kato–Majda (BKM) vorticity criterion. With $u(t) = \mathrm{Ev}(W(t))$ and $\omega(t) = \operatorname{curl} u(t)$, the push-down identity and Cauchy–Schwarz give a frame constant $C_{\mathrm{curl}E} < \infty$ with

$$\|\omega(t)\|_{L^\infty(\mathbb{T}^3)} \leq \sqrt{C_{\mathrm{curl}E}}\, \sqrt{\Gamma(W(t))(\mathrm{id})}, \tag{5}$$

so that a uniform-in-time bound on $\Gamma(W(t))(\mathrm{id})$ makes the BKM integral $\int_0^T \|\omega(t)\|_{L^\infty}\, dt$ finite on every finite interval. By the log chain rule $\Gamma(\log \Phi) = \Gamma(\Phi)/\Phi^2$ and the maximum principle $\Phi(t, \mathrm{id}) \geq m_\varphi$ (v7, Prop. 7.1),

$$\Gamma(W(t))(\mathrm{id}) = 4\nu^2\, \Gamma(\log \Phi(t))(\mathrm{id}) \leq \frac{4\nu^2}{m_\varphi^2}\, \Gamma(P_t\varphi)(\mathrm{id}), \tag{6}$$

which reduces the whole program to a single uniform bound on $\Gamma(P_t\varphi)(\mathrm{id})$.

A.2 Reduction to a local energy bound, and hypothesis (H1)

Rather than invoking a global Bakry–Émery Γ_2 curvature lower bound, v7 obtains that single bound from a *local* package. Let $\kappa : U \to V \subset H^m_\sigma(\mathbb{T}^3)$ be the chart of a small SDiff^m neighborhood of the identity ($m \geq 5$ integer, with the Γ-Hilbert structure at id equivalent to the H^m_σ norm), let χ be the cutoff defining W_0 and φ, and set the cutoff region

$$K := \kappa^{-1}(\operatorname{supp} \chi) \subset U.$$

Outside K one has $W_0 \equiv 0$, hence $\varphi \equiv 1$ and $\Gamma(\varphi) \equiv 0$. On K, v7 assumes two local hypotheses (v7, §5.3 and §6, displayed at Lemma D.18):

$$\text{(H1) Adjoint bounded on support:} \quad \exists\, C_{\mathrm{Ad}} < \infty :\ \|\mathrm{Ad}_g\|_{H^m \to H^m} \le C_{\mathrm{Ad}} \quad \text{for all } g \in K, \tag{7}$$

$$\text{(H2) Finite cutoff energy:} \quad \exists\, C_\varphi < \infty :\ \Gamma(\varphi)(g) \le C_\varphi^2 \quad \text{for all } g \in K. \tag{8}$$

From (7)–(8) v7 derives the key estimate via a conjugation differentiation identity (v7, Prop. 6.5, proved as Lemma F.7): for all $t \ge 0$,

$$\Gamma(P_t\varphi)(\mathrm{id}) \ \le\ C_{\mathrm{Ad}}^2\, C_\varphi^2. \tag{9}$$

The mechanism is transparent. In the flow representation $P_tF(g) = \mathbb{E}[F(G_t \circ g)]$, differentiating along a conjugation curve $C_{t,i}(\varepsilon) = g \circ \exp(\varepsilon E_i) \circ g^{-1}$ produces the direction $\mathrm{Ad}_g E_i$, and Cauchy–Schwarz in the Γ-Hilbert structure gives the pointwise bound

$$\left|D_{\mathrm{Ad}_g E_i}\varphi(g)\right|^2 \le \|\mathrm{Ad}_g\|_{\mathrm{op}}^2\, \Gamma(\varphi)(g) \le C_{\mathrm{Ad}}^2\, C_\varphi^2\, \mathbf{1}_{\{g \in K\}}, \tag{10}$$

after which a Jensen/energy reduction yields (9). Chaining (9) through (6) and (5) closes BKM. Hypothesis (H1) is therefore the single load-bearing analytic input of the route: *every* subsequent estimate inherits the constant C_{Ad} from (7), and the program establishes global regularity if and only if (7) holds.

A.3 How v7 attempts to verify (H1)

v7 verifies (H1) at Lemma 6.1 (and again at Lemma D.18). The adjoint / push-forward action is written, for $g \in \mathrm{SDiff}^m$ and $v \in H^m_\sigma$, as

$$\mathrm{Ad}_g v := (Dg)\,(v \circ g^{-1}), \tag{11}$$

and the argument is that, for $m > 5/2$, composition and inversion are continuous (indeed smooth) operations on SDiff^m, so the map $(g, v) \mapsto \mathrm{Ad}_g v$ is continuous from $\mathrm{SDiff}^m \times H^m_\sigma$ into H^m_σ; hence $g \mapsto \|\mathrm{Ad}_g\|_{\mathrm{op};\, H^m \to H^m}$ is locally bounded near id, and one shrinks K until (7) holds.

The gap is in (11) itself. The factor Dg is the Jacobian of g, and a map $g \in H^m$ has Jacobian $Dg \in H^{m-1}$ only. The adjoint thus carries a *derivative loss*: Ad_g maps H^m into H^{m-1}, and there is no uniform $H^m \to H^m$ operator bound near the identity in the H^m topology. This is the classical Ebin–Marsden phenomenon: right translation on SDiff^m is smooth, but the adjoint of the right-invariant frame is not a bounded H^m operator uniformly over an H^m-neighborhood. The local-boundedness argument conflates topological-group continuity of composition with the operator-norm estimate (7) that the route actually needs.

B The Fourier-mode shear model and the checked ¬(H1) theorems

We make the derivative-loss obstruction precise and machine-checkable in a clean analytic model, on the normalized two-torus, of an area-preserving shear family

$$g_k(x, y) = \big(x,\ y + a_k \sin(kx)\big), \qquad \delta_k := g_k - \mathrm{id} = \big(0,\ a_k \sin(kx)\big).$$

The amplitude a_k is chosen so that the displacement has *fixed* H^m size ε for every wavenumber k; thus the whole family sits inside any prescribed chart radius. Applying the push-forward (11) to the constant horizontal field $v = (1, 0)$ gives

$$\mathrm{Ad}_{g_k} v = (1,\ a_k\, k\, \cos(kx)),$$

and, with the same H^m normalization, the operator ratio is

$$\frac{\|\mathrm{Ad}_{g_k} v\|_{H^m}}{\|v\|_{H^m}} = \sqrt{1 + (\varepsilon k)^2}.$$

The displacement stays at size ε while the ratio grows without bound: no finite C_{Ad} can dominate it on any positive chart radius.

This is formalized, with no `sorry` and by elementary means, in the Lean module `Mettapedia.FluidDynamics.NavierStokes.BenH1Break`. The model quantities are defined as follows.

```
/-- In the normalized two-torus shear family, every selected displacement has
fixed `H^m` size `epsilon`. -/
def benH1ModeDisplacementHNorm (epsilon : Real) (_k : Nat) : Real :=
  epsilon

/-- Lower-bound ratio obtained by applying the shear push-forward to the
constant horizontal vector field. -/
def benH1ModeAdjointRatio (epsilon : Real) (k : Nat) : Real :=
  Real.sqrt (1 + (epsilon * (k : Real)) ^ 2)

/-- The linearized commutator part alone grows like `epsilon * k`. -/
def benH1ModeCommutatorRatio (epsilon : Real) (k : Nat) : Real :=
  epsilon * (k : Real)
```

Hypothesis (H1), restricted to this normalized family inside a chart radius ρ (taking $\varepsilon = \rho/2$ so the family is strictly inside the ball), is the predicate:

```
/-- H1 restricted to the normalized Fourier-mode shear family inside chart
radius `rho`. -/
def benH1FourierModeUniformBoundWithinRadius (rho C : Real) : Prop :=
  forall k : Nat,
    benH1ModeDisplacementHNorm (rho / 2) k <= rho ->
      benH1ModeAdjointRatio (rho / 2) k <= C
```

The core fact is that the adjoint ratio dominates the linear commutator part εk, which is unbounded over k. The proof is fully elementary: it picks $k > C/\varepsilon$, lower-bounds $\sqrt{1 + (\varepsilon k)^2}$ by εk, and derives a contradiction.

```
theorem benH1ModeAdjointRatio_unbounded {epsilon C : Real}
    (hepsilon : 0 < epsilon) :
    Not (forall k : Nat, benH1ModeAdjointRatio epsilon k <= C) := by
  intro hbound
  obtain ⟨k, hk⟩ := exists_nat_gt (C / epsilon)
  have hC_lt : C < epsilon * (k : Real) := by
    have hmul := mul_lt_mul_of_pos_left hk hepsilon
    field_simp [ne_of_gt hepsilon] at hmul
    exact hmul
```

```
  have hnonneg : 0 <= epsilon * (k : Real) := by positivity
  have hlinear_le_ratio :
      epsilon * (k : Real) <= benH1ModeAdjointRatio epsilon k := by
    unfold benH1ModeAdjointRatio
    have hsquare :
        (epsilon * (k : Real)) ^ 2 <= 1 + (epsilon * (k : Real)) ^ 2 := by
      linarith
    calc
      epsilon * (k : Real)
          = Real.sqrt ((epsilon * (k : Real)) ^ 2) := by
              rw [Real.sqrt_sq_eq_abs, abs_of_nonneg hnonneg]
      _ <= Real.sqrt (1 + (epsilon * (k : Real)) ^ 2) :=
              Real.sqrt_le_sqrt hsquare
  exact not_lt_of_ge (le_trans hlinear_le_ratio (hbound k)) hC_lt
```

From this the refutation of (H1) on the model follows directly. Any positive chart radius ρ admits the normalized family at $\varepsilon = \rho/2$ (whose displacements all have H^m size $\rho/2 \le \rho$), and its adjoint ratios are unbounded, contradicting any proposed finite C.

```
/-- Any positive chart radius contains the normalized shear family with
‘epsilon = rho / 2‘, and its adjoint ratios are unbounded. -/
theorem not_benH1FourierModeUniformBoundWithinRadius {rho C : Real}
    (hrho : 0 < rho) :
    Not (benH1FourierModeUniformBoundWithinRadius rho C) := by
  intro hbound
  have hepsilon : 0 < rho / 2 := by positivity
  have hmode_bound : forall k : Nat, benH1ModeAdjointRatio (rho / 2) k <= C := by
    intro k
    apply hbound k
    unfold benH1ModeDisplacementHNorm
    linarith
  exact benH1ModeAdjointRatio_unbounded hepsilon hmode_bound

/-- Compact public verdict: Ben’s H1, as a uniform ‘H^m -> H^m‘ adjoint bound,
is false on the normalized Fourier-mode shear model for every positive chart
radius and every proposed finite constant. -/
theorem benGoertzelH1_false_in_fourierModeShearModel :
    forall rho C : Real,
      0 < rho -> Not (benH1FourierModeUniformBoundWithinRadius rho C) := by
  intro rho C hrho
  exact not_benH1FourierModeUniformBoundWithinRadius (rho := rho) (C := C) hrho
```

The three checked statements are `benH1ModeAdjointRatio_unbounded`, `not_benH1FourierModeUniformBoundWithinRadius`, and the compact verdict `benGoertzelH1_false_in_fourierModeShearModel`: for every positive chart radius and every proposed finite constant, the modeled (H1) uniform $H^m \to H^m$ adjoint bound fails. (A companion lemma `benH1ModeCommutatorRatio_unbounded` records that the linearized commutator part εk is already unbounded on its own.)

C The numeric lab: the norms are derived, not assumed

The Lean development encodes the two normalized quantities $\|\delta_k\|_{H^m} = \varepsilon$ and $\|\mathrm{Ad}_{g_k} v\|_{H^m}/\|v\|_{H^m} = \sqrt{1+(\varepsilon k)^2}$ as definitions. To confirm that these are not circular — that the factor k in the adjoint ratio is a genuine consequence of the Sobolev-weight structure and the derivative in (11), rather than an assumption — a companion numeric lab (`ns_ben_h1_fourier_mode_lab.py`, in the MeTTapedia validation lab) *computes* the norms from first quantities. With H^m Fourier weight $w_k = (1+k^2)^{m/2}$:

- the amplitude is set to $a_k = \sqrt{2}\,\varepsilon/w_k$, so the displacement norm is $\|\delta_k\|_{H^m} = |a_k|\,w_k/\sqrt{2} = \varepsilon$ (the $\sqrt{2}$ is the L^2 normalization of $\sin(kx)$);
- differentiating, $\partial_x(a_k \sin(kx)) = a_k k \cos(kx)$, whose H^m norm is $|a_k|\,k\,w_k/\sqrt{2} = \varepsilon k$. The weight w_k *cancels* the amplitude, leaving ε; the surviving factor k is the one derivative gained by Dg_k;
- the constant field has $\|v\|_{H^m} = 1$, so the adjoint norm is $\sqrt{1^2+(\varepsilon k)^2}$ and the ratio is $\sqrt{1+(\varepsilon k)^2}$.

This is exactly the Ebin–Marsden derivative loss of (11) made quantitative: $g_k \in H^m$ has $Dg_k \in H^{m-1}$, and the action of Dg_k multiplies the mode's high-frequency content by its wavenumber. Table 1 lists the lab output for $\varepsilon = 0.01$, $m = 3$. The displacement norm stays fixed at ε across four decades of k while the adjoint ratio grows.

k	amplitude a_k	$\|\delta_k\|_{H^m}$	commutator εk	ratio $\sqrt{1+(\varepsilon k)^2}$
8	2.699×10^{-5}	0.0100	0.080	1.0032
16	3.433×10^{-6}	0.0100	0.160	1.0127
32	4.310×10^{-7}	0.0100	0.320	1.0500
64	5.393×10^{-8}	0.0100	0.640	1.1873
128	6.743×10^{-9}	0.0100	1.280	1.6243
256	8.429×10^{-10}	0.0100	2.560	2.7484
512	1.054×10^{-10}	0.0100	5.120	5.2167
1024	1.317×10^{-11}	0.0100	10.240	10.2887

Table 1: Fourier-mode lab output ($\varepsilon = 0.01$, $m = 3$). The displacement H^m norm is pinned at ε for every wavenumber, while the adjoint operator ratio $\sqrt{1+(\varepsilon k)^2}$ grows without bound. For any proposed constant C the ratio first exceeds it near $k \approx \sqrt{C^2-1}/\varepsilon$ (e.g. $C = 2$ at $k = 174$, $C = 10$ at $k = 995$, $C = 100$ at $k = 10000$): no finite C_{Ad} survives.

D The repaired stochastic-Lagrangian route: the full formalized layer

After the H1 obstruction was isolated, the formalization effort tested a different repair strategy. The repaired route abandons the false $H^m \to H^m$ adjoint estimate and moves the load-bearing estimate to the stochastic-Lagrangian derivative flow of Constantin–Iyer (*Comm. Pure Appl. Math.* **61** (2008), 330–345): the state is a momentum one-form $\mu = u^\flat + dW$ transported along a noisy flow, the velocity is recovered by averaging and projection, and the continuation estimate is a vorticity-stretching gate feeding the BKM criterion. In the Lean development the gate appears

first as the coarse input `StochasticStretchingEstimate`, and in the dyadic refinement as the scale-local predicate `ScaleLocalPancakeStrainBudget`.

This section records the machine-checked surface of the earlier repair campaign, including its conditional interfaces. The listings record the declaration names, hypotheses and conclusions at that stage, with Unicode notation transliterated to ASCII for typesetting. Current Lean source and its audit are normative. Table 2 summarizes these layers.

Layer (module)	Headline results	Scope label
H1 model (`BenH1Break`)	adjoint ratio unbounded; $\neg$(H1)	model refutation
Evasion (`ShearJacobianBounds`)	same family: pointwise Jacobian $\leq \varepsilon$	model contrast
One-forms (`OneFormFrameCalculus`)	frame-Laplacian identity $\sum_i L_{E_i}^2 = \Delta$	exact identity
Exactness (`OneFormExactness`)	div-free smooth non-exact witness	exact witness
Gradients (`GradientOneForm`)	Cartan formula; self-transport identity	exact identities
Pushdown (`ConstantinIyerRepresentation`)	NS with explicit pressure from transport	conditional (interface)
Instances (`DecayingShearInstance`, regression)	nonzero exact inhabitant; rest canaries	exact instances
Gate (`VorticityStretchingGate`)	pointwise common envelope from named pin	conditional
Countermodel (`FrozenStrainModel`)	exact pancake solution; no uniform depletion	model refutation
Aperture (`ApertureCoverageDichotomy`)	dichotomy $\iff$ strain budget; gap failures	model equivalence
Null structure (`PlaneWaveSelfStrain`)	single-mode Biot–Savart null stretching	exact identity
Dyadic closure (`DyadicPancakeClosure`)	absorption algebra; sharpened pin; route	conditional
Coherence (`PancakeCoherentPairEstimate`)	pairwise aperture gain	exact inequality
Refined pin (`MisalignmentRefinedPin`)	misalignment budget implies pancake budget	conditional
Direction dynamics (`DirectionEvolutionTilting`)	direction laws; gap-free tilting no-go	exact/refutation

Table 2: The formalized surface of the repair campaign (all modules under `Mettapedia.FluidDynamics.NavierStokes`, September 2026). "Model" statements are proved in named analytic models; "exact" statements are unconditional identities or witnesses on $\mathbb{R}^3$-coordinate space; "conditional" statements are machine-checked implications whose analytic inputs are named hypotheses.

D.1 Why a pointwise transport route can evade the H1 countermodel

The first question for any repair is whether the Section B countermodel already defeats it. It does not, and this is itself a theorem. On *the same* H^m-normalized shear family that refutes (H1), the module `ShearJacobianBounds` proves (`shearFamily_pointwise_bounded_while_sobolev_ratio_unbounded`):

1. every pointwise Jacobian deviation is bounded by ε uniformly in the frequency k (`shearFamily_jacobian_uniform_bound`: $\|Dg_k(x)v - v\| \le \varepsilon\|v\|$ for all x, v); while

2. the $H^m \to H^m$ operator ratio $\sqrt{1 + (\varepsilon k)^2}$ of the same family admits no finite uniform bound (Section B).

The high-frequency-shear derivative-loss mechanism therefore lives entirely in the Sobolev operator topology: the quantity feeding a pointwise gate $\int_0^T \mathbb{E}\|DX_t\|_\infty \, dt$ is not stressed at all by this family. The checked statement is deliberately labeled: it does not prove any pointwise gate — it proves this family cannot refute one. That is what licenses moving the route from adjoint bounds to Lagrangian transport.

D.2 The one-form layer: non-exact momenta and the frame Laplacian

The repaired state is a momentum one-form with a gauge, $\mu = u^\flat + dW$, not a scalar potential. Two checked facts pin down why this shape is forced and why the frame generator acts correctly on it.

Non-exact momenta are necessary. Exact one-forms are closed (`oneFormExtDerivAt_eq_zero_of_exact`), and there is a smooth, everywhere divergence-free field that is not exact — the transverse sine mode:

```
theorem exists_divergenceFree_smooth_not_exact_dim3 :
    exists u0 : EuclideanSpace Real (Fin 3) -> EuclideanSpace Real (Fin 3),
      ContDiff Real 2 u0 /\
      (forall x, coordDivergence u0 x = 0) /\
      Not (IsExactOneForm u0)
```

A scalar-potential (pure Cole–Hopf) ansatz therefore cannot carry a general divergence-free momentum: vorticity lives exactly in the non-exact part. This is the constructive face of the Section A diagnosis, and it is what the additive gauge dW is for.

The frame reproduces the Laplacian. The route's generator is a sum of squared frame derivatives (v7's $L = \sum_i D_i^2$). On the truncated trigonometric coordinate frame (three axes, transverse polarizations, both phases) the squared Lie derivatives of a one-form sum *exactly* to the coordinate Laplacian: with amplitude $1/\sqrt{2}$ in dimension 3,

$$\sum_j \sum_{p \neq j} \Big(L^2_{\cos\text{-mode}(j,p)} + L^2_{\sin\text{-mode}(j,p)} \Big) \mu = \Delta \mu$$

(`frameLaplacian_oneForm_normalized`, from the general $(\operatorname{card}\iota - 1)a^2$ identity `frameLaplacian_oneForm` and the phase-pair cancellation `lieOneForm_sq_cos_add_sin`). The supporting calculus (`GradientOneForm`) includes the coordinate Cartan formula for exact forms (`lieOneForm_gradField`: the Lie derivative of dV along u is exact with potential $\langle u, \nabla V\rangle$, i.e. $L_u\, dV = d\langle u, \nabla V\rangle$ pointwise) and the self-transport identity (`lieOneForm_self`: $L_u u^\flat = (u \cdot \nabla)u + \frac{1}{2}\nabla|u|^2$). These two identities are what turn transport equations into Navier–Stokes with a pressure, next.

D.3 The pushdown theorem: Navier–Stokes with an explicit pressure

The structure `TransportedMomentumData` packages a velocity u, a gauge W, their certified time derivatives, the gradient/Laplacian exchange facts, and one transport hypothesis — the Itô-

averaged frame-transport equation for the momentum one-form:

$$\partial_t u + \nabla \dot{W} + L_u(u^\flat + dW) = \nu\, \Sigma_{\text{frame}}(u^\flat + dW),$$

where Σ_{frame} is the normalized frame second-order sum of Section D.2 (the hypothesis is stated in averaged form; no stochastic calculus is formalized, and the field is labeled accordingly in the source). From this single hypothesis the pushdown theorem derives the Navier–Stokes equations with a fully explicit pressure:

```
/-- Pushdown theorem: the averaged frame-transport equation for the
momentum one-form u^flat + dW forces u to satisfy the Navier-Stokes
equations with the explicit pressure
p = dW/dt + (1/2)<u,u> + <u, grad W> - nu * Lap W. -/
theorem navierStokes_of_transportedMomentum (t : Real) (x : R3) :
    deriv (fun s => D.u s x) t + fderiv Real (D.u t) x (D.u t x) +
      gradField (D.pressure t) x =
    D.nu smul coordLaplacian (D.u t) x
```

with

$$p \;=\; \partial_t W \;+\; \tfrac{1}{2}\langle u, u\rangle \;+\; \langle u, \nabla W\rangle \;-\; \nu\, \Delta W.$$

The pressure is not posited: it is assembled from the exact terms of the transport (gauge drift, kinetic energy via the self-transport identity, gauge advection via the Cartan formula, viscous gauge term), and the theorem is unconditional given the data. This is the repaired route's replacement for v7's identity-energy pushdown: the same endpoint (a genuine Navier–Stokes solution with pressure), reached without the false adjoint bound.

D.4 The vorticity-stretching gate and the conditional route

Continuation is fed by the averaged Cauchy formula $\omega(t) = \mathbb{E}[(DX_t)(\omega_0 \circ A_t)]$, which bounds $\|\omega(t)\|_\infty$ by moments of the flow Jacobian. The gate is the named open input of the coarse route, exactly:

```
/-- The vorticity-stretching gate (named analytic input; OPEN). -/
def StochasticStretchingEstimate
    (D : StochasticCauchyVorticityData Omega) (T : Real) : Prop :=
  IntervalIntegrable D.expectedJacobianBound volume 0 T
```

Its consumers are mechanical and checked: a time-uniform Jacobian bound gives a time-uniform vorticity bound (`vorticity_uniform_bound_of_jacobian_bound`), and under the gate the vorticity norm $t \mapsto \|\omega(t,x)\|$ at each fixed point is interval-integrable with $\int_0^T \|\omega\| \le \|\omega_0\|_\infty \int_0^T \mathbb{E}\|DX_t\|_\infty$ (`bkm_integrand_intervalIntegrable`, `bkm_integral_bound`). The assembled conditional route (`stochasticLagrangian_conditional_route`) states: given transported momentum data, averaged-Cauchy vorticity data linked by $\omega = \operatorname{curl} u$, and the gate on $[0,T]$, the velocity satisfies Navier–Stokes with the explicit pressure *and* the BKM integrand is controlled. Every step is machine-checked; the only analytic input is the gate.

The route is inhabited. Non-vacuity is itself checked, twice. The rest state instantiates the route end-to-end (`conditional_route_rest`, with the gate discharged by a constant envelope). Beyond the rest state, the module `DecayingShearInstance` provides a nonzero exact solution of the full transported-momentum system: the decaying shear $u(t,y) = e^{-t}\sin(y_1)\, e_0$ with gauge amplitude $b(t) = (e^{-2t} - e^{-4t})/8$ on the frequency-doubled mode, whose balance $b' = -4b + \frac{1}{4}e^{-2t}$

(`gaugeAmp_balance`) absorbs the kinetic self-transport term; `decayingShear_navierStokes` confirms the pair satisfies Navier–Stokes with the representation pressure at $\nu = 1$. So the conditional machinery is not vacuously true: its hypotheses have nonzero exact witnesses.

D.5 The frozen-strain countermodel and the aperture geometry

Any proof of the gate must use more than local strain geometry plus stochastic averaging. The frozen-strain model makes this a theorem. For the prescribed strain $A = \mathrm{diag}(2\gamma, -\gamma, -\gamma)$ and the pancake packet (vorticity along the expanding axis, wave vector along a compressive axis), the module `FrozenStrainModel` proves that

$$\omega(t,x) = a(t)\,\sin\big(k(t)\,x_1\big)\,e_0, \qquad k(t) = k_0 e^{\gamma t}, \qquad a(t) = a_0 \exp\Big(2\gamma t - \nu k_0^2 \tfrac{e^{2\gamma t}-1}{2\gamma}\Big)$$

is an *exact* solution of the frozen vorticity equation $\partial_t\omega + (Ax\cdot\nabla)\omega = A\omega + \nu\Delta\omega$ (`frozenStrainVorticity_solves`; the drift, stretching, phase and viscous terms are all checked, not modeled). Writing $g = 2\gamma/(\nu k_0^2)$ for the normalized strain, the amplification peaks at $t^* = \log g/(2\gamma)$ with the exact value

$$\frac{a(t^*)}{a_0} = g\,e^{1/g-1} \qquad (\texttt{frozenStrainAmplification_peak}),$$

which is unbounded over the strain family: no uniform depletion bound of the form "pancake alignment always acquires diffusive decay at rate $\nu\lambda^2$" survives (`frozenStrain_no_uniform_amplification_bound`). Any correct gate proof must therefore use the *self-consistency* of the Navier–Stokes strain — a statement quantified over arbitrary imposed strains is false.

The angular version is the aperture-coverage geometry. Anisotropic Bernstein estimates control cone apertures up to $\lambda^{-1/2}$; stretch–diffusion covariance controls apertures above $\sqrt{\gamma/\nu}/\lambda$. The band claimed by neither,

$$(\lambda^{-1/2},\ \sqrt{\gamma/\nu}/\lambda), \qquad \text{is nonempty} \iff \nu\lambda < \gamma \qquad (\texttt{anisotropic_coverage_gap}).$$

D.6 July additions: the aperture dichotomy is the strain budget

The July campaign sharpened the coverage gap into an equivalence and closed the remaining escape routes. In `ApertureCoverageDichotomy.lean`:

```
theorem pancakeApertureCovered_iff_strain_le
    (hnu : 0 < nu) (hlam : 0 < lam) :
    (forall theta : Real,
        theta <= (Real.sqrt lam)^{-1} \/ Real.sqrt (gamma / nu) / lam <= theta)
      <-> gamma <= nu * lam
```

That is: the "narrow ⇒ enstrophy-controlled, else ⇒ angularly coercive" dichotomy tiles the whole aperture range *exactly when* the scale-local strain budget $\gamma \le \nu\lambda_q$ holds. On the gap band both mechanisms fail quantitatively: `gapAperture_diffusionRate_lt_strain` proves $\nu(\theta\lambda)^2 < \gamma$ below the stretch–diffusion threshold (the diffusive rate is strictly beaten by the strain), and `gapAperture_bernstein_volume_exceeds_planar_budget` proves $\lambda^2 < \theta^2\lambda^3$ above the Bernstein threshold (the cone's Fourier-support volume exceeds the planar budget behind the improved $L^2 \to L^\infty$ constant).

The time-averaged escape route is closed by a Grönwall-saturation computation: over the coherent window the packet consumes an integrated strain budget of exactly $\int_0^{t^*} 2\gamma\,dt = \log g$ (`frozenStrain_integratedStrain_peakWindow`) while amplifying to $g\,e^{1/g-1} \geq e^{\log g-1}$, i.e. amplification at least $e^{\,\mathrm{budget}-1}$ (`frozenStrainAmplification_peak_ge_exp_integratedStrain_sub_one`). An averaged strain hypothesis therefore bounds amplification only exponentially in itself and cannot substitute for the pointwise-in-time aperture condition in a dyadic relaxation estimate.

D.7 The dyadic pancake closure: verified pieces and the isolated pin

The July repair proposal claims a complete closure of the gate: decompose the weighted stretching sum by angular sector, prove the pancake-sector estimate

$$\int_0^T \sum_{q\geq Q} \lambda_q^{-2} \big\| A_q\, \omega_q^{\mathrm{pan}} \big\|_\infty \, dt \;\leq\; C(T,\nu,u_0,Q) + \varepsilon \int_0^T D_Q(t)\,dt, \qquad D_Q(t) := \sum_{q\geq Q} \|\Delta_q \omega(t)\|_\infty,$$

with $A_q = S(S_{q-L}u)$ the low-frequency strain and ω_q^{pan} the projection onto the pancake cone of aperture $\delta_q = \lambda_q^{-1/2}$, then absorb $\varepsilon \int D_Q$ and conclude $\int_0^T D_Q < \infty$, hence BKM. The formalization adjudicates this claim piece by piece.

The genuinely new ingredient is verified. The proposal's Step 1 — a pancake vorticity mode does not generate the strain that stretches itself — is checked exactly, not at symbol level. In `PlaneWaveSelfStrain`: for $\omega(x) = \alpha \sin(\kappa x_1) e_0$ with its Biot–Savart velocity (the shear $u(x) = -(\alpha/\kappa)\cos(\kappa x_1)e_2$), `curlField_planeWaveVelocityMode` proves $\operatorname{curl} u = \omega$ exactly, and `strainRateApply_planeWaveVelocityMode_vorticity` proves $S(u)\,\omega = \frac{1}{2}(\nabla u + (\nabla u)^{\mathsf{T}})\omega = 0$ pointwise, with a non-vacuity witness (`planeWaveVorticityMode_ne_zero`). The scope is honest: the identity is for one exact mode; a pancake *sector* aggregates a cone of modes, and charging the cross-mode strain is precisely the open budget below.

The absorption skeleton is checked with explicit constants. The structure `DyadicVorticityStretchingData` carries the four time functionals (D_Q; pancake, angularly coercive, and residual stretching sums), their nonnegativity and integrability, and the transported dyadic relaxation inequality. The three sector estimates are explicit predicates, e.g.

```
def PancakeSectorEstimate (C eps : Real) : Prop :=
  (Integral of D.pancakeStretchSum over 0..D.T) <=
    C + eps * (Integral of D.highBlockSum over 0..D.T)
```

(coercive and residual analogous; both are *interfaces* here — established in the source analysis by good-unknown/stretch–diffusion and Coifman–Meyer arguments respectively, consumed as named hypotheses, not re-proved). The absorption theorem `highBlockIntegral_le_of_sectorEstimates` proves: sector estimates with total gain $\varepsilon_p + \varepsilon_c + \varepsilon_r < 1$ imply

$$\int_0^T D_Q(t)\,dt \;\leq\; \frac{C_{\mathrm{relax}} + C_p + C_c + C_r}{1 - (\varepsilon_p + \varepsilon_c + \varepsilon_r)},$$

with the a-priori finiteness carried by the integrability fields (supplied in the analytic setting by local-in-time regularity and continuation, which is standard and not re-derived). The theorem `pancakeSectorEstimate_of_scaleLocalStrainBudget` derives the pancake-sector estimate from the pointwise scale-local budget by time integration.

Where the claimed proof fails. The proposal's high-low split charges the whole non-pancake complement to the angularly coercive estimate. By Section D.6 that charge is *equivalent* to the

scale-local budget $\gamma \le \nu\lambda_q$ — which is exactly the bound the proposal's own conditional section had identified as "the remaining burden" before the claimed proof. The claimed proof therefore consumes, at its high-low split, the statement it set out to prove; the closure is circular at precisely that point, and nothing else in it repairs the circle.

The remaining pin is a precise predicate, structurally verbatim:

```
/-- The scale-local pancake strain budget (named analytic input; OPEN --
the first load-bearing missing lemma of the claimed pancake closure). -/
def ScaleLocalPancakeStrainBudget (D : DyadicVorticityStretchingData)
    (B eps : Real) : Prop :=
  exists budget : Real -> Real,
    IntervalIntegrable budget volume 0 D.T /\
    (Integral of budget over 0..D.T) <= B /\
    forall t in Set.Icc 0 D.T,
      D.pancakeStretchSum t <= budget t + eps * D.highBlockSum t
```

Here `pancakeStretchSum` represents $\sum_{q\ge Q}\lambda_q^{-2}\|A_q\omega_q^{\mathrm{pan}}(t)\|_\infty$. The obligation is to establish this pointwise envelope for the *self-consistent* Navier–Stokes strain (generated from the vorticity by the Biot–Savart law), not for an externally imposed frozen strain — via a genuinely Navier–Stokes-specific mechanism (pressure/Hessian feedback, multi-scale Biot–Savart self-depletion, or frequency-cascade accounting, in the proposal's own taxonomy). The verified plane-wave null structure is evidence in the right direction; the aperture equivalence states exactly what quantitative multi-scale estimate is still missing.

The sharpened conditional route. The headline theorem `stochasticLagrangian_conditional_route_of_pancakeBudget` closes the loop: given transported momentum data, dyadic relaxation data whose block sum dominates the high-frequency part of $\|\operatorname{curl} u\|$ (the Littlewood–Paley decomposition interface), the scale-local pancake strain budget, and the coercive/residual interfaces with total gain below 1, the velocity satisfies Navier–Stokes with the explicit pressure and, for each fixed spatial point, the vorticity norm is interval-integrable with an x-independent bound. This reaches the same pointwise endpoint as the coarse conditional route with the gate replaced by the sharper geometric pin; the two pins are cross-referenced in the sources and both remain open. The module `SpatialBKMIntegrand` now promotes that common time majorant to the actual spatial BKM quantity

$$\Omega_\infty(t) = \operatorname*{ess\,sup}_x \|\operatorname{curl} u(t,x)\| \in [0,\infty].$$

It proves exact equality with the library's L^∞ seminorm. Spatial continuity and positivity of volume on nonempty open sets identify this essential supremum with the supremum over a countable dense spatial set. Thus pointwise a.e. time measurability suffices to prove a.e. measurability of the spatial supremum without interchanging an uncountable family of exceptional time sets. For the transported velocity data the result is stronger: explicit spatial difference quotients, together with the recorded time derivatives of velocity and its C^2 spatial regularity, prove time measurability of curl itself. The constructed spatial supremum is therefore measurable with no additional curl-measurability hypothesis.

The common envelope proves that $\Omega_\infty(t) < \infty$ on the relevant slab before conversion to its real representative $\Omega(t)$. The theorem `spatialBKM_of_misalignmentBudget` gives

$$\int_0^T \Omega(t)\,dt \le \int_0^T D_{\mathrm{low}}(t)\,dt + \frac{C_{\mathrm{relax}} + B + C_c + C_r}{1 - (g_{\mathrm{coherent}} + \varepsilon + \varepsilon_c + \varepsilon_r)},$$

under the existing decomposition, misalignment-budget, coercive-sector, and residual-sector hypotheses and the strict total-gain condition. The extended-valued integral is also bounded directly, so finiteness is not an artifact of the convention that converts infinity to real zero.

Finally, an increasing exhaustion of $(0, T_*)$ proves that a single bound on the envelope integrals for all shorter horizons yields a finite spatial BKM integral on that open terminal interval. The envelopes may vary with the shorter horizon, but their integral bound must be uniform. This constructs the spatial endpoint and its limiting step; it does not prove that uniform analytic bound, construct the stochastic data from arbitrary initial data, or invoke a formal continuation theorem. An end-to-end rest-state canary (`rest_conditional_route`) checks that the refined hypotheses are simultaneously satisfiable.

E The direction-coherence refinement of the pin

The September 2026 campaign adjudicated the Biot–Savart self-depletion mechanism of the pancake budget by pre-registered kinematic experiment, then distilled and proved the surviving content, and committed a strictly sharper pin. All statements below are machine-checked with the standard three axioms; the labs are `ns_pancake_kinematic_depletion_lab.py` and `ns_pancake_coherent_aggregate_lab.py` in the validation lab, with pre-registered verdict criteria recorded in the files before execution.

The dichotomy. Wavevector-cone confinement alone does not deplete pairwise stretching. In the exact lattice interaction model (the amplitude-level Fourier symbol of $S(u)\omega$ with Biot–Savart velocity), the aggregate cone gain equals the cone measure (measured $D \sim \delta^{2.06}$ across dyadic shells), while the worst pair saturates the generic maximum at every aperture. The distilled exact statement (`PancakeConeDirectionDichotomy`): the axial source $k_1 = (0, 0, m)$ with vorticity e_x stretches the *perpendicular* in-plane vorticity direction e_y of the cone receiver $k_2 = (1, 0, M)$ with $\sigma^2 = 1/4$ exactly, for all m, M; the *aligned* polarization $(-M, 0, 1)$ of the same wavevector pair is depleted at exactly $\sigma^2 = 1/(4(1 + M^2))$. The operative small parameter of the self-depletion mechanism is therefore the vorticity-direction misalignment inside the pancake — the Constantin–Fefferman coherence quantity — not the wavevector cone.

The coherent pair estimate (proved). Give every mode the direction-coherent polarization: the fixed in-plane reference e_x projected off the wavevector. Then (`PancakeCoherentPairEstimate.sigmaSq_coherent_le`) for wavevectors inside pancake cones of squared apertures $d_1, d_2 \le 1/4$,

$$\sigma^2 \le 6\,(d_1 + d_2),$$

with the mechanism carried by the exact identity $k \times (|k|^2 e_x - (k \cdot e_x)k) = |k|^2(0, k_z, -k_y)$: the Biot–Savart velocity of a coherent mode is perpendicular to the reference direction. The bound aggregates over any finite family of coherent sources (`sum_sigmaSq_coherent_le`), matching the measured coherent-aggregate gain of the second lab.

Amplitude and field-transfer layer (proved, but not yet complete). The module `PancakeCoherentWeightedTransfer` inserts the actual scalar Fourier amplitudes into the pair estimate. It proves the cardinality-free finite convolution bound

$$\Big\| \sum_{k \in E} T_k(a_k p_k, b p_\ell) \Big\|_{\ell^1(\mathbb{C}^3)} \le 6(\delta_1 + \delta_2)\Big(\sum_{k \in E} |a_k| \Big) |b|,$$

and its Cauchy–Schwarz energy form, where $p_k = P_k e_0$. The exact theorem `PancakeConeLatticeCount.card_le_pancakeBox` shows that, at $\lambda = N^2$ and aperture $1/N$, the cone lies in a box of cardinality $(4N+1)^2(4N^2+1) = O(\lambda^2)$, rather than the ambient $O(\lambda^3)$ count.

The modules `PancakePeriodicCoherentSplit` and `PancakePeriodicComplexStretch` then move from the rational symbol model to the genuine Fourier lattice $\mathbb{Z}^3$ and arbitrary complex coefficients. For every cone mode they construct the exact orthogonal split

$$\widehat{\omega}(k) = \frac{\langle P_k e_0, \widehat{\omega}(k)\rangle}{\langle P_k e_0, P_k e_0\rangle} P_k e_0 + \widehat{\omega}_{\mathrm{mis}}(k), \qquad \langle P_k e_0, \widehat{\omega}_{\mathrm{mis}}(k)\rangle = 0,$$

prove that both terms remain divergence-free, define the complexified Biot–Savart strain symbol, and prove its agreement with the rational symbol. The aperture estimate and the $O(\lambda^2)$ energy bound therefore hold for arbitrary complex coherent amplitudes, not only rational test vectors.

The module `PancakeRealCoherentPairEstimate` removes the rational-grid restriction needed for a smooth multiplier. It proves the same cancellation for all real frequencies, constructs the rank-one orthogonal coherent projection of an arbitrary amplitude, proves that this projection does not increase Euclidean energy, and obtains the field-facing pointwise estimate against the energies of the original input amplitudes.

The first attempted receiver-only endpoint has now been fenced off rather than promoted. The generic modules `PancakeConeKernelScaling` and `PancakeConeOperatorKernel` correctly prove that $K_\delta(x_\perp, z) = \delta^3 K(\delta x_\perp, z)$ has $\|K_\delta\|_{L^1} = \delta \|K\|_{L^1}$, but one of those three powers must already be a genuine symbol gain; only two come from the transverse Jacobian. The periodic theorem `PancakePeriodicOperatorKernel` now proves the corresponding no-go: if an exact projection fixes a unit Fourier mode, its kernel has L^1 mass at least one. Thus the smooth family in `PancakeSmoothCoherentAperture`, whose symbol was explicitly attenuated by an additional factor δ, is a useful analytic model but is not the desired coherent field projection.

The genuine aperture factor has instead been located in the complete bilinear interaction. For $k_1 = (\delta a, \delta b, c)$ and $k_2 = (\delta d, \delta e, f)$, `PancakeCoherentStretchFactorization` proves exactly, both for the canonical coherent vectors and for coherent projections of arbitrary input amplitudes,

$$T(k_1, k_2) = \delta\, T_{\mathrm{norm}}(\delta, a, b, c, d, e, f),$$

where T_{norm} is smooth through $\delta = 0$ before the projection denominators are introduced. The module `PancakeSmoothCoherentStretchSymbol` supplies nested cutoffs keeping both axial frequencies away from zero, constructs all 27 entries of the localized normalized tensor as compactly supported Schwartz symbols on six frequency variables, proves the tensor is nonzero, and gives an integrable inverse-Fourier kernel with exact entrywise and tensor Fourier recovery for each fixed δ. Finally, `PancakeBilinearKernelScaling` proves the correct four-transverse variable scaling

$$K_\delta(y_\perp, z_\perp; y_\parallel, z_\parallel) = \delta^5 K(\delta y_\perp, \delta z_\perp; y_\parallel, z_\parallel),$$

whose four-dimensional Jacobian leaves one power of δ, together with the resulting bilinear $L^\infty \times L^\infty \to L^\infty$ Young bound. The module `PancakeUniformKernelFamily` differentiates the symbol jointly in δ and all six frequency variables, uses its fixed compact support to obtain uniform Schwartz seminorms, converts those bounds into uniform polynomial decay of every inverse-Fourier entry, and sums the 27 entries to prove

$$\sup_{0 \le \delta \le 1/2} \int_{\mathbb{R}^6} \|K_{\mathrm{norm},\delta}(y, z)\|\, dy\, dz < \infty.$$

Thus the normalized Euclidean kernel-mass obligation is discharged. The module `PancakeBilinearPeriodization` then carries this kernel to the standard six-torus. It proves the

fundamental-domain unfolding lemma, matches the torus characters to the Euclidean Fourier normalization, and shows for every $n \in \mathbb{Z}^6$ that

$$\widehat{K^{\mathrm{per}}_{\mathrm{norm},\delta}}(n) = M_{\mathrm{norm},\delta}(n).$$

It also proves that the resulting six-dimensional kernel acts on a pair of three-torus Fourier monomials through this exact coefficient. Restoring the algebraically extracted factor gives a periodic kernel realizing the localized coherent-stretching symbol and one constant C, independent of $0 \le \delta \le 1/2$, such that

$$\|B_\delta(f,g)\|_{L^\infty(\mathbb{T}^3)} \le 9\,\delta C\,\|f\|_{L^\infty(\mathbb{T}^3)}\|g\|_{L^\infty(\mathbb{T}^3)}.$$

Thus Euclidean-to-torus transference and the frequency-uniform frozen-axis coherent aperture bound are discharged. The module `PancakeFiniteFourierBilinearTransfer` distributes the bilinear action over arbitrary finite trigonometric polynomials and proves the exact double Fourier sum without a mode-cardinality loss. It also identifies the assembled 27-entry symbol, on the retained chart, with the complex-bilinear extension of the coherently projected normalized Biot–Savart stretching interaction; off that chart both sides vanish. Thus the periodic kernel now acts exactly as the localized coherent PDE interaction on finite Fourier fields.

The module `PancakeDyadicKernelScaling` carries this base chart to actual square-dyadic shells. For every positive r it proves

$$\int_{\mathbb{R}^6} \|r^6 K(rx)\|\,dx = \int_{\mathbb{R}^6} \|K(x)\|\,dx, \qquad \widehat{r^6 K(r\,\cdot)}(\xi) = \widehat{K}(\xi/r),$$

and proves the corresponding exact fundamental-cell coefficient after periodization. It then constructs a volume-preserving regrouping of the six variables into four transverse and two axial coordinates and proves

$$\int_{\mathbb{R}^6} \|\delta^5 K(\delta x_\perp, x_{\mathrm{ax}})\|\,dx = \delta \int_{\mathbb{R}^6} \|K(x)\|\,dx, \qquad \widehat{K_\delta}(\xi) = \delta \widehat{K}(\xi_\perp/\delta, \xi_{\mathrm{ax}}).$$

At aperture $\delta = 1/N$ and shell scale $r = N^2$, the reciprocal argument is machine-checked to be precisely the source/receiver chart $(k_1/N, k_2/N, k_3/N^2)$ (and similarly for the second input). Consequently the periodized square-dyadic kernel has both the exact coefficient

$$\widehat{K^{\mathrm{per}}_N}(k,l) = \frac{1}{N} M_{\mathrm{norm},1/N}\left(\frac{k_1}{N}, \frac{k_2}{N}, \frac{k_3}{N^2}; \frac{l_1}{N}, \frac{l_2}{N}, \frac{l_3}{N^2}\right)$$

and the uniform fundamental-cell estimate $\int \|K^{\mathrm{per}}_N\| \le C/N$ for every $N \ge 2$. Thus the frozen-axis anisotropic square-dyadic transfer is discharged. The full-annulus and variable-frame assembly is described below.

The module `PancakeSquareDyadicFiniteTransfer` descends this final kernel through the canonical half-open representative of the unit six-torus. It proves that torus Fourier coefficients and torus L^1 mass are exactly the corresponding fundamental-cell quantities, and hence that arbitrary finite Fourier fields are acted on by the exact double sum of square-dyadic coherent interactions. The endpoint estimate is cardinality-free:

$$\|B_N(f,g)\|_{L^\infty(\mathbb{T}^3)} \le 9\frac{C}{N}\,\|f\|_{L^\infty(\mathbb{T}^3)}\|g\|_{L^\infty(\mathbb{T}^3)}, \qquad N \ge 2.$$

Thus the finite-field frozen-axis square-dyadic transfer is also discharged.

The module `PancakeFrozenFrameKernelTransfer` performs the frozen-frame step at the kernel level. Because a general rotation does not preserve $\mathbb{Z}^3$, it first rotates the Euclidean cover kernel

and only then periodizes on the fixed standard lattice. Orthogonal change of variables preserves the exact cover L^1 mass, while Fourier covariance evaluates the symbol at the rotated reciprocal frequency. The module constructs a genuine Euclidean linear isometry from every algebraic `OrientedFrameEquiv`, lifts it diagonally to the two interacting frequencies, and proves the exact finite-Fourier identity with the retained coherent Biot–Savart interaction in the resulting oriented square-dyadic chart. One constant works for all such frames:

$$\|B_{N,F}(f,g)\|_{L^\infty(\mathbb{T}^3)} \le 9\frac{C}{N}\,\|f\|_{L^\infty(\mathbb{T}^3)}\|g\|_{L^\infty(\mathbb{T}^3)}, \qquad N \ge 2.$$

Thus frozen oriented-frame transference of the exact square-dyadic kernel is discharged. Physical vector conjugation, pointwise frame variation, and the replacement of the initial demonstration chart by the full annulus are handled next.

The module `PancakePhysicalFrameKernelTransfer` supplies the missing input/output conjugation. It complexifies the algebraic real frame, proves that complexification commutes with inversion, and bounds both the forward and inverse maps by the dimension-only coordinate constant 3. Applying the frequency-rotated kernel to the two frame-coordinate inputs and rotating the output back therefore gives

$$\|B^{\mathrm{phys}}_{N,F}(f,g)\|_{L^\infty} \le 243\frac{C}{N}\|f\|_{L^\infty}\|g\|_{L^\infty}.$$

The exact coherent finite-Fourier formula is proved with these three frame maps in place. More importantly, the same estimate is proved for arbitrary continuous bounded fields and then for an arbitrary pointwise frame field $F = F(x)$, simply by instantiating the frame-uniform kernel estimate at the observation point. No derivative of F occurs. This removes the variable-frame commutator as an analytic loss at this particular pointwise L^∞ endpoint.

The modules `PancakeAnnularChartCutoff`, `PancakeAnnularUniformKernel`, and `PancakeAnnularMultiplierAssembly` remove the remaining demonstration-chart restriction. On a square-dyadic annulus the cone inequality forces each normalized axial coordinate into $[-2,-3/4]\cup[3/4,2]$, while the transverse coordinates lie in $[-2,2]$. There are therefore exactly four source/receiver sign combinations. Smooth compact buffers are constructed for them, their coherent denominators are proved globally nonzero, and their inner cutoffs satisfy the exact partition identity

$$\chi_{++}+\chi_{+-}+\chi_{-+}+\chi_{--}=1$$

throughout the normalized annulus. The parametric Schwartz estimate is proved for every buffered cutoff and yields one aperture-uniform inverse Fourier L^1 bound for the assembled four-chart kernel. Its Fourier symbol is machine-checked to equal the *uncut* coherent Biot–Savart stretching interaction on the whole annulus.

Finally, `PancakeAnnularFrozenFrameTransfer` and `PancakeAnnularPhysicalFrameTransfer` carry this assembled multiplier through the rotated Euclidean cover, square-dyadic rescaling, periodization, physical input/output conjugation, and arbitrary pointwise frame fields. The exact finite-Fourier identity and the cardinality-free estimate

$$\|B^{\mathrm{ann}}_{N,F(x)}(f,g)(x)\| \le 243\frac{C}{N}\|f\|_{L^\infty}\|g\|_{L^\infty}$$

hold uniformly for $N \ge 2$. Thus finite sign-chart coverage and partition assembly are discharged. The module `PancakeAnnularSectorProjector` then defines the concrete finite Fourier projector obtained by filtering a given support with the normalized pancake-annulus predicate.

It proves, coordinate by coordinate, that the six-dimensional frozen square-dyadic chart of a selected pair is exactly the pair of their independently selected three-dimensional normalized modes. Consequently the full-annulus premise of the exact physical kernel formula is automatic on these projected fields; it is no longer a free support hypothesis. The subsequent module `PancakeSquareDyadicSectorProjector` narrows this enclosing selector to the genuine microlocal sector: radial frequency in $[N^2, 2N^2)$ and squared transverse aperture at most N^{-2} after rotation into the chosen frame. From these physical inequalities it proves the transverse bounds and the nontrivial axial lower bound $3/4$, hence inclusion in the four-chart annulus. The exact coherent-action formula therefore applies directly to the concrete finite shell-and-cone projections. The modules `PancakeAxialFrameConstruction` and `PancakeStrainAxialFrame` then remove the caller-supplied pointwise frame. The first constructs, by two explicit reflections, an orientation-preserving orthogonal map sending any unit direction to the axial vector; the second feeds it the smallest-eigenvalue direction supplied by the ordered spectral theorem for the symmetric strain. Thus the pointwise C/N endpoint now applies directly to a frame constructed from the strain. Measurable dependence of the spectral choice through eigenvalue collisions, and identification of these concrete finite projectors with the actual Littlewood–Paley coherent share, are still required.

The modules `PancakeComplexMisalignmentExpansion`, `PancakeComplexCoherentPDEIdentification`, and `PancakePhysicalMisalignmentSplit` close the algebraic split behind that remaining analytic task. The former expands the actual complex Biot–Savart stretching interaction into the coherent–coherent term and the three terms containing at least one misalignment factor, first modewise and then under arbitrary finite source/receiver Fourier sums. Thus the finite-field norm analogue of `split_le` follows from an exact identity and one triangle inequality. The latter proves that N^{-1} times the normalized annular symbol is literally the complexified PDE stretching operator applied to the two coherent chart projections. On the genuine rotated square-dyadic sector, the exact physical kernel formula is therefore rewritten directly as a sum of coherent–coherent PDE interactions; no analogy between two separately defined expressions remains. The third module performs the sum-level substitution: for arbitrary finite Fourier data it proves the exact physical identity

$$\mathcal{S}_{N,F} = \mathcal{K}^{\mathrm{coh}}_{N,F} + \mathcal{R}^{\mathrm{mis}}_{N,F},$$

where the remainder is precisely the sum of the coherent–misaligned, misaligned–coherent, and misaligned–misaligned interactions. A triangle inequality and the annular kernel theorem give

$$|\mathcal{S}_{N,F}(x)| \le 243(C/N)MP + |\mathcal{R}^{\mathrm{mis}}_{N,F}(x)|,$$

with no dependence on the two Fourier support cardinalities. The same theorem is specialized to the pointwise frame built from the most compressive strain eigenvector, so this statement no longer accepts a caller-supplied frame. The follow-up module `PancakeFullStrainFrame` fixes the residual rotation around the compressive axis by an explicit orientation-preserving planar rotation. It proves that the inverse complex image of e_x is exactly the most expanding strain eigenvector and repeats the exact split and C/N bound in this full eigenline-aligned frame. Consequently the Fourier remainder's coherent reference is now the same expanding line used by the invariant alignment-energy dynamics below.

The module `PancakeMisalignmentEnergyBridge` makes that compatibility quantitative. If a complex Fourier amplitude u is divergence free at frequency k, its exact coherent-line remainder $u - P_k u$ satisfies

$$|u - P_k u|^2 \le |u_y|^2 + |u_z|^2$$

in the full strain chart, where the chart's x-axis is the expanding eigenline. The proof applies the real orthogonal-projection inequality to the real and imaginary parts and proves that complexification

of the oriented frame preserves the divergence pairing. Summing over the actual integer modes selected by the physical square-dyadic projector introduces no cardinality factor. An exact finite Parseval theorem then gives

$$\sum_{k\in\Gamma_{N,F}} |u_k - P_k u_k|^2 \le \int_{\mathbb{T}^3} \big(|U_y(x)|^2 + |U_z(x)|^2\big)\, dx,$$

where U is the finite reconstruction in one frozen full strain frame. This closes the coefficient-to-L^2 bridge; it does not by itself control a frame varying with x, nor does an L^2 estimate alone supply the L^∞-based BKM integrand.

The follow-up module `PancakeTransverseEnergyFreezing` supplies the local varying-frame comparison without choosing eigenvector signs. For symmetric strains A, B with $\|A - B\| \le \varepsilon$, it first proves the top-eigenvalue Weyl bound

$$|\lambda_0(A) - \lambda_0(B)| \le \|A - B\|.$$

It then applies the matched-eigenline projector estimate. For every positive threshold, either a frozen top gap of B is smaller than threshold $+\,\varepsilon$, or

$$\sum_{k\in\Gamma_{N,A}} |u_k - P_{A,k} u_k|^2 \le \int_{\mathbb{T}^3} |U_{\perp B}(x)|^2\, dx + 4\frac{\varepsilon}{\text{threshold}} \int_{\mathbb{T}^3} |U(x)|^2\, dx.$$

Thus the eigenvector-sign and eigenvalue-collision problems are no longer hidden in a continuity premise. The remaining task is to aggregate this cellwise alternative with the same-strain small-gap coercivity and the space–time Littlewood–Paley budget.

The module `PancakeTransverseEnergyCoercivity` closes the separated cell algebra. If S is symmetric, e_0 is its top eigenvector, and both top gaps dominate g, then for every vector u—including $u = 0$—

$$g\big(|u|^2 - |u \cdot e_0|^2\big) \le \lambda_0(S)|u|^2 - u \cdot Su.$$

The proof is the three-dimensional spectral expansion and is deliberately homogeneous, so it never divides by vorticity magnitude. Applying it to the real and imaginary parts, integrating the finite reconstruction, and composing with the freezing theorem yields the checked cellwise dichotomy: either a frozen top gap is below threshold $+\,\varepsilon$, or the gap-weighted exact Fourier remainder is bounded by the integrated complex Rayleigh defect plus $(\text{threshold} + \varepsilon)4\varepsilon/\text{threshold}$ times the full sector energy. The remaining issue is no longer the separated-cell estimate itself, but its bounded-overlap space–time aggregation and endpoint scaling.

The finite combinatorics of that aggregation is now checked in `PancakeCellAggregation`. It packages one frozen comparison datum, filters an arbitrary finite family into collision and separated cells, and sums the separated coercive estimates with no support-cardinality constant. Nonnegativity of both the complex Rayleigh defect and full energy allows the right-hand sums to be enlarged to the whole family. Two generic analytic lemmas prove the corresponding bounded-overlap transfer: for measurable hard cells of multiplicity at most M,

$$\sum_i \int_{Q_i} f \le M \int f,$$

and for nonnegative smooth weights χ_i satisfying $\sum_i \chi_i \le M$, the same conclusion holds for varying local densities dominated by one global density. This does *not* yet identify the global Parseval energy in each frozen comparison with a hard- or smooth-localized cell energy. A spatial cutoff changes

Fourier support, so that step requires a genuine localization/commutator or almost-orthogonality estimate; bounded overlap alone cannot supply it.

The first part of that localization junction is now exact rather than an interface. The module `PancakeFiniteFourierLocalization` proves that for finite Fourier cutoff χ and vector field U,

$$\chi(x)U(x) = \sum_{p,k} e^{2\pi i(p+k)\cdot x}\, \widehat{\chi}(p)\widehat{U}(k).$$

Thus the localized output support is precisely contained in the Minkowski sum of the cutoff and field supports, and every output mode differs from an original sector mode by an actual cutoff frequency. It also proves directly in physical space that $\|\chi U\|_{L^2}^2 \le C^2\|U\|_{L^2}^2$ whenever $\|\chi\|_{L^\infty} \le C$; there is no mode-count loss. Additivity of the normalized oriented-frame chart then proves that cutoff support bounded by transverse and axial margins $(\rho_\perp, \rho_\parallel)$ moves the original annulus into the explicit buffered box

$$|v_{1,2}| \le 2 + \rho_\perp, \qquad \tfrac{3}{4} - \rho_\parallel \le |v_3| \le 2 + \rho_\parallel.$$

For the fixed margins $(\rho_\perp, \rho_\parallel) = (1/4, 1/8)$, the module `PancakeBufferedAnnularMultiplier` now completes the next part of this junction. It enlarges the signed retained axial gates from radius $3/4$ to $7/8$, while leaving their support radius 1 and the old denominator-safe outer gates unchanged. The resulting four cutoffs form an exact partition on the whole leaked box

$$|v_{1,2}| \le \frac{9}{4}, \qquad \frac{5}{8} \le |v_3| \le \frac{17}{8}.$$

The assembled tensor symbol therefore equals the uncut coherent stretching symbol on every permitted localized output pair. Applying the generic buffered-cutoff theorem chartwise and summing gives one aperture-uniform L^1 bound for the assembled inverse kernel.

The modules `PancakeBufferedAnnularFrozenFrameTransfer` and `PancakeBufferedAnnularPhysicalFrameTransfer` carry this enlarged operator through anisotropic aperture scaling, square-dyadic rescaling, frozen rotation, periodization, and physical input/output conjugation. They prove the exact torus coefficient formula and retain the cardinality-free C/N endpoint, uniformly even when a different frozen frame is selected at each output point. Finally, `PancakeLocalizedBufferedTransfer` collects the repeated cutoff/field pairs by their output frequency and proves the exact physical coherent identity directly on products χU and ψV. Its premises mention only the original annular modes and the finite-band cutoff margin; the leaked-support condition is derived rather than assumed. It also proves the expected endpoint with losses only $\|\chi\|_\infty\|U\|_\infty$ and $\|\psi\|_\infty\|V\|_\infty$, never a support cardinality.

This closes the finite-band localization junction. A nonconstant smooth spatial cutoff still has an infinite Fourier tail, so the operator side of its finite approximation is checked in `PancakeBilinearActionStability` and `PancakeBufferedLocalizationStability`: for the physical buffered operator $\mathcal{B}_N$,

$$\|\mathcal{B}_N(f,g) - \mathcal{B}_N(f_0,g_0)\| \le \frac{243C}{N}(\|f-f_0\|_\infty\|g\|_\infty + \|f_0\|_\infty\|g-g_0\|_\infty).$$

A further specialization transfers uniform scalar cutoff-approximation errors for $\chi - \chi_0$ and $\psi - \psi_0$ directly into this bound. Thus only the cutoff-specific Fourier decay input remained after that module. More precisely, `PancakeSmoothCutoffFourierTail` proves that every continuous torus cutoff with absolutely summable Fourier coefficients has, for each $\varepsilon > 0$, a finite Fourier approximation within ε uniformly in space. It further proves that one such finite support lies in the

fixed normalized cutoff-margin box at all sufficiently large square-dyadic scales. The scale threshold is uniform over every oriented frame because rotations preserve the Euclidean norm.

That remaining cutoff construction is now discharged in `PancakeConcreteSmoothCellCutoff`. A fixed compactly supported smooth real bump is periodized on the unit circle and tensored over three coordinates. Poisson transference identifies its Fourier coefficients with integer samples of a Schwartz Fourier transform, proving absolute summability. Translating this cutoff over an explicit $5^3 = 125$ quarter-grid produces a finite smooth cover: at every torus point at least one cutoff equals one, the summed quadratic weights have explicit uniform lower and upper bounds, and every translated cutoff retains absolute Fourier summability. One common scale therefore supplies simultaneous finite-Fourier approximations of all 125 cells. The module `PancakeConcreteSpatialCoverTransfer` feeds those approximations through the buffered stability theorem. For every pair of actual smooth cells and every sufficiently large scale, the localized physical interaction differs from the named coherent finite convolution by the explicit $O(\varepsilon/N)$ error, uniformly in the cells, frame, and support sizes. At this junction only the passage from the hard annular selector to the route's chosen smooth Littlewood–Paley projector remains.

The new module `PancakeArbitraryDataBlocks` removes the separate finite-ambient-support restriction. It enumerates complete hard radial shells directly in the integer lattice, proves that every nonzero mode belongs to a unique ordinary dyadic shell, and shows that each ordinary shell is covered without a gap by two adjacent square-dyadic multiplier shells. The lower and upper pieces split exactly into pancake and nonpancake modes, both as Fourier polynomials and as Parseval energies. Applied to an arbitrary periodic field, the pancake pieces satisfy the exact coherent multiplier formula and the actual smooth-cell $O(\varepsilon/N)$ transfer above; the ambient field need not have finite Fourier support. For a jointly continuous time-dependent field, every shell-energy profile is continuous and interval-integrable. At the natural L^2 level, vector Parseval and an explicit reindexing prove

$$\|u\|_{L^2}^2 = |\widehat{u}(0)|^2 + \sum_{q\ge 0} \|\Delta_q u\|_{L^2}^2,$$

with summability of the nonnegative shell series; for mean-zero data the first term vanishes. This is the complete hard Littlewood–Paley energy ledger, not a finite-support surrogate.

The companion module `PancakeArbitraryDataMisalignment` connects these complete blocks to the exact PDE stretching split. On every complete square-dyadic pancake block, the full physical stretching sum is the coherent kernel action plus the exact three-term misalignment remainder, and the coherent term keeps its support-cardinality-free C/N bound. For divergence-free data, the squared misalignment coefficients are controlled by the physical transverse energy in the expanding strain frame; their frozen-frame time profile is continuous and interval-integrable whenever the Fourier coefficients are continuous. The collision-safe frozen-strain dichotomy therefore now applies to complete arbitrary-data blocks rather than an externally supplied finite support.

Finally, `PancakeConcreteCoverEnergy` performs the first actual cell-aggregation step. For every continuous nonnegative density f,

$$\int f \le \sum_{j=1}^{125} \int |\chi_j|^2 f \le 125 \|\chi\|_\infty^2 \int f.$$

This is specialized to complete pancake-block energy, transverse misalignment energy, and frozen-strain Rayleigh defect. Spectral-gap coercivity is also proved after multiplication by each genuine smooth-cell weight, and summing the weighted inequalities introduces no cell-count factor beyond the explicit fixed cover comparison. The module `PancakeConcreteWeightedFreezing` then allows

the frozen strain B_j to vary with the cell. If every B_j is within ε of the observation strain and its top gap dominates threshold $+\,\varepsilon$, the complete arbitrary-data misalignment energy satisfies

$$(\text{threshold}+\varepsilon)\,M \le \sum_{j=1}^{125}\int |\chi_j|^2 D_{B_j} + (\text{threshold}+\varepsilon)\frac{4\varepsilon}{\text{threshold}}125\|\chi\|_\infty^2\int |u_{\text{pan}}|^2.$$

Thus varying the frozen strain from cell to cell introduces neither a frequency-support cardinality nor an additional cell-count loss. What remains is to construct these B_j from the true space–time strain with the required oscillation bound and to treat cells entering the small-gap branch.

The module `PancakeSpectralProjectorRegularity` addresses the measurability issue on the complementary separated branch. Writing $P_{\max}(S)$ for the rank-one projector onto the expanding line and $g(S) = \lambda_0(S) - \lambda_1(S)$, it proves that if $\|A - B\| < g(B)/2$, then

$$\|P_{\max}(A) - P_{\max}(B)\| \le \frac{8}{g(B)}\|A - B\|.$$

Consequently $P_{\max}(S(x,t))$, and the complex transverse-energy density formed from it, are continuous and measurable on every region where a continuous strain field has positive top gap. This avoids both arbitrary eigenvector signs and any false continuity claim through a collision.

The module `PancakeSpectralClusterDichotomy` then resolves the topology of the top-gap collision branch. It proves the bottom-eigenvalue Weyl bound and local Lipschitz regularity of the projector onto the expanding two-plane (the complement of the bottom eigenline) wherever $g_{\text{bot}} = \lambda_1 - \lambda_2 > 0$. It also proves the exact defect identity

$$D_S(u) = g_{\text{top}}\,|u_1|^2 + (g_{\text{top}} + g_{\text{bot}})\,|u_2|^2,$$

with the corresponding real/imaginary complex-coordinate energies. Finally, for trace-free strain, if both adjacent gaps are below τ, then $|\lambda_i| < 2\tau$ for all three eigenvalues and

$$\langle u_{\text{re}}, Su_{\text{re}}\rangle + \langle u_{\text{im}}, Su_{\text{im}}\rangle \le 2\tau|u|^2.$$

Thus a small top gap now has two explicit subbranches: a stable expanding plane, or uniformly small stretching. The unresolved collision geometry is specifically the dynamical coherence problem inside that stable plane.

The module `PancakeExpandingPlaneResidence` supplies the first dynamic estimate on that branch. For a unit vorticity direction ξ and a unit bottom eigenvector e_2, define the sign-invariant contracting-line energy $C = \langle \xi, e_2\rangle^2$. This is exactly the energy escaping the stable expanding-plane projector. The checked spectral identity gives

$$g_{\text{bot}}(1 - C) \le \langle \xi, S\xi\rangle - \lambda_2.$$

Consequently, while $C \le 1/2$, the direction equation yields the integrated residence estimate

$$\int_0^T g_{\text{bot}}(t)C(t)\,dt \le C(0) + 2\int_0^T |\langle d, e_2\rangle + \langle \xi, \dot{e}_2\rangle|\,dt,$$

where d is the non-strain part of the direction rate. This proves that strain damps leakage into the contracting line even when the two expanding eigenvalues collide. It deliberately does not replace the two right-hand forcing channels by an a priori Navier–Stokes bound; doing so, and controlling coherence between directions inside the expanding plane, is the remaining dynamic content rather than a consequence of spectral algebra alone.

There is a second field-level issue which a fixed-axis multiplier does not address. The actual pancake frame is the eigensystem of the low-frequency strain and therefore depends on (x,t). The module `PancakeFrameCovariance` proves that the continuous coherent estimate is rotation-covariant for every frozen oriented orthonormal frame, including arbitrary projected amplitudes. The module `PancakeVariableFrameCommutator` then gives the exact finite-Fourier identity

$$T_{m(x,\cdot)}f = T_{m(x_0,\cdot)}f + T_{m(x,\cdot)-m(x_0,\cdot)}f$$

and bounds the second term by the frequencywise symbol oscillation times the actual Fourier-amplitude mass. The uniform frozen-axis periodic kernel bound is now supplied by `PancakeBilinearPeriodization`; exact finite-field summation and the full anisotropic square-dyadic scaling are supplied by the two subsequent modules above; frozen oriented-frame kernel transference is supplied by `PancakeFrozenFrameKernelTransfer`; the full four-sign annulus and its physical variable-frame endpoint are supplied by the annular modules above, and the pointwise frame is now constructed from the strain. The hard dyadic time profiles, their pointwise Parseval identities, and their interval integrability for jointly continuous fields are now explicit. The remaining field-transfer pieces are aggregation of the proved frozen-cell/small-gap alternative over a physical space–time cover (the separated line and stable-plane projectors are now measurable, and the triple-collision branch has small stretching), passage to the intended smooth projectors, and identification of the hard pancake/nonpancake split with the route's dynamically defined coherent/misalignment functional. The older frequencywise commutator estimate remains correct, but the new uniform physical-kernel endpoint shows that no frame-oscillation estimate is needed merely to obtain the pointwise L^∞ bound. They are not silently included in the finite-sector estimate, and `coherent_le` has not yet been instantiated for the time-dependent Navier–Stokes data structure.

There is also a concrete scale warning at this junction. The present 5^3 smooth cover is a fixed-scale localization device. It gives a valid bounded-overlap energy comparison and uniform Fourier-cutoff transfer, but it cannot by itself force an arbitrarily oscillatory strain field to be ε-close to a constant on each cell. Instantiating the weighted freezing theorem therefore requires a scale-refined cover with overlap bounds independent of the refinement (and compatible cutoff-frequency margins), or a pointwise variable-strain argument. Treating the fixed 125 cells as if they automatically supplied arbitrary strain oscillation would be an invalid closure. The module `PancakeAdaptiveStrainCover` now supplies the correct topological replacement. For every continuous spatial strain, and also for every continuous strain on the compact space–time cylinder, it constructs a finite field- and tolerance-dependent cover together with a continuous partition of unity subordinate to the sets

$$\{z : \|S(z) - S(z_j)\| < \varepsilon\}.$$

Consequently a nonzero cell weight certifies the local freezing estimate at that point. The weighted linear and quadratic freezing errors are proved bounded by ε and ε^2, respectively, with no cell-count factor; applying the operator error to a vector costs only its norm. This closes the finite adaptive-cover existence and aggregation problem.

The module `PancakeAdaptiveFourierPartition` supplies the next splice. Multivariate Stone–Weierstrass density of torus characters is converted into finitely supported Fourier coefficients approximating every continuous adaptive weight, simultaneously over the finite family. More robustly, it approximates the square roots of the nonnegative partition weights and uses the squared complex norms of the resulting trigonometric polynomials. These quadratic weights are nonnegative, differ pointwise from the original weights by at most $\delta(\delta+2)$, and their total differs from one by at most $N\delta(\delta+2)$ for N adaptive cells. If this last quantity is at most $1/2$, the total Fourier weight

stays between 1/2 and 3/2 everywhere. Exact spatial support is necessarily lost under Fourier approximation, but where an original cell weight vanishes the corresponding quadratic tail is proved smaller than δ^2.

The module `PancakeQuadraticFourierSupport` then collects the quadratic weight itself as a finite Fourier polynomial. If p has coefficient support K, the collected support is proved contained in $-K+K$, has cardinality at most $|K|^2$, and evaluates exactly to $|p(x)|^2$; thus the analytic weight and its multiplier coefficients are not separate surrogates. The finite difference set eventually fits the fixed normalized localization-margin box at every sufficiently large square-dyadic scale, uniformly over every oriented frozen frame. For a finite adaptive family, one common scale threshold works for all cells. The integrated existence theorem supplies, for one and the same coefficient family, the 1/2–3/2 coverage bound, the δ^2 off-cell leakage, the exact quadratic coefficient identity, and the common buffered-frequency threshold. Consequently the remaining localization burden is the analytic estimate that absorbs the quantified tails after nonlinear material boundary transfer, uniformly across dyadic scale and adaptive aggregation; the small-gap time branch also remains open. The module `PancakeAdaptiveFourierFreezing` makes the first such cost explicit at the frozen-strain level. If the Fourier quadratic weights differ from the subordinate partition by at most $\eta = \delta(\delta+2)$ and every center error is globally bounded by M, then

$$\sum_i |p_i(x)|^2 \, \|(S(x) - S(x_i))v\| \le (\varepsilon + N\eta M)\|v\|.$$

The same coefficient family simultaneously retains coverage, leakage, and the common frequency margin. This removes any hidden claim that Fourier replacement preserves exact subordination. It also exposes the remaining uniformity issue: on each compact regular interval δ may be chosen after N and M are known, but continuation up to a putative singular time requires a noncircular way to prevent the resulting choices and transfer costs from degenerating.

The refined pin (committed). The module `MisalignmentRefinedPin` splits the pancake-sector functional into a direction-coherent share, carrying the interface form of the proved aperture-gain kinematics, and a misalignment share; the new pin `MisalignmentStrainBudget` demands the time-integrable envelope only for the misalignment share, and the sealed reduction `scaleLocalPancakeStrainBudget_of_misalignmentBudget` reproduces the original budget with gain coherentGain + ε. The refinement is strict: the open burden moved from the full pancake stretching sum to its misalignment share, whose controlling quantity is the direction variation. The vorticity-direction equation exposes a dynamical handle on this quantity, but does not by itself give the required a priori estimate.

Exact unproven boundary. (i) Completion of the field-level transfer: the modewise complex split, amplitude estimate, lattice count, exact bilinear aperture factor, fixed-δ smooth tensor kernel, bilinear scaling/Young bound, exact finite torus realization, frozen-frame covariance, and variable/frozen split are now proved. Uniform normalized Euclidean bilinear-kernel mass, exact standard-six-torus Fourier realization, and the frequency-uniform periodic $O(\delta)$ bilinear endpoint bound are also proved. Exact finite-Fourier summation, inner-chart identification with the coherent PDE interaction, mass-preserving isotropic dyadic rescaling, exact four-transverse/two-axial scaling, square-dyadic chart identification, its periodized coefficient formula, its uniform C/N mass bound, and uniform frozen oriented-frame transference, physical input/output conjugation, and the pointwise endpoint for an arbitrary frame field are now proved as well. Four-sign coverage of the full normalized annulus, exact partition assembly, one uniform inverse-Fourier mass bound for the assembled multiplier, its exact uncut coherent-symbol identity on the annulus, and its physical variable-frame C/N endpoint are also proved. The finite-support annular sector projector is now constructed, its six-dimensional chart is identified with the two independently filtered mode

charts, and the genuine rotated radial-shell/angle-cone selector is proved to lie in that annulus. The exact physical coherent formula follows without an additional support premise. The pointwise strain-derived axial frame and its expanding-line orientation are also constructed. Finally, the full physical finite-sector stretching sum is proved to be exactly that annular coherent action plus the three-term misalignment remainder, with a uniform C/N bound on the first term, including in the strain-derived frame. The complex coefficient remainder is now bounded by transverse expanding-line energy, first modewise, then after finite-sector summation, and finally by an exact physical-space L^2 integral via finite Parseval, with no support-cardinality loss. A sign-invariant line-projector theorem controls the frozen-to-varying-frame error by 4ε/threshold times the full L^2 energy, with the ordinary frozen top-gap band returned as the alternative. On its separated branch, homogeneous spectral-gap coercivity now bounds the transverse energy by the integrated top Rayleigh defect. The separated-cell finite sum, generic bounded-overlap integration, the actual 125-cell weighted sum with cell-dependent frozen strains, and continuity/measurability of the sign-invariant projector on every positive-gap region are now checked. On the small-top-gap branch the bottom-separated expanding-plane projector is also measurable, while simultaneous collapse of both adjacent gaps gives a direct small stretching-rate bound for trace-free strain. The localization/commutator connection from global reconstructions to spatial cells is now checked through the concrete smooth cover. Exact finite cutoff convolution, Minkowski-sum support leakage, and the sharp physical L^2 cutoff bound are now proved, as is the exact buffered annulus containing every shifted output mode. A denominator-safe four-chart multiplier is now constructed on that buffer, proved exactly coherent there, and given one aperture-uniform inverse-kernel L^1 bound. Its complete square-dyadic, rotated, periodized, and physical transfer is proved, and finite-band cutoff products are connected to it with no support-cardinality loss. The actual spatial cutoff is now a periodized compactly supported smooth bump; its Fourier coefficients are proved absolutely summable by Schwartz decay and Poisson transference. Its explicit 125 translates cover the torus, admit a common scale-compatible finite approximation, and satisfy the uniform smooth-cell-to-coherent $O(\varepsilon/N)$ transfer. Complete ordinary dyadic shells for arbitrary Fourier data are also constructed, each is covered by two adjacent square-dyadic scales, and each has an exact four-way lower/upper and pancake/nonpancake split. For componentwise L^2 data these shell energies are summable and, together with the zero mode, equal the full physical energy.

The remaining field-level steps are the smooth-projector collar, dynamical coherence inside the bottom-separated expanding plane, aggregation of these branches over space and time, construction of the cellwise frozen strains from the actual strain field with the needed oscillation, conversion of the resulting L^2/defect estimate to the required scale-local endpoint budget, and assembly of the hard dyadic pancake split into `coherent_le` for the actual Navier–Stokes functionals. (ii) The misalignment budget itself: dynamical, time-integrated control of the misalignment share; the checked fences of Sections B, D.6 and the static no-go apply to it verbatim.

F Opening the dynamical stage: direction rate and the tilting gap

The first dynamical audit separates two notions that should not be conflated: rotation of the vorticity direction and rotation of the strain eigenframe. Both are relevant to a pancake configuration, but they obey different equations.

For a nonzero vorticity vector written $\omega = r\xi$, with $|\xi| = 1$, and a material forcing F, define the transverse projection

$$P_\xi^\perp F = F - (\xi \cdot F)\xi, \qquad \dot{\xi} = r^{-1} P_\xi^\perp F.$$

The module `DirectionEvolutionTilting` verifies this formula from a differentiable factorization

$\omega = r\xi$, including tangency $\xi \cdot \dot{\xi} = 0$. If the pointwise vorticity equation is written as $F = rS\xi + d$, where S is the strain and d is the diffusive remainder, the exact split is

$$\dot{\xi} = P_\xi^\perp(S\xi) + r^{-1}P_\xi^\perp d.$$

For two differentiable direction paths, the same module proves

$$\frac{d}{dt}|\xi_1 - \xi_2|^2 = 2(\xi_1 - \xi_2) \cdot (\dot{\xi}_1 - \dot{\xi}_2),$$

together with its Cauchy–Schwarz bound by $2|\xi_1 - \xi_2|(|\dot{\xi}_1| + |\dot{\xi}_2|)$. Thus total direction turning is an exact dynamical input to misalignment production; obtaining the scale-critical time integral required by `MisalignmentStrainBudget` remains analytic work.

The module `PancakeDyadicDirectionEvolution` makes this kinematics apply to the same hard block used by the Fourier misalignment estimate. For the square-dyadic pancake selector $P_{N,F}$ at a fixed spatial point, it proves coefficientwise that

$$\partial_t \Re(P_{N,F}\omega) = \Re(P_{N,F}\dot{\omega}).$$

If this real block is $r\xi$, the exact direction equation is

$$\dot{\xi} = P_\xi^\perp(S\xi) + r^{-1}P_\xi^\perp R_{N,F}, \qquad R_{N,F} = \Re(P_{N,F}\dot{\omega}) - rS\xi.$$

Thus the previously separate physical-direction and Fourier-block statements now meet at a single explicit residual. This is not yet a bound on that residual: its Navier–Stokes decomposition and scale-critical time estimate are the next dynamical obligations. For any proposed coefficient field G representing the stretching part, the same module further proves the exact split

$$R_{N,F} = \Re P_{N,F}(\dot{\omega} - G) + (\Re P_{N,F}G - rS\xi).$$

The first term is the projected equation remainder (transport and viscosity for the vorticity equation); the second is the projection/variable-strain defect once G is identified with the coefficients of $S\omega$. This separates the two estimates needed for closure instead of charging both to an undifferentiated direction-coherence premise. The companion module `PancakeFourierStretchingEvolution` constructs the canonical phase-suppressed coefficient field: every retained source/receiver pair is collected in the fiber $k+\ell = q$, and finite reconstruction of the resulting output coefficients is proved exactly equal to the previously defined pair-indexed complex stretching polynomial. The module `PancakePeriodicVorticityEquation` then fixes the unit-torus constants. Writing $D = 2\pi i$, it defines the physical Biot–Savart coefficient as $-D^{-1}|k|^{-2}k \times \widehat{\omega}_k$ and proves exactly that physical strain stretching is the negative of the suppressed symbol. It also derives the right-hand-side transport sign, includes the multiplier $\nu D^2|q|^2$, and packages the finite coefficient equation. Under that equation the first block residual is proved to be precisely projected transport plus viscosity; the second is the physical stretching projection/variable-strain defect. What remains is a scale-critical estimate for those two concrete channels and a controlled passage from finite convolutions to the full periodic solution, not a normalization ambiguity. The companion module `PancakeBlockReality` proves that both the radial shell and oriented squared-aperture predicate are invariant under frequency negation. Reindexing the finite Fourier sum by $k \mapsto -k$, using character and coefficient conjugate symmetry, proves $\overline{P_{N,F}\omega} = P_{N,F}\omega$ coordinatewise. It follows that the complex block has zero imaginary part and complexifying the real block used above returns the full reconstruction exactly. The same module then checks symbol-by-symbol conjugate covariance of the physical unit-torus equation and reindexes both convolution supports simultaneously. For negation-closed

supports and real coefficient fields it proves $\mathrm{RHS}(-q) = \overline{\mathrm{RHS}(q)}$, so the finite dynamics is internally consistent with the real block on which the direction law is stated. The next exact cancellation is material transport. If B is the hard pancake mode set, then

$$P_B[-(u\cdot\nabla)\omega] + (u\cdot\nabla)(P_B\omega) = [u\cdot\nabla, P_B]\omega$$

is checked coefficientwise for finite supports. For an output $q \in B$ the surviving sum contains only receiver frequencies initially outside B (incoming transfer); for $q \notin B$ it is minus the sum from receivers inside B (outgoing transfer). Thus following the fluid removes bulk advection and leaves an explicit frequency-boundary flux. The physical stretching defect has the same algebra: the checked coefficient of $P_B(S\omega) - S(P_B\omega)$ is incoming stretching when $q \in B$ and minus outgoing stretching otherwise. Combining the two shows that the entire material nonlinear commutator consists only of transport-plus-stretching interactions crossing the boundary of B; interactions staying within the same hard block cancel from this residual exactly. The module — now completed as `PancakeMovingBlockEvolution` — verifies that this commutator is the one actually produced by a moving observation point, rather than merely a formal coefficient subtraction. It differentiates both factors in

$$\sum_{k\in B} e^{2\pi i k\cdot X(t)}\widehat{\omega}_k(t).$$

For the finite velocity-generated character rate, the character derivative is proved exactly equal to positive advection of $P_B\omega$, including the $2\pi i$ normalization. Combining this product rule with the finite vorticity equation gives the material hard-block direction law

$$\dot{\xi} = P_\xi^\perp(S\xi) + r^{-1}P_\xi^\perp(R_{\partial B,\nu} + R_{\mathrm{proj}})\,.$$

Here $R_{\partial B,\nu}$ is the material nonlinear boundary-transfer field plus projected viscosity, and R_{proj} is the discrepancy between physical stretching of the projected block and $rS\xi$. At coefficient level, Lean proves that $R_{\partial B,\nu}$ contains only incoming/outgoing frequency-boundary interactions and the viscous multiplier. This is still an exact finite-dimensional identity, not the scale-critical estimate needed for global regularity. In this theorem the observation point moves, but the frame F and the hard mode set are fixed in time.

The module `PancakeHardBoundaryObstruction` tests that residual at the actual angular boundary. For every integer $N \geq 2$, set

$$k = (-1,0,0), \qquad \ell = (N+1,0,N^2), \qquad q = k+\ell = (N,0,N^2), \qquad w_k = w_\ell = (0,1,0).$$

The receiver ℓ lies outside the identity-frame hard pancake sector, while q lies inside its complete square-dyadic shell-and-cone selector. Both input amplitudes are transverse and have norm one. The physical transport symbol is exactly $-N^2(0,1,0)$, and the physical stretching symbol is zero. Lean checks that the corresponding incoming material boundary coefficient has norm exactly N^2 and hence admits no scale-independent instantaneous bilinear bound by the product of the two input amplitude norms. Adding the conjugate partner modes $-k, -\ell$ leaves the coefficient at q unchanged, so the calculation also applies to real trigonometric inputs. This is a bilinear obstruction, not a self-consistent solution counterexample or a refutation of a time-integrated budget with viscosity.

The module `PancakeEvolvingFilter` supplies the derivative term needed when the filter itself changes. For differentiable weights $\gamma(t,k)$ on a fixed finite ambient support, it differentiates the filter, vorticity coefficient, and moving Fourier character. At a fixed time the exact material residual coefficient is

$$R_q = \dot{\gamma}(q)\widehat{\eta}_q + \sum_{k+\ell=q}\big(\gamma(q)-\gamma(\ell)\big)\big(T_{k,\ell}+S_{k,\ell}\big) + \gamma(q)V_q,$$

where T, S, V are the physical transport, strain-stretching, and viscous symbols with the same finite supports as the vorticity equation. The nonlinear formula specializes exactly to the previously constructed hard commutator when γ is the indicator of its block. A changing hard membership predicate need not be differentiable at a crossing; the product rule does not supply that missing hypothesis. An adaptive argument must control the filter-motion term as well as the nonlinear boundary transfer. Standard smooth Littlewood–Paley transport estimates provide useful comparison machinery, but their velocity-gradient assumptions must also be discharged: for example, the regularity-preserving estimates in R. Danchin, *Estimates in Besov spaces for transport and transport-diffusion equations with almost Lipschitz coefficients*, *Rev. Mat. Iberoamericana* **21** (2005), 863–888, Section 2, require time-integrated velocity-gradient control.

The module `PancakeSignedTransportEnergy` now retains signed quadratic transfer before taking absolute values. For a negation-closed finite source set K and conjugate-symmetric source coefficients, let

$$b_{q\ell} = \mathbf{1}_{q-\ell\in K}\, \ell \cdot \big(|q-\ell|^{-2}(q-\ell) \times \widehat{\omega}_{q-\ell}\big).$$

This is proved to be the matrix of the existing physical negative-advection convolution, with its defined zero-mode value, and $b_{\ell q} = -\overline{b_{q\ell}}$. Consequently $F_{q\ell} = \operatorname{Re}\langle \widehat{\eta}_q, b_{q\ell}\widehat{\eta}_\ell\rangle$ is antisymmetric. On any finite receiver/output mode set M, a real filter γ satisfies the exact transport-energy identity

$$2\sum_{q\in M} \gamma_q^2 \operatorname{Re}\langle \widehat{\eta}_q, (T\widehat{\eta})_q\rangle = \sum_{q,\ell\in M} (\gamma_q^2 - \gamma_\ell^2) F_{q\ell}.$$

Thus an exact quadratic frequency partition $\sum_i \gamma_i(q)^2 = 1$ cancels the total signed transport contribution. Differentiating a differentiable such partition also proves $\sum_i 2\gamma_i\dot{\gamma}_i = 0$, rather than assuming the filter-rate cancellation. These are identities for exact frequency partitions, not a claim that the previously constructed approximate spatial partitions already satisfy them. The full filtered coefficient-energy derivative is checked with filter motion, transport, stretching, and viscous dissipation all retained.

Directional energy has an additional term. For self-adjoint complex-linear mode weights A_q, the same argument leaves

$$2\sum_{q\in M} \operatorname{Re}\langle A_q\widehat{\eta}_q, (T\widehat{\eta})_q\rangle = \sum_{q,\ell\in M} \operatorname{Re}\langle \widehat{\eta}_q, b_{q\ell}(A_q - A_\ell)\widehat{\eta}_\ell\rangle.$$

A common frozen weight cancels; different directional weights need not. The module proves self-adjointness and idempotence of the route's actual `misalignmentComponent` projector Q_q, including its degenerate reference case, and specializes this identity to that projector. Under the physical finite vorticity coefficient equation it then proves

$$\begin{aligned}\frac{d}{dt}\sum_{q\in M} |Q_q\widehat{\eta}_q|^2 = &\sum_{q,\ell\in M} \operatorname{Re}\langle \widehat{\eta}_q, b_{q\ell}(Q_q - Q_\ell)\widehat{\eta}_\ell\rangle \\ &+ 2\sum_{q\in M} \operatorname{Re}\langle Q_q\widehat{\eta}_q, (S\widehat{\eta})_q\rangle - 2\nu(2\pi)^2 \sum_{q\in M} |q|^2 |Q_q\widehat{\eta}_q|^2.\end{aligned}$$

Here the coherent reference is fixed, $S\widehat{\eta}$ is the already constructed physical stretching convolution, and the energy normalization is the one used by the finite Parseval theorem. This is a differential identity for the existing projector and coefficient equation, not a constructed solution or a dynamical budget. The remaining estimate must control the projector-mismatch and stretching terms together with dissipation; moving strain references, spatial localization, and the scale-critical time sum still require additional work. In particular, scalar energy conservation does not by itself close the misalignment pin.

The next module, `PancakeFrequencyProjectorCommutator`, estimates this specific projector-mismatch channel. In a complex Hilbert space, normalization and rank-one projection give the checked bound

$$\|Q_q - Q_\ell\| \le \frac{16|q-\ell|}{|q|\,|p(q)|}, \qquad p(q) = P_q e_x,$$

provided both frequencies and both coherent references are nonzero. The Hilbert-space construction is identified exactly with the periodic `misalignmentComponent`; it is not a surrogate projector. The actual identity-frame pancake cone at $N \ge 2$ gives $|p(q)| \ge 1/2$ and $N^2 \le |q| < 2N^2$. A separate checked physical-symbol estimate gives

$$|b_{q\ell}| \le \frac{6|\ell|}{|q-\ell|}\,|\widehat{\omega}_{q-\ell}|$$

for a retained nonzero source frequency. Thus the source-frequency factors cancel. For q, ℓ in the same pancake shell,

$$|b_{q\ell}(Q_q - Q_\ell)v| \le 384\,|\widehat{\omega}_{q-\ell}|\,|v|.$$

The more general theorem needs only $|\ell| \le 2|q|$ and the two coherent-reference norms at least $1/2$, so it also applies to buffered frequencies satisfying those conditions. The coincident-frequency and absent-source cases are included explicitly.

Injectivity of the frequency shifts bounds both row and column sums of the source convolution matrix. A checked finite Schur estimate then yields

$$\left|\sum_{q,\ell\in M} \mathrm{Re}\langle \widehat{\eta}_q, b_{q\ell}(Q_q - Q_\ell)\widehat{\eta}_\ell\rangle\right| \le 384\left(\sum_{k\in K}|\widehat{\omega}_k|\right)\left(\sum_{q\in M}|\widehat{\eta}_q|^2\right),$$

with no additional mode-count or frequency factor. It plugs directly into the signed misalignment-energy derivative above. This removes a derivative loss from the same-shell projector-mismatch channel, but it does not prove the time budget: the source Fourier ℓ^1 vorticity factor still needs dynamical control, and the right-hand energy is full receiver energy, not misalignment energy alone. Nor does this estimate control a hard-selector crossing, transfers from other sectors, or moving strain-frame terms. Uniform physical-frame localization and the scale-critical time sum remain required before this estimate can discharge any part of `MisalignmentStrainBudget` for actual solution data.

Projected-amplitude absorption and the remaining source cutoff. The modules `PancakeViscousMisalignmentAbsorption` and `PancakeSameSectorDissipation` strengthen this estimate by retaining the misaligned output amplitude. Write

$$E_Q = \sum_{q\in M}|Q_q\widehat{\eta}_q|^2, \qquad E = \sum_{q\in M}|\widehat{\eta}_q|^2, \qquad D_Q = \sum_{q\in M}|q|^2|Q_q\widehat{\eta}_q|^2, \qquad A_K = \sum_{k\in K}|\widehat{\omega}_k|.$$

For the same finite source set K, closed under negation, and conjugate-symmetric source coefficients, transport of $Q\eta$ has zero total energy rate. Subtracting that identity gives the exact expression

$$T_Q = 2\,\mathrm{Re}\sum_{q,\ell\in M}\langle Q_q\widehat{\eta}_q, b_{q\ell}(Q_q - Q_\ell)\widehat{\eta}_\ell\rangle.$$

A weighted finite Schur estimate, with an arbitrary parameter $r > 0$, yields

$$|T_Q| \le 384A_K(rE_Q + E/r).$$

In particular, for every $\delta > 0$,

$$|T_Q| \le \delta E_Q + \frac{(384A_K)^2}{\delta} E.$$

The zero-source case is proved separately, so this optimization does not divide by an amplitude assumed positive.

Stretching can be treated with the same projected amplitude. The exact complex Lagrange identity for a real lattice frequency is

$$|k \times w|^2 = |k|^2|w|^2 - |k \cdot w|^2.$$

It gives $|k|\,|\widetilde{u}_k| \le |w|$ for the phase-suppressed Biot–Savart symbol $\widetilde{u}_k$. Restoring the physical Fourier phases and estimating both terms of the symmetric strain gives the checked, non-sharp bound

$$|S_k(w)v| \le 3|w|\,|v|, \qquad |S_Q| \le 3A_K(rE_Q + E/r).$$

This includes the zero source frequency. The second inequality follows by reindexing the actual physical stretching convolution by its receiver frequency and applying the same row/column estimate.

Transport and stretching are therefore absorbed *together*, spending the physical diffusion only once. Since $N^4 E_Q \le D_Q$, their combined constant $387 = 384 + 3$ gives

$$E_Q' + \nu(2\pi)^2 D_Q \le C_{K,N}, \qquad C_{K,N} := \frac{(387A_K)^2}{\nu(2\pi)^2 N^4} E. \tag{12}$$

Lean identifies the derivative here under the previously defined physical finite coefficient equation. With the displayed rates explicitly interval-integrable on $[0, T]$, it also proves

$$E_Q(T) + \nu(2\pi)^2 \int_0^T D_Q(t)\,dt \le E_Q(0) + \int_0^T C_{K,N}(t)\,dt.$$

Neither statement constructs a solution of that coefficient equation or asserts that a sector of a full Navier–Stokes solution is an autonomous subsystem. Incoming receiver-sector interactions and moving-projector terms must be restored for that application.

The remaining amplitude cost has a precise physical energy interpretation. Let

$$U_K = \sum_{k \in K} |u_k|^2, \qquad V_M = \sum_{q \in M} |v_q|^2, \qquad H_K = \sum_{k \in K} |k|^2,$$

where u_k and v_q are the physical unit-torus Biot–Savart velocities of $\widehat{\omega}_k$ and $\widehat{\eta}_q$. These sums use the squared Parseval norm, without the conventional kinetic-energy factor $1/2$. For nonzero transverse source modes, the checked exact identity $(2\pi)^2|k|^2|u_k|^2 = |\widehat{\omega}_k|^2$ and Cauchy–Schwarz give

$$A_K^2 \le (2\pi)^2 H_K U_K.$$

For transverse receiver modes in the actual pancake shell, its upper frequency bound gives $E \le 4(2\pi)^2 N^4 V_M$. Consequently

$$C_{K,N} \le \frac{4 \cdot 387^2 (2\pi)^2}{\nu} H_K U_K V_M. \tag{13}$$

For a finite family of disjoint receiver sectors of a single coefficient field, with the same source set K, Lean sums this estimate with V_M replaced by the velocity energy on the union. There is no extra receiver-sector count or receiver-frequency factor.

The explicit factor H_K is essential to the present bound. A fixed finite source set permits low-source energy accounting, but allowing its cutoff to grow is not free. Source-cutoff-uniform control, incoming cross-sector interactions, evolving strain frames, physical localization, and the conversion to the scale-critical misalignment strain budget remain unproved. Equations (12) and (13) do not establish the arbitrary-data continuation theorem.

Incoming low sources: full receivers and a fixed-cutoff time bound. `PancakeIncomingLowSourceEnergy` removes the same-receiver-sector restriction for the low-source channel. Let S be the finite ambient source support, R the full finite receiver support, and M the selected output modes. They need not coincide. For a fixed cutoff L, set

$$K_L = \{k \in S : |k| \le L\}, \quad K_L^+ = S \setminus K_L, \quad B_L = \sum_{k \in K_L} |u_k|, \quad E_R = \sum_{\ell \in R} |\widehat{\eta}_\ell|^2.$$

Assume $|q| \ge L$ for $q \in M$. For an actual pancake output shell this follows from $L \le N^2$, with no restriction on receiver modes.

The physical Biot–Savart velocity satisfies

$$\ell \cdot u_{q-\ell} = q \cdot u_{q-\ell}.$$

Thus the transport derivative can be estimated at the output frequency, where diffusion is available. Lean checks

$$|T_{q\ell}| \le 3(2\pi)|q|\,|u_{q-\ell}|\,|\widehat{\eta}_\ell|, \qquad |S_k\widehat{\eta}_\ell| \le 3(2\pi)|k|\,|u_k|\,|\widehat{\eta}_\ell|.$$

For low sources, $|k| \le L \le |q|$, so both terms have the same output-frequency bound. Pairing with $Q_q\widehat{\eta}_q$ and using a rectangular Schur estimate over $M \times R$ gives

$$|I_L| \le 6(2\pi)B_L(rD_Q + E_R/r), \qquad r > 0,$$

where I_L is the actual low-source contribution to the derivative of E_Q. Optimizing the parameter proves

$$|I_L| \le \nu(2\pi)^2 D_Q + C_L, \qquad C_L = \frac{36B_L^2}{\nu}E_R.$$

Every incoming receiver mode remains in R, including modes outside the hard pancake boundary. No scalar cutoff smoothing or cancellation between an inside-only set of receivers is assumed.

The source split is also an exact identity for the physical nonlinear coefficient and its energy rate. Under the full finite coefficient equation, the checked differential inequality is therefore

$$E_Q' + \nu(2\pi)^2 D_Q \le I_L^+ + C_L, \tag{14}$$

where I_L^+ is the explicitly defined physical contribution from $K_L^+ \times R$, not an abstract error variable. Lean integrates this inequality with all displayed rates interval-integrable. The high-source term is retained, not declared controlled.

There is now a fixed-cutoff, horizon-uniform bound for the low-source *cost*, conditional on physical energy estimates. Writing $U_L(t) = \sum_{k \in K_L} |u_k(t)|^2$, Cauchy–Schwarz gives $B_L^2 \le \#K_L\, U_L$. For a natural cutoff L, an explicitly constructed integer cube gives $\#K_L \le (2L+1)^3$, independently of S. If

$$U_L(t) \le U \quad (0 \le t \le T), \qquad 2\nu \int_0^T E_R(t)\,dt \le U, \qquad U \ge 0,$$

then, with the stated integrability assumptions, Lean proves

$$\int_0^T C_L(t)\,dt \le \frac{18(2L+1)^3U^2}{\nu^2}. \tag{15}$$

This bound has no time-horizon, output-frequency, output-mode-count, or ambient-support factor. It does depend on the *fixed* source cutoff L; choosing L to grow with output frequency does not preserve a uniform constant. The kinetic and enstrophy bounds above remain explicit inputs; this module does not derive them from an arbitrary-data solution construction. It also leaves I_L^+, moving-frame terms, physical localization, the strain-budget conversion, and continuation open. The previous instantaneous hard-boundary obstruction is consistent with this estimate: its output-frequency growth is paid for by diffusion rather than denied.

Exchange-symmetric refinement: the complete low-touch budget. The modules `PancakeExchangeSymmetricCurl` and `PancakeLowTouchEnergy` now impose physical self-consistency: the receiver is the same transverse vorticity field as the source, and the finite support S excludes the zero mode. The existing physical Biot–Savart formula then satisfies $\widehat{\omega}_k = (2\pi i)k \times u_k$. With $q = k + \ell$, Lean checks

$$T(k,\ell) + S(k,\ell) = (2\pi i)q \times (u_k \times \widehat{\omega}_\ell) - \tfrac{1}{2}\widehat{\omega}_k \times \widehat{\omega}_\ell.$$

The last term sums to zero on every finite ordered pair family closed under exchanging k and ℓ. The curl-recovery condition is essential: transversality alone does not imply it at zero frequency, as an explicit checked counterexample records.

Write $H_L = S \setminus K_L$. The disjoint physical partition is

$$F_{S,S} = F_L^{\text{touch}} + F_L^{HH}, \qquad F_L^{\text{touch}} = F_{K_L,S} + F_{H_L,K_L}, \qquad F_L^{HH} = F_{H_L,H_L}.$$

Both pair families are exchange-symmetric. Thus

$$F_L^{\text{touch}} = \nabla \times (u_{K_L} \times \omega + u_{H_L} \times \omega_{K_L}), \qquad F_L^{HH} = \nabla \times (u_{H_L} \times \omega_{H_L}).$$

This repartitions the *full* nonlinearity. The old high-source term alone is $I_L^+ = I_{H_L,K_L} + I_{H_L,H_L}$; it is not the sum of the complete low-touch and high-high contributions.

Let $m_L = \#K_L$, $U(t) = \sum_{k\in S} |u_k(t)|^2$, and $\Omega(t) = \sum_{k\in S} |\widehat{\omega}_k(t)|^2$. The complex Lagrange identity gives the sharp Euclidean estimate $|a \times b| \le |a|\,|b|$. A rectangular Schur estimate for the output-curl convolution, applied once in each input orientation, proves for every $\theta > 0$ that

$$|I_L^{\text{touch}}| \le \theta\nu(2\pi)^2 D_Q + C_L^{\text{touch}}, \qquad C_L^{\text{touch}} = \frac{4m_L}{\theta\nu}U\Omega. \tag{16}$$

No lower bound on output frequency, same-sector receiver confinement, or moving-frame cancellation is used. The output projector remains the actual fixed misalignment projector.

If $U(t) \le U_0$, $U_0 \ge 0$, and $2\nu\int_0^T \Omega(t)\,dt \le U_0$, with the displayed rates interval-integrable, Lean proves

$$\int_0^T C_L^{\text{touch}}(t)\,dt \le \frac{2m_L U_0^2}{\theta\nu^2} \le \frac{2(2L+1)^3U_0^2}{\theta\nu^2} \quad (L \in \mathbb{N}). \tag{17}$$

It also integrates the absolute flux in (16), retaining its viscous payment. Under the full self-consistent finite vorticity equation, the resulting checked evolution estimate is

$$E_Q(T) + (2-\theta)\nu(2\pi)^2\int_0^T D_Q(t)\,dt \le E_Q(0) + \int_0^T I_L^{HH}(t)\,dt + \frac{2m_L U_0^2}{\theta\nu^2}. \tag{18}$$

Choosing $0 < \theta < 2$ retains positive output dissipation. The cost is uniform in time horizon, output frequency and output count, and in the ambient retained support, at a *fixed* low cutoff and uniform U_0. The kinetic/enstrophy estimates, integrability, and solution realization remain explicit hypotheses. The high-high dynamics, moving-frame terms, field-level strain transfer, and arbitrary-data continuation remain open.

Physical velocity realization of the energy and vorticity inputs. `PancakeGalerkinKineticEnergy` and `PancakeGalerkinVorticityBridge` now discharge the independent kinetic/enstrophy, vorticity-evolution, and integrability inputs above for coefficient paths satisfying the physical finite velocity equation. Let S be a finite nonzero mode set closed under negation and let $u_k(t)$ be conjugate-symmetric and transverse, $k \cdot u_k(t) = 0$, on $[0, T]$. Write $c = 2\pi i$ and define

$$\widehat{\omega}_k = c\, k \times u_k, \qquad V_q(u) = -cP_q \sum_{\substack{k,\ell \in S \\ k+\ell=q}} (\ell \cdot u_k) u_\ell - \nu(2\pi)^2 |q|^2 u_q.$$

The source velocity recovered from $\widehat{\omega}$ is exactly u on S. This identifies the implemented RHS with V, including the unit-torus normalization. For every path with $\dot{u}_q = V_q(u)$ on the retained modes, pressure does no kinetic work and signed transport cancels. Lean proves

$$U(T) + 2\nu \int_0^T \Omega(t)\, dt = U(0). \tag{19}$$

The coefficient equation itself implies continuity of the finite enstrophy sum and hence its time integrability; that integrability is not a premise.

For the curl bridge, exchanging the two inputs cancels the gradient defect in the identity between $u \times \operatorname{curl} u$ and negative velocity convection. Pressure has zero curl and viscosity commutes with curl. Consequently the exact implemented coefficient identity is

$$c\, q \times V_q(u) = F_{S,S}(\widehat{\omega}, \widehat{\omega})_q - \nu(2\pi)^2 |q|^2 \widehat{\omega}_q.$$

The pathwise vorticity ODE follows by differentiating the finite linear curl map. Continuity of the fixed misalignment projector and finite bilinear convolution supplies all remaining rate-integrability facts.

Thus `integrated_highHigh_remainder_of_velocityRHS` proves (18) for every output subset $M \subseteq S$ using $U_0 = U(0)$, without independent kinetic, enstrophy, vorticity-ODE, or rate-integrability assumptions. Its natural-cutoff version replaces m_L by $(2L+1)^3$. The output dynamics still use the full retained source and receiver sets; the selected output sector is not declared autonomous.

These results are a priori theorems about paths satisfying the finite ODE. They do not construct such paths, prove preservation of the real transverse subspace from initial data, or pass to an infinite-dimensional solution. They also do not replace the fixed misalignment projector by the moving strain-eigenframe projector. High-high control, that moving-frame passage, the scale-critical strain budget, and BKM continuation remain open.

Moving localization: positive energy and material projector tangency. `PancakeLocalizedQuadraticEnergy` implements the noncommuting localized energy

$$A = C^* P C, \qquad \mathcal{E}_A = \langle \omega, A\omega \rangle = \|PC\omega\|_2^2$$

for an orthogonal projector P. No commutation of C and P is required. Its coherent/complementary partner is $C^*(I-P)C$, and the two operators sum exactly to C^*C. The unsandwiched expression C^2P is not used as though it were self-adjoint.

For $\dot\omega = -B\omega + s + \nu\Delta\omega$, the checked identity is

$$\begin{aligned}\dot{\mathcal{E}}_A = \langle\omega, \dot A\omega\rangle + \langle\omega, [B,A]\omega\rangle + 2\langle A\omega, s\rangle \\ + \nu\langle\omega, (A\Delta + \Delta A)\omega\rangle.\end{aligned}$$

The abstract statement requires skew symmetry of B and symmetry of Δ on an explicit common core containing ω and $A\omega$; neither unbounded operator is declared bounded on ambient L^2. Nor is advection treated as an endomorphism of divergence-free fields. Finite complete operator partitions cancel the summed motion and signed transport-transfer terms, not the corresponding individual sector terms.

Writing $z = C\omega$, the sum of the motion and transport terms is exactly

$$2\langle Pz, (\dot C + [B,C])\omega\rangle + \langle z, (\dot P + [B,P])z\rangle.$$

The common-core hypotheses now include z and Pz. The diffusion anticommutator remains explicit; positivity of A alone does not turn it into a commuting negative dissipation term.

`PancakeMaterialProjectorTangent` sharpens the second term. For a differentiable path of orthogonal projectors and bounded skew-adjoint B, differentiation of $P^2 = P$ proves that

$$F := \dot P + [B,P], \qquad F^* = F, \qquad FP + PF = F, \qquad PFP = 0.$$

Consequently Lean checks the exact off-diagonal identity and bounds

$$\begin{aligned}\langle z, Fz\rangle &= 2\langle Pz, F(I-P)z\rangle, \\ |\langle z, Fz\rangle| &\le 2\|F\|\,\|Pz\|_2\,\|(I-P)z\|_2 \\ &\le \rho\|(I-P)z\|_2^2 + \frac{\|F\|^2}{\rho}\|Pz\|_2^2 \qquad (\rho > 0).\end{aligned}$$

For the complementary energy the projector-motion term has the opposite sign, so the same absolute estimate applies. This retains a misalignment factor; it does not assert that the squared material-rate cost is integrable. Vanishing of the diagonal block is instantaneous and does not imply persistence of a pure coherent or misaligned sector.

`PancakeFiniteLocalizedEvolution` constructs the ambient retained Hilbert space $\mathbb{C}^{S\times\{0,1,2\}}$, its restriction/extension maps, and the actual advection and Laplacian matrices. Their real inner product agrees with the previous Fourier Hermitian sum, and their squared norm with the previous coefficient energy. Advection is exactly the negative of the physical RHS transport; its skew-adjointness follows from source reality. The Laplacian is the physical $(2\pi i)^2|q|^2$ multiplier. The moving-energy identity follows directly from a physical velocity path by the preceding curl bridge, with no separate abstract vorticity ODE or transport identity assumed. The off-diagonal material-rate estimate is also specialized to this actual finite advection operator.

The localizer and projector paths remain explicit inputs. Constructing them from the actual evolving strain, bounding their material rates and diffusion errors uniformly in truncation and dyadic scale, controlling the high-high channel, and reaching the pointwise strain/BKM budget remain open. The bounded-operator tangent theorem does not by itself extend to unbounded advection on the infinite-dimensional ambient space.

The pressure Hessian enters through the strain-eigenframe equation instead. Differentiating $Se_i = \lambda_i e_i$ and pairing with e_j, $i \neq j$, gives

$$\langle e_j, \dot{e}_i \rangle = \frac{\langle e_j, \dot{S} e_i \rangle}{\lambda_i - \lambda_j}.$$

The anisotropic pressure Hessian contributes to the numerator. The checked theorem `no_uniformGapFreePressureTiltingBound` records the necessary denominator: no nonnegative constant bounds the angular rate from a bounded off-diagonal torque independently of the strain eigenvalue gap. Unit torque with gap $1/(C+1)$ already defeats a proposed constant C.

This rules out a gap-free angular-rate envelope based only on a raw pressure-Hessian norm. A viable dynamical continuation must instead control a gap-weighted pressure torque, or prove cancellation/residence-time control over near-degenerate intervals, and then transfer that control to the Littlewood–Paley misalignment share.

An intrinsic defect identity through collisions. The new `PancakeTopEigenvalueDerivative`, `PancakeSpectralDefectEvolution`, and `PancakeSpectralDefectBudget` modules work directly with

$$\lambda = \lambda_{\max}(S), \qquad D = \lambda \|w\|^2 - \langle w, Sw \rangle \geq 0, \qquad z = (\lambda I - S)w.$$

These are still strain-alignment quantities: in the constructed ordered eigenbasis,

$$D = (\lambda_0 - \lambda_1)|w_1|^2 + (\lambda_0 - \lambda_2)|w_2|^2,$$
$$\|z\|^2 = (\lambda_0 - \lambda_1)^2|w_1|^2 + (\lambda_0 - \lambda_2)^2|w_2|^2.$$

For real Hilbert-space paths obeying

$$\dot{w} = Sw + f, \qquad \dot{S} = -S^2 + R,$$

and a unit top eigenvector e at the time in question, the exact checked cancellation is

$$\dot{D} = -\|z\|^2 + \mathcal{A}_R(e, w) + 2\langle z, f \rangle, \qquad \mathcal{A}_R(e, w) = \langle e, Re \rangle \|w\|^2 - \langle w, Rw \rangle. \tag{20}$$

In particular no derivative of a chosen eigenvector appears. On a simple top-eigenvalue branch, the derivative $\dot{\lambda} = \langle e, \dot{S} e \rangle$ is proved using the continuous spectral projector applied to one fixed reference eigenvector, not an assumed differentiable eigenframe.

The time-integrated statement requires no simple-gap hypothesis at all. The top eigenvalue is globally 1-Lipschitz as a function of the symmetric operator. Along a C^1 strain path on a compact interval it is absolutely continuous. At every time where its derivative exists, the Rayleigh variational minimum proves the same derivative formula for every unit top eigenvector, including at repeated eigenvalues. The resulting almost-everywhere identity and absolute continuity of D yield

$$D(b) + \int_a^b \|z\|^2 = D(a) + \int_a^b \mathcal{A}_R(e, w) + 2\int_a^b \langle z, f \rangle.$$

For $a \leq b$ and $\theta > 0$, Young's inequality gives

$$D(b) + (1 - \theta)\int_a^b \|z\|^2 \leq D(a) + \int_a^b \mathcal{A}_R(e, w) + \theta^{-1}\int_a^b \|f\|^2.$$

The formal hypotheses are C^1 strain and vector paths, continuous f, and the two displayed differential equations on the compact interval. Integrability of the scalar anisotropy is derived from the almost-everywhere identity, not assumed for a potentially discontinuous eigenvector selection. No bound on that integral from the initial data is obtained.

Isotropic forcing cancels exactly: $\mathcal{A}_{R+\alpha I} = \mathcal{A}_R$. For skew-adjoint spin W,

$$\mathcal{A}_{-W^2}(e, w) = \|We\|^2\|w\|^2 - \|Ww\|^2.$$

Thus when $Ww = 0$, its contribution is nonnegative and cannot be discarded as damping. For a high-frequency receiver and low-frequency spin, $W_{<L}w$ need not vanish; the full two-term identity must be retained. In the full strain equation, R includes $-W^2$, the negative pressure Hessian, and viscous strain forcing. A filtered equation also has the actual subgrid forcing; see the gradient equation (2.4b) in Tom–Carbone–Bragg. Their coefficient-level and real-material-path identifications are now checked below; a uniform estimate of their signed combination remains required. Moreover each corresponding squared-gap weight vanishes at a collision: this budget is not an unweighted line-misalignment estimate or a completed Littlewood–Paley/BKM bridge.

Physical pressure and the filtered strain equation. The `PancakeFourierPressureStrain` module constructs pressure from the same finite convection convolution as the physical velocity RHS. Writing $c = 2\pi i$, $A_{q,ij} = cq_j u_{q,i}$ and $C_q = \sum_{k+\ell=q}(\ell \cdot u_k)u_\ell$, it sets

$$p_q = -\frac{q \cdot C_q}{|q|^2}, \qquad p_0 = 0, \qquad H_{q,ij} = c^2 q_i q_j p_q.$$

The Poisson equation, pressure-gradient identity, and exact material gradient coefficient equation are proved from these definitions and the physical velocity RHS. Strain and spin are the transpose-symmetric and transpose-antisymmetric parts of A_q; individual complex Fourier coefficients are not incorrectly declared Hermitian.

For a static multiplier χ, `PancakeFilteredStrainDynamics` retains the explicit stress-divergence coefficient

$$T^\chi_q = \sum_{k+\ell=q} [\chi(q) - \chi(k)\chi(\ell)](\ell \cdot u_k)u_\ell.$$

Thus $\chi C(u, u) = C(\chi u, \chi u) + T^\chi$ exactly. Differentiating convection and symmetrizing after exchanging the ordered Fourier pair gives the coefficient form of

$$\partial_t S_\chi + (u_\chi \cdot \nabla)S_\chi = -S_\chi^2 - W_\chi^2 - \chi H + \nu\Delta S_\chi + \operatorname{sym}\nabla(-cT^\chi).$$

Products and advection here mean the explicitly defined finite Fourier convolutions. Their coefficient time derivatives are derived from the original retained-mode velocity ODE. When χ vanishes outside the retained modes, the derivative theorem holds at every output frequency without assuming an equation for discarded modes. Reconstruction along real material paths is supplied by the next bridge below. A moving multiplier would also require its time-derivative term, which is not suppressed by this static-filter theorem.

Low-output pressure is paid by kinetic energy. The `PancakeLowPressureBudget` module uses incompressibility to move both numerator derivatives to the output mode:

$$q \cdot C_q = \sum_{k+\ell=q} (q \cdot u_k)(q \cdot u_\ell), \qquad |p_q| \le U, \qquad U = \sum_k |u_k|^2.$$

The constant in the last bound is exactly one, independent of the source support and all input frequencies. Each convolution fiber uses each source and receiver at most once, so Cauchy–Schwarz incurs no cardinality factor. For a fixed finite pressure output set K, define

$$C_K = (2\pi)^2 \sum_{q\in K} |q|^2, \qquad H_K(x) = \sum_{q\in K} e^{2\pi i q\cdot x} H_q.$$

The constructed Hessian operators satisfy $\|H_K(x)\| \le \sum_{q\in K} \|H_q\| \le C_K U$, uniformly in space. Let $\Omega = \sum_k |\omega_k|^2$. The physical energy identity yields

$$\int_0^T 2\sum_{q\in K} \|H_q(t)\|\,\Omega(t)\,dt \le \frac{C_K U(0)^2}{\nu}.$$

The `PancakePressureAnisotropyBudget` module connects this envelope to the actual spatial quadratic term. For unit directions $e(t,x)$ and the reconstructed vorticity, finite Parseval gives

$$\int_0^T \int_{\mathbb{T}^3} |\mathcal{A}_{H_K(t,x)}(e(t,x),\omega(t,x))|\,dx\,dt \le \frac{C_K U(0)^2}{\nu}.$$

The bound is proved on complex amplitudes regarded as a real Hilbert space, so real fields are included. Spatial integrability follows for measurable unit directions. The time theorem explicitly assumes measurability of the spatial integral and derives its time integrability and bound; it does not assume a pressure or enstrophy budget. All these dynamic statements assume a retained-mode velocity solution with preserved reality and incompressibility; existence and preservation are not proved by the estimates themselves.

This pays all pressure inputs generating a fixed low output, including high–high inputs. It does not pay high pressure outputs: C_K grows with the output cutoff. The signed high-pressure/spin/subgrid combination, the remaining high–high transfer, and the scale-uniform BKM closure are still open.

The defect identity for actual material fields. The `PancakeFourierMaterialPaths` and `PancakeSpatialStrainEvolution` modules now construct that bridge. For a differentiable real coordinate path $X(t)$, the Fourier character rate is proved from the exponential formula:

$$\frac{d}{dt} e^{2\pi i q\cdot X(t)} = 2\pi i\,(q\cdot \dot X(t)) e^{2\pi i q\cdot X(t)}.$$

The real trajectory equation $\dot X = u_\chi(t,X)$ therefore supplies the fluid-generated phase rate; no independent phase-derivative hypothesis is needed. The reconstructed Fourier gradient is also proved to give the directional derivative of the real velocity field along every such path. Its real operator splits into self-adjoint strain and skew-adjoint spin. Conjugate symmetry justifies multiplication of reconstructed real matrices; multiplicativity of real part is not presumed.

The output set is constructed as $Q = S \cup (S+S)$ for retained modes S. Thus reconstruction includes every linear and quadratic output. Finite convolution becomes the full matrix product, and positive advection cancels the trajectory's character-motion contribution exactly. The resulting real material equation is

$$\dot S_\chi = -S_\chi^2 + R_\chi, \qquad R_\chi = -W_\chi^2 + F_\chi,$$

where F_χ is the reconstructed pressure/viscous/subgrid expression above, not an assumed forcing operator. The checked spin contribution retains both terms:

$$\mathcal{A}_{R_\chi}(e,w) = \|W_\chi e\|^2\|w\|^2 - \|W_\chi w\|^2 + \mathcal{A}_{F_\chi}(e,w).$$

The `PancakePhysicalSpectralDefect` module constructs $w = \omega(t, X(t))$ from the retained curl coefficients and its exact material rate from the physical vorticity RHS and the derived character rate. Subtracting $S_\chi w$ defines the explicit mismatch f_χ; there is no time derivative hidden in that definition, and no smallness claim is attached to it. Both evolution hypotheses of (20) are thereby discharged from the original velocity and real trajectory equations.

Finally, `PancakePhysicalDefectODEBudget` derives C^1 regularity of the retained velocity coefficients and real path from those ODEs. Finite reconstruction transfers it to strain and vorticity. The largest eigenvalue is absolutely continuous across collisions, and the actual vector forcing is continuous because it equals $\dot{w} - S_\chi w$ on the compact interval. The endpoint argument uses the within-interval derivative and handles the singleton interval separately. The resulting exact physical identity is

$$D(b) + \int_a^b \|z\|^2 = D(a) + \int_a^b \mathcal{A}_{R_\chi}(e, w) + 2\int_a^b \langle z, f_\chi \rangle.$$

Integrability of the signed anisotropy is a conclusion. No separate strain ODE, phase ODE, C^1 field assumption, positive eigenvalue gap, or forcing-integrability budget occurs in the final theorem. A Young inequality version is also checked, but the exact signed pairing is retained for the next estimate: bounding the viscous part of $\|f_\chi\|^2$ term by term would introduce second-derivative costs that have not been paid.

These theorems still assume an existing retained-mode velocity solution and an existing trajectory, together with the stated support, conjugate-symmetry and incompressibility conditions. They prove neither arbitrary-data existence nor a uniform bound for the remaining signed forcing. Passage from individual paths to spatial and scale-uniform budgets, the high–high channel, and BKM continuation remain open.

Diffusion is not free damping of the intrinsic defect. The signed viscous term cannot be discarded on the strength of incompressibility alone. The modules `PancakePlanarSpectralDefect`, `PancakePlanarDiffusion`, and `PancakePeriodicDiffusionWitness` give an explicit pointwise test. For a planar incompressible velocity, write

$$S = \begin{pmatrix} a & b & 0 \\ b & -a & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \omega = (0, 0, w).$$

The constructed top eigenvalue is identified with $\sqrt{a^2 + b^2}$ on the branch $a > 0$, and the actual defect is $D = \sqrt{a^2 + b^2}\, w^2$. At a normalized point $a = w = 1$, $b = 0$, the checked second-derivative formula in one spatial direction is

$$D'' = a'' + (b')^2 + 4a'w' + 2(w')^2 + 2w''.$$

The mixed term has no fixed sign. More generally, at $b = 0$, $a > 0$, the corresponding directional heat-chain-rule production has the completed-square algebra

$$\begin{aligned} Q &= -\frac{w^2(b')^2}{a} - 4wa'w' - 2a(w')^2 \\ &= \frac{2w^2(a')^2}{a} - \frac{w^2(b')^2}{a} - 2a\left(w' + \frac{wa'}{a}\right)^2 \le \frac{2w^2(a')^2}{a} - \frac{w^2(b')^2}{a}. \end{aligned}$$

The algebraic equality and bound are checked; they expose a gradient cost rather than supplying a scale-uniform estimate of it.

On the 2π-periodic torus, take the smooth velocity

$$u_1(x,y) = -\tfrac{1}{2}\sin y + \sin x\cos y - 9\cos x\cos y + 16\cos y,$$
$$u_2(x,y) = \tfrac{1}{2}\sin x - \cos x\sin y + 8\cos x - 9\sin x\sin y, \qquad u_3 = 0.$$

Lean differentiates this velocity and verifies divergence zero, its strain, its curl, and periodicity. At the origin the values are

$$(a,b,w) = (1,0,1), \qquad (\Delta a, \Delta b, \Delta w) = (-2,0,-1),$$
$$\nabla a = (9,0), \qquad \nabla b = (-4,-8), \qquad \nabla w = (-8,-2).$$

For the actual constructed spectral defect, the checked values are

$$D_{xx} = -146, \qquad D_{yy} = 70, \qquad \Delta D = -76.$$

The directional derivative of D in the physical viscous direction $(\nu\Delta S, \nu\Delta\omega)$ is -4ν. An explicit affine jet realizes this derivative; it is not asserted to solve the heat equation at nonzero time. Thus the instantaneous viscous chain-rule production is

$$D'[\nu\Delta S, \nu\Delta\omega] - \nu\Delta D = 72\nu > 0 \qquad (\nu > 0).$$

This refutes a pointwise heat-subsolution claim for this defect. It does *not* refute a spatially integrated estimate, the repaired alignment route, or Navier–Stokes regularity.

Two-shear obstruction to the integrated sign. A simpler constructed heat evolution shows that spatial integration does not restore free monotonicity. Its initial velocity is

$$u = (2\varepsilon N^{-1}\sin(Ny), -2\sin x, 0), \quad f = \varepsilon\cos(Ny), \quad g = \cos x.$$

The module `PancakeTwoShearDiffusion` identifies the actual off-diagonal-strain defect with $4|f-g|(f+g)^2$ and proves the exact opposite-sign pairing, for nonnegative amplitudes F, G:

$$P(F,G) = \tfrac{1}{2}\{D(F,G) + D(-F,G)\} = \begin{cases} 4G(G^2-F^2), & F \le G, \\ 4F(F^2-G^2), & G \le F. \end{cases}$$

Its fast logarithmic amplitude rate away from $F = G$ is

$$A(F,G) = \begin{cases} -8GF^2, & F < G, \\ 12F^3 - 4FG^2, & G < F. \end{cases}$$

Assigning zero at the cusp preserves the checked bound $A \le -8GF^2 + 16F^3$. The actual two heat rates $F' = -\nu N^2 F$, $G' = -\nu G$ give $P' = -\nu[3P + (N^2-1)A]$ away from the crossing; this derivative identity is checked. The negative leading amplitude term explains why removing a small fast shear can increase this alignment defect.

For $\varepsilon = 1/16$, $N = 64$, $\nu = 1$, numerical periodic quadrature at resolutions 256, 512, 1024, and 2048 gives a normalized mean heat rate between 35.433 and 35.439. These computations are *diagnostic, not certified quadrature or a formal integral theorem.* The script `ns_two_shear_diffusion_sanity.py` reproduces the diagnostic and a finite-difference check under exact Fourier heat decay.

A separate exact finite-time theorem now certifies an integrated counterexample, with $N = 128$. The modules `PancakePeriodicMean`, `PancakeTwoShearFiniteTime`, and `PancakeTwoShearSpatialMean` use the actual normalized Lebesgue integral over $[0, 2\pi]^2$. Half-period translation performs the sign pairing, and integer-period repetition removes N from the spatial average, but not from the heat rate. Write

$$M(p, q) = \operatorname*{mean}_{x,y} P(p|\cos y|, q|\cos x|).$$

The checked finite-time comparison, valid for all $F, G \geq 0$ and $0 \leq \alpha \leq 1$, is

$$P(\alpha F, G) - P(F, G) \geq (1 - \alpha^2)(4G - 8F)F^2.$$

Its proof treats all three amplitude-ordering regions, including their boundaries. Thus no derivative through a crossing or null-set hypothesis is needed. The exact moment $\operatorname{mean}(\cos^2) = 1/2$ and the bound $\operatorname{mean}(|\cos|) \geq 1/2$ give

$$M(\alpha\varepsilon, 1) - M(\varepsilon, 1) \geq \tfrac{1}{2}(1 - \alpha^2)(2 - 8\varepsilon)\varepsilon^2.$$

In `PancakeTwoShearHeatWitness`, the actual velocity is constructed as

$$u(t, x, y) = \left(\frac{2\varepsilon}{N} e^{-\nu N^2 t} \sin(Ny), -2e^{-\nu t} \sin x, 0\right).$$

Its component heat equations, divergence zero, periodicity, strain and curl are verified by differentiation. For their constructed top spectral defect $D(t)$, at $\nu = 1$, $N = 128$, $\varepsilon = 1/16$, Lean proves

$$\operatorname{mean} D(1/16384) - \operatorname{mean} D(0) \geq \frac{24567}{33554432} > 0.$$

This uses cubic homogeneity, $M(1/16, 1) \leq 4$, and elementary exponential bounds; it is not certified numerical quadrature. The evolution solves the heat equation, not the nonlinear NS equation. It refutes free integrated heat monotonicity of this alignment defect, not the repaired route or global regularity.

A regularized paid gradient bound. The module `PancakeRegularizedDiffusion` constructs $K_\delta = (\lambda_{\max}(S) + \delta)I - S$. Its quadratic form equals $D + \delta|\omega|^2$ and is bounded below by $\delta|\omega|^2$. For $G_j = \partial_j K_\delta$, $v_j = \partial_j \omega$, the two mixed diffusion product-rule terms have the checked algebraic bound

$$Q_j = -2\langle v_j, K_\delta v_j\rangle - 4\langle v_j, G_j\omega\rangle,$$
$$Q_j + \delta|v_j|^2 \leq \frac{4}{\delta}\|G_j\|^2|\omega|^2 \leq \frac{16}{\delta}\|\partial_j S\|^2|\omega|^2, \qquad \delta > 0.$$

The derivative of K_δ is constructed along actual differentiable strain paths whenever the top eigenvalue is differentiable; its norm is at most twice that of the strain derivative. This uses the unit Rayleigh derivative, not an inverse eigenvalue gap. The finite sum over spatial directions and multiplication by $\nu \geq 0$ are also checked. This theorem proves the gradient algebra only. The spatial coordinate diffusion bridge is proved below by finite differences, without a smooth eigenframe. The full material balance, a scale-uniform bound for the time-integrated strain-gradient cost, and control of the regularization in the complete alignment budget remain obligations. In particular, δ is not silently sent to zero and the estimate is not scale-uniform.

Physical payment for a fixed low strain cutoff. The modules `PancakeFourierStrainGradient` and `PancakeLowDiffusionBudget` now pay the preceding gradient cost when strain is restricted to a fixed finite frequency set K. Let χ be its multiplier, let $\mathcal{M} \supseteq K$ be the retained velocity modes, and let $U = \sum_{q\in\mathcal{M}} |\widehat{u}_q|^2$ and $\Omega = \sum_{q\in\mathcal{M}} |\widehat{\omega}_q|^2$. The strain is the actual symmetric gradient of the reconstructed filtered velocity; its coordinate derivatives are proved by differentiating the Fourier characters. A deliberately coarse explicit bound is

$$\sum_{j=1}^{3} \|\partial_j S_K(x)\|^2 \le C_{K,\chi} U, \qquad C_{K,\chi} = 243(2\pi)^4 \sum_{q\in K} |\chi(q)|^2 |q|^4.$$

The fixed factor 243 comes from a checked elementary matrix-to-operator bound and the three spatial directions, not a source-frequency count. Cauchy–Schwarz supplies the fourth frequency moment. Restriction identities identify this K-reconstruction with the full retained reconstruction whenever χ vanishes outside K.

The actual spatial density $G_K(x) = \sum_j \|\partial_j S_K(x)\|^2 |\omega_{\mathcal{M}}(x)|^2$ satisfies $\int G_K \le C_{K,\chi} U \Omega$ by finite Parseval, without a receiver-cardinality factor. For coefficient paths satisfying the retained velocity ODE, conjugate symmetry and incompressibility, the previously checked kinetic-energy identity gives

$$\int_0^T \int G_K \le \frac{C_{K,\chi} U(0)^2}{2\nu}, \qquad \frac{16\nu}{\delta} \int_0^T \int G_K \le \frac{8 C_{K,\chi} U(0)^2}{\delta}.$$

Time continuity and integrability of the spatial integral are derived from the coefficient paths, not supplied as extra budget hypotheses. These bounds are independent of the larger retained cutoff at fixed K, χ and initial energy, but not uniform as K expands. They do not construct the coefficient trajectories or close the full alignment budget.

Finite differences through spectral collisions. The module `PancakeSpectralFiniteDifference` supplies an exact precursor to a weak spatial chain rule. For symmetric strains S, T, let e_S be a constructed unit top eigenvector, $E_\delta(S, w) = \langle w, K_\delta(S) w\rangle$, and

$$L = \big(\langle e_S, (T-S)e_S\rangle |w|^2 - \langle w, (T-S)w\rangle\big) + 2\langle K_\delta(S)w, v\rangle.$$

The supporting Rayleigh inequality and the global Lipschitz estimate for the top eigenvalue yield, for every $\delta > 0$,

$$L + \frac{\delta}{2}|v|^2 \le E_\delta(T, w+v) - E_\delta(S, w) + \frac{8}{\delta}\|T-S\|^2 |w|^2.$$

There is no spectral-gap assumption. Summing with $T = S_{\sigma(i)}$, $v = w_{\sigma(i)} - w_i$ for a permutation of a finite index set cancels the energy differences exactly; this telescoping inequality is also checked. The next result supplies a spatial continuum limit for actual finite Fourier fields, with its full-rate envelope and measure-theoretic hypotheses stated explicitly.

Collision-safe Haar diffusion limit. Write

$$\mathcal{L}_\delta(S, w; A, b) = \big(\langle e_S, A e_S\rangle |w|^2 - \langle w, Aw\rangle\big) + 2\langle K_\delta(S)w, b\rangle.$$

For the actual finite Fourier reconstructions S_K and $w = \omega_{\mathcal{M}}$, set $A_j = \partial_j^2 S_K$ and $b_j = \partial_j^2 w$. The module `PancakePhysicalDiffusionLimit` constructs these derivatives from the Fourier coefficients and checks their derivative identities. For $\delta > 0$, $\nu \ge 0$, and integrable real fields F, B, the almost-everywhere envelope

$$F(x) \le \nu \mathcal{L}_\delta(S_K(x), w(x); A_j(x), b_j(x)) + B(x)$$

implies, for normalized Haar measure on the unit torus,

$$\int F \le \frac{16\nu}{\delta}\, B_{K,\chi}\, U\Omega + \int B, \qquad B_{K,\chi} = \sum_{q\in K} \left(9(2\pi)^2|\chi(q)|\,|q|^2\right)^2 = \frac{C_{K,\chi}}{3}.$$

There is no retained-high-mode cardinality factor. This is a coordinate diffusion estimate for any integrable full-rate envelope, not an assumption that the frame-dependent response is separately integrable.

The proof pairs translations by $\pm h$ in coordinate j. Haar invariance cancels both translated energy differences exactly. The reconstructed Fourier fields define twice continuously differentiable curves in the normed space of continuous spatial fields, so a proved vector-valued Taylor argument makes their centered second differences converge uniformly in space. Real continuous linear maps transfer this construction to the real strain operator and vorticity. The first strain increment is bounded by $|h|\sum_{q\in K} 9(2\pi)^2|\chi(q)||q|^2|\widehat{u}_q|$; Cauchy–Schwarz and finite Parseval then give the displayed price. The consistency errors are multiplied only by the continuous envelopes $|w|^2$ and $|K_\delta(S_K)w|$, whose integrals are finite. No twice differentiable top eigenvalue and no measurable selection of e_S is required.

Separately, `PancakeRegularizedMaterialRate` identifies $\mathcal{L}_\delta(S, w; \dot{S}, \dot{w})$ with the derivative of the actual energy wherever the top eigenvalue is time-differentiable. For $\dot{S} = -S^2 + R + \nu A$ and $\dot{w} = Sw + f + \nu b$, the exact split is

$$\dot{E}_\delta = -|z|^2 + \mathcal{A}_R + 2\langle z, f\rangle + 2\delta\langle w, Sw + f\rangle + \nu\mathcal{L}_\delta(S, w; A, b), \qquad z = (\lambda_{\max} I - S)w.$$

The extra regularization term is retained. The next construction combines the coordinates before integration and identifies a measurable material derivative; it never separately integrates the selected-frame coordinate responses.

Joint Laplacian and the constructed material derivative. `PancakeLaplacianDiffusion` sums the linear responses before taking the spatial integral. The summed continuous finite-difference envelopes converge with uniform consistency errors, giving

$$\int F \le \frac{16\nu}{\delta} C_{K,\chi} U\Omega + \int B \quad \text{if} \quad F \le \nu\mathcal{L}_\delta(S_K, w; \Delta S_K, \Delta w) + B \quad \text{a.e.}$$

Here F, B are integrable. `PancakePhysicalLaplacian` constructs both Laplacians by summing the actual three second Fourier derivatives. Their coefficient multiplier is $(2\pi i)^2|q|^2$. It also proves that the reconstructed viscous coefficients equal $\nu\Delta S_K$ and $\nu\Delta w$. Thus these are the viscous directions of the retained equation, not assigned operator fields.

Let R_0, f_0 denote the previously constructed strain remainder and vorticity forcing with their viscous coefficient terms omitted, evaluated at the current velocity. Setting the coefficient parameter to zero here does not replace the trajectory by an Euler solution. `PancakePhysicalViscousSplit` proves the exact identities $R_\nu = R_0 + \nu\Delta S_K$ and $f_\nu = f_0 + \nu\Delta w$, including the finite-support restrictions. The material equations therefore have the form used above. Young's inequality gives the pointwise bound

$$\dot{E}_\delta + \tfrac{1}{2}|z|^2 \le \nu\mathcal{L}_\delta(S_K, w; \Delta S_K, \Delta w) + \mathcal{B}_\delta, \qquad \mathcal{B}_\delta = 2\|R_0\||w|^2 + 2|f_0|^2 + 2\delta\langle w, S_K w + f_0\rangle.$$

The last term keeps its sign; it is not discarded as if it were negative.

To define the rate without a measurable eigenframe, write $v_K(t,x)$ for the actual filtered real velocity and set

$$\mathfrak{m}_\delta(t,x) = \left.\frac{d}{dh}\right|_{h=0} E_\delta\big(S_K(t+h, x+hv_K(t,x)), w(t+h, x+hv_K(t,x))\big),$$

with addition in the unit torus. Lean's derivative takes the value zero where it does not exist; the subsequent AE agreement theorem is essential. `PancakeMeasurableMaterialRate` proves joint measurability of this constructed scalar from continuous retained coefficients. `PancakeFrozenMaterialDerivative` derives its physical rate using the retained velocity equation and the straight path with the actual instantaneous material tangent. Only that tangent is needed for the chain rule, not a separately supplied global flow.

`PancakeSpacetimeSpectralDifferentiability` and `PancakeAEMaterialRate` remove the spectral differentiability hypothesis for C^1 coefficient paths. The real spacetime Fourier strain is proved C^1. On each compact ball its operator field is Lipschitz; the largest eigenvalue is globally 1-Lipschitz in that operator. Rademacher's theorem and countable ball exhaustion give real spacetime Fréchet differentiability almost everywhere. Every frozen material direction is then differentiable at these points. The quotient map of the measured fundamental cube to the torus is proved measure-preserving; measurability of the differentiability set and Fubini give the required AE-space statement at AE times. No simple eigenvalue or smooth eigenvector assumption is used, including at spectral collisions.

Finally, `PancakeMaterialSpatialEstimate` derives C^1 coefficient regularity from a supplied finite velocity solution on $\mathbb{R}$, using the explicit continuity of its RHS. Under the retained reality, incompressibility, mode-symmetry and finite-output closure hypotheses, it proves for almost every time

$$\int_{\mathbb{T}^3} \mathfrak{m}_\delta + \frac{1}{2}\int_{\mathbb{T}^3} |z|^2 \le \frac{16\nu}{\delta} C_{K,\chi} U\Omega + \int_{\mathbb{T}^3} \mathcal{B}_\delta.$$

Spatial integrability of the rate is a conclusion: its AE physical representative has a constructed continuous absolute envelope on the compact torus. No extra full-rate integrability budget is assumed. The finite velocity solution itself is still supplied, not constructed from arbitrary admissible data by this theorem.

Eulerian transport and time integration. The spatial estimate now feeds a checked time-integrated balance. `PancakeLocallyLipschitzEnergy` proves that the actual regularized alignment energy is locally Lipschitz on real spacetime, including at eigenvalue collisions. Compact time intervals and bounded representatives of the unit torus give spatially uniform time-Lipschitz bounds. The same construction supplies uniform coordinate-translation bounds. `PancakePeriodicWeakDerivative` derives periodic integration by parts for these locally Lipschitz scalar fields: Rademacher gives actual coordinate derivatives almost everywhere, the measured fundamental cube transfers them to Haar measure, and differentiation of a translated spatial mean gives zero. Product differentiation then proves

$$\int_{\mathbb{T}^3} (\partial_j F) G = -\int_{\mathbb{T}^3} F(\partial_j G).$$

`PancakePhysicalTransportCancellation` identifies the coordinate derivatives of the actual filtered velocity with their Fourier reconstructions. Retained incompressibility gives $\operatorname{div} v_K = 0$, hence $\int_{\mathbb{T}^3} v_K \cdot \nabla E_\delta = 0$.

`PancakeMaterialChainRule` and `PancakeAEMaterialChainRule` identify the actual frozen material derivative almost everywhere as $\mathfrak{m}_\delta = \partial_t E_\delta + v_K \cdot \nabla E_\delta$. `PancakeCompactMeanDerivative` justifies differentiation under the spatial integral using the uniform local Lipschitz bound. The mean energy is absolutely continuous on compact time intervals, so `PancakePhysicalMeanBalance` proves

$$\int_a^b \int_{\mathbb{T}^3} \mathfrak{m}_\delta \, dx\, dt = \overline{E}_\delta(b) - \overline{E}_\delta(a), \qquad \overline{E}_\delta(t) = \int_{\mathbb{T}^3} E_\delta(t,x)\, dx.$$

Time integrability of the mean material rate is a conclusion, not an assumption. No smooth eigenframe or separately supplied volume-preserving flow is used.

`PancakeParametricForcing` proves joint continuity of the actual nonviscous envelope from continuous retained coefficients. At viscosity parameter zero, the strain forcing depends only on finite quadratic expressions in those coefficients, even at output modes outside the retained set. The vorticity RHS is the actual curl of the velocity RHS. Spatial integration therefore gives continuous residual and forcing means. In `PancakeEulerianAlignmentBudget`, the integrated spatial estimate and the retained kinetic-energy identity yield, for $\delta > 0$, $\nu > 0$, and $T \geq 0$,

$$\boxed{\overline{E}_\delta(T) + \frac{1}{2}\int_0^T \int_{\mathbb{T}^3} |z|^2 \leq \overline{E}_\delta(0) + \frac{8C_{K,\chi}U(0)^2}{\delta} + \int_0^T \int_{\mathbb{T}^3} \mathcal{B}_\delta.}$$

The intermediate estimate pays $(16\nu/\delta)C_{K,\chi}\int_0^T U\Omega$; kinetic dissipation bounds $\int_0^T U\Omega \leq U(0)^2/(2\nu)$. This is a finite-cutoff Eulerian alignment budget for the supplied finite solution on $\mathbb{R}$, not arbitrary-data global regularity. The nonviscous forcing remains unpaid at all scales, the fixed-low constant grows as K expands, and neither $K \to \infty$ nor $\delta \to 0$ is justified. Construction from arbitrary data, the required stochastic objects, scale-uniform field transfer and continuation remain open.

Signed transport before Young's inequality. The vector forcing in the last estimate can be split before its norm is squared. Write v_M, S_M for the full retained velocity and strain, and let $f_{M,0}$ be the previously defined nonviscous vorticity forcing with velocity multiplier identically one. Define

$$a = v_K - v_M, \qquad g = f_{M,0} + (S_M - S_K)w.$$

`PancakeVorticityTransportSplit` proves for the actual Fourier reconstructions

$$f_{K,0} = g + (a \cdot \nabla)w.$$

The character-motion term is identified with the actual coordinate derivatives of w. Reality of the reconstructed velocity is used when passing from complex Fourier products to real transport; it is not assumed that real-part extraction preserves arbitrary products. The term $f_{M,0}$ remains explicit: finite-output truncation has not been discarded, and this term is not asserted to vanish.

For a symmetric matrix weight $K(x)$ with locally Lipschitz entries, `PancakeWeightedWeakTransport` and `PancakeWeightedMatrixTransport` prove the signed identity

$$2\int_{\mathbb{T}^3} \langle Kw, (a\cdot\nabla)w\rangle = -\int_{\mathbb{T}^3} \sum_{i,j,k=1}^{3} a_k(\partial_k K_{ij})w_i w_j, \qquad \operatorname{div} a = 0.$$

The proof differentiates scalar products almost everywhere, integrates by parts, and uses symmetry to identify the two transported-field terms. `PancakeWeightedOperatorTransport` constructs the entries in the canonical Euclidean basis and identifies the matrix sum with the actual operator inner product. No operator derivative is supplied independently.

`PancakePhysicalWeightedTransport` applies the identity to $K = K_\delta(S_K) = (\lambda_{\max}(S_K)+\delta)I - S_K$ and to the actual filtered velocity difference. The weight is locally Lipschitz even at spectral collisions: the already proved estimate $\|K_\delta(S) - K_\delta(T)\| \leq 2\|S - T\|$ suffices. Set

$$\mathcal{W}_\delta = \sum_{i,j,k=1}^{3} a_k\, \partial_k(K_\delta(S_K))_{ij}\, w_i w_j.$$

The weak entry derivatives are actual coordinate derivatives, with the AE interpretation established above. The module also proves the exact identity $\mathcal{W}_\delta = \mathcal{W}_0$: adding a constant multiple of the identity matrix does not change its spatial transport derivative. The signed forcing identity is therefore

$$\int_{\mathbb{T}^3} 2\langle K_\delta w, f_{K,0}\rangle = \int_{\mathbb{T}^3} 2\langle K_\delta w, g\rangle - \int_{\mathbb{T}^3} \mathcal{W}_0.$$

It contains no squared derivative of w from the transport mismatch. It does not assign a favorable sign to $\mathcal{W}_0$.

Define the reduced envelope

$$\widetilde{\mathcal{B}}_\delta = 2\|R_0\||w|^2 + 2|g|^2 + 2\delta\langle w, S_K w + g\rangle.$$

`PancakeReducedMaterialBudget` applies Young's inequality only to g, retains the signed transport pairing, and pays the same joint diffusion term as before. The spatial mean of $\mathcal{W}_\delta$ is continuous in time even though its pointwise weak derivatives need not be: `PancakeReducedForcingContinuity` identifies that mean with a difference of continuous physical forcing pairings. Thus time integrability is proved without an extra budget assumption. Under the same supplied finite-solution hypotheses, `PancakeReducedEulerianBudget` obtains

$$\overline{E}_\delta(T) + \frac{1}{2}\int_0^T \int_{\mathbb{T}^3} |z|^2 \le \overline{E}_\delta(0) + \frac{8C_{K,\chi}U(0)^2}{\delta} + \int_0^T \int_{\mathbb{T}^3} \widetilde{\mathcal{B}}_\delta - \int_0^T \int_{\mathbb{T}^3} \mathcal{W}_0.$$

This resolves the mismatch transport algebraically, not its all-scale budget. Uniform bounds for the signed weight variation, the reduced nonlinear and projection forcing, the growing cutoff constant and the limiting construction remain necessary. No frequency-uniform regularity theorem follows from this finite-cutoff identity alone.

Actual weight-gradient bound and its cutoff price. `PancakeWeightVariationGradient` identifies the weak coordinate derivative of $K_\delta(S_K)$ almost everywhere with

$$G_j = \langle e, \partial_j S_K e\rangle I - \partial_j S_K, \qquad \|G_j\| \le 2\|\partial_j S_K\|,$$

where e is a constructed unit top eigenvector. Only the scalar largest eigenvalue is differentiated, almost everywhere; e is not differentiated. The actual entrywise transport sum is proved to equal $\mathcal{W}_\delta = \sum_j a_j\langle w, G_j w\rangle$ almost everywhere. Consequently,

$$|\mathcal{W}_\delta| \le 2|a|\Big(\sum_j \|\partial_j S_K\|^2\Big)^{1/2}|w|^2.$$

The operator norm is used for each coordinate derivative. This estimate is independent of δ and contains no inverse eigenvalue gap.

For an arbitrary transporting multiplier β, put

$$H_{M,\beta} = \sum_{q\in M} |\beta(q)|^2, \qquad D_{K,M,\chi,\beta} = 2\sqrt{H_{M,\beta}}\sqrt{C_{K,\chi}}.$$

`PancakeWeightVariationCutoff` proves the physical velocity bound $|v_\beta| \le \sqrt{H_{M,\beta}}\sqrt{U}$ and then, by finite Parseval, $\int |\mathcal{W}_\delta| \le D_{K,M,\chi,\beta}U\Omega$. For the mismatch $\beta = \chi - 1$, write $D = D_{K,M,\chi,\chi-1}$.

Under the supplied retained-solution hypotheses and $\nu > 0$, the checked time payment and its insertion into the alignment budget are

$$\int_0^T \left| \int_{\mathbb{T}^3} \mathcal{W}_\delta \right| \le \frac{D\,U(0)^2}{2\nu},$$
$$\overline{E}_\delta(T) + \frac{1}{2}\int_0^T \int |z|^2 \le \overline{E}_\delta(0) + \left(\frac{8C_{K,\chi}}{\delta} + \frac{D}{2\nu}\right) U(0)^2 + \int_0^T \int \widetilde{\mathcal{B}}_\delta.$$

This is a finite-cutoff payment, not a uniform one. The same module proves the exact identity

$$H_{M,\chi-1} = |M \setminus K| + \sum_{q\in K} |\chi(q) - 1|^2, \qquad K \subseteq M, \quad \operatorname{supp}\chi \subseteq K.$$

Thus the estimate itself exposes a retained-high-mode count. This is a limitation of this bound, not a lower bound on the actual transport term and not a refutation of a finer cancellation.

Alignment depletion, including the full top eigenspace. `PancakeAnisotropyDepletion` retains the transverse component $\eta = w - \langle e, w\rangle e$. For any real continuous linear operator A and unit e, it proves

$$\left|\langle e, Ae\rangle |w|^2 - \langle w, Aw\rangle\right| \le 4\|A\|\,|w|\,|\eta|.$$

The exact expansion into the transverse quadratic term and two mixed terms gives the bound. Applied to the physical weight variation, it yields

$$|\mathcal{W}_\delta| \le 4\Big(\sum_j |a_j| \|\partial_j S_K\|\Big)|w|\,|\eta| \quad \text{almost everywhere.}$$

No time-integrated estimate of this depleted expression is asserted here.

At a repeated largest eigenvalue, the selected line is not the full aligned space. `PancakeTopSpaceTransport` proves the stronger exact vanishing statement: if $K_0(S_K)w = 0$, then $\mathcal{W}_\delta = 0$ almost everywhere. Indeed, along a differentiable operator path, the nonnegative scalar function $\langle w, K_0(S_K)w\rangle$ has derivative zero whenever its value is zero. This applies to every vector in the entire top eigenspace, without choosing a differentiable eigenvector or dividing by a spectral gap. Quantitative control near that space, uniform summation, the remaining reduced forcing, and arbitrary-data continuation are still open.

The actual projection defect and its signed oscillation cost. Let $M \subseteq O$ and assume $M + M \subseteq O$, with the preceding nonzero retained frequencies, incompressibility, conjugate reality and negation symmetry. Write N_q for the actual nonviscous convolution coefficient and define the discarded output field

$$D(x) = \operatorname{Re} \sum_{q\in O\setminus M} e^{2\pi i q\cdot x} N_q.$$

`PancakePhysicalNonlinearReconstruction` identifies the full-output reconstruction with $S_M w - (v_M \cdot \nabla)w$. It uses physical curl inversion, the exact matrix–vector convolution and real reconstruction of the velocity and strain. Consequently `PancakeRetainedProjectionDefect` proves

$$f_{M,0} = -D, \qquad g = (S_M - S_K)w - D.$$

These are instantaneous identities for the same coefficients u; setting the viscosity parameter to zero in this expression does not replace the viscous solution by an Euler solution.

`PancakeFiniteFourierOrthogonality` proves orthogonality of disjoint finite Fourier supports, including the real vector pairing when the first reconstructed field is real. No reality assumption on the second output set is needed. The retained curl has conjugate symmetry, and `PancakeProjectionWeightOscillation` extends the cancellation to every constant real linear operator A:

$$\int_{\mathbb{T}^3} \langle Aw, D\rangle = 0.$$

For any continuous operator weight $L(x)$ it follows that

$$\int \langle L(x)w, D\rangle = \int \langle (L(x) - A)w, D\rangle.$$

This cancellation is global. Multiplying by a spatial cell cutoff can mix frequencies, so the same cancellation is not asserted on each cell.

`PancakeProjectionEnergyCancellation` first removes the scalar regularization from the integrated projection pairing. Taking $A = K_0(S_K(x_0))$ then gives the checked exact identity

$$\int 2\langle K_\delta(S_K)w, f_{M,0}\rangle = -2\int \langle [K_0(S_K(x)) - K_0(S_K(x_0))]w, D\rangle.$$

If $\|S_K(x) - S_K(x_0)\| \le \varepsilon$ for all x, the already proved gap-operator Lipschitz bound yields

$$\left|\int 2\langle K_\delta(S_K)w, f_{M,0}\rangle\right| \le 4\varepsilon \int |w|\,|D|.$$

There is no inverse eigenvalue gap, inverse δ or mode count in this inequality. The size and time integral of D are still unbounded here; the displayed estimate alone is not a cutoff-uniform budget.

`PancakeProjectionMaterialBudget` preserves this cancellation inside the material estimate. Set

$$h = (S_M - S_K)w, \qquad \mathcal{B}_\delta^h = 2\|R_0\||w|^2 + 2|h|^2 + 2\delta\langle w, S_K w + h\rangle.$$

For integrable F dominated almost everywhere by the actual material rate, $\delta > 0$ and $\nu \ge 0$, the checked spatial estimate is

$$\int F + \frac{1}{2}\int |z|^2 \le \frac{16\nu}{\delta} C_{K,\chi} U\Omega + \int \mathcal{B}_\delta^h - \int \mathcal{W}_\delta \\ - 2\int \langle [K_0(S_K(x)) - K_0(S_K(x_0))]w, D\rangle.$$

Only h, not $h - D$, is squared by Young's inequality. Thus the projection defect has a signed linear cost instead of an artificial squared cost. The spatially varying projection term, high-strain term, strain remainder, transport variation, growing diffusion constant and arbitrary-data limiting construction remain to be controlled. No new time-integrated projection budget is claimed here.

Localizing the cancellation: the exact escaping field. For finite scalar cutoff frequencies P and coefficients c_p, let $\rho = \sum_{p\in P} c_p e^{2\pi i p\cdot x}$ be real-valued. Let $b_q = \sum_{p+k=q,\ p\in P,\ k\in M} c_p\omega_k$ be the already constructed cutoff convolution, and define

$$R_\rho = \mathrm{Re}\sum_{q\in M} e^{2\pi i q\cdot x} b_q, \qquad E_\rho = \mathrm{Re}\sum_{q\in (P+M)\setminus M} e^{2\pi i q\cdot x} b_q.$$

Here E_ρ is escaped *localized vorticity*, not the discarded *nonlinear output* D above. `PancakeLocalizedProjectionBoundary` proves

$$\rho w = R_\rho + E_\rho, \qquad \int \langle A(\rho w), D\rangle = \int \langle AE_\rho, D\rangle$$

for every constant real operator A and negation-closed M. `PancakeRealFourierProjection` removes the need for conjugate symmetry of the localized coefficients in this step: on a negation-closed set, replacing b_q by $(b_q+\overline{b_{-q}})/2$ leaves the real reconstruction unchanged and makes it real as a complex reconstruction. Thus reflection of frequencies under real-part extraction is explicitly accounted for.

The source of E_ρ is narrowed further by `PancakeCutoffBoundarySupport`. Define the actual finite boundary receiver set

$$\partial_P M = \{k \in M : \ \exists p \in P, \ p + k \notin M\}.$$

At every exterior output $q \notin M$, replacing receiver set M by $\partial_P M$ leaves b_q unchanged. Accordingly E_ρ reconstructs only outputs in $(P + \partial_P M) \setminus M$. If every shift of every active receiver lies inside M, then $E_\rho = 0$ exactly. This is a buffered-support theorem, not a claim that the full unbuffered field has no boundary receivers.

Quadratic weights, coverage error, and a count-free energy bound. Take the previously constructed weights $\rho_i = |p_i|^2 \geq 0$ with their actual collected Fourier coefficients. `PancakeQuadraticProjectionLocalization` identifies their scalar reconstructions with ρ_i and sets $E_i = E_{\rho_i}$. The support of each quadratic coefficient family lies in the proved difference set $\operatorname{supp} p_i - \operatorname{supp} p_i$; a buffer containing all these candidate shifts makes the corresponding E_i zero. For a continuous operator field $L(x)$ and constant operators A_i, the exact integrated decomposition is

$$\int \langle Lw, D\rangle = \int \sum_i \rho_i \langle (L - A_i)w, D\rangle + \sum_i \int \langle A_i E_i, D\rangle + \int \left(1 - \sum_i \rho_i\right) \langle Lw, D\rangle.$$

The three terms are local freezing error, boundary-frequency coupling, and coverage defect. The approximate Fourier partition is not silently normalized or treated as exact.

`PancakeBoundaryEnergy` proves the orthogonal energy splitting

$$\int |\rho w|^2 = \int |R_\rho|^2 + \int |E_\rho|^2.$$

In particular, if $\sum_i \rho_i(x) \leq C$ everywhere, positivity and finite Parseval give

$$\sum_i \int |E_i|^2 \leq \int \sum_i \rho_i^2 |w|^2 \leq C^2 \int |w|^2 \leq C^2 \Omega.$$

There is no cell-count or retained-mode-count factor. This controls the *sum of individual squared norms*, not the squared norm of an unweighted sum of exterior fields. The latter does not follow without additional cross-term control.

For a fixed quadratic family and a supplied retained velocity solution, `PancakeBoundaryTimeBudget` constructs continuity of the actual parameterized cutoff convolution and its mean energy. Time integrability is therefore proved, not supplied as a budget assumption. With $\nu > 0$, $T \geq 0$, the preceding retained reality/incompressibility hypotheses and the retained ODE supplied on $\mathbb{R}$, kinetic dissipation gives

$$\boxed{\int_0^T \sum_i \int_{\mathbb{T}^3} |E_i(t,x)|^2 \leq \frac{C^2}{2\nu} U(0).}$$

For the constructed coverage bound $C = 3/2$, the price is $9U(0)/(8\nu)$, independent of cell count, cutoff bandwidth and retained mode count. The weights are fixed in time in this theorem. Moving their supports or coefficients requires further time-regularity work. Most importantly, this pays only the escaped-vorticity side of $\sum_i \int \langle A_i E_i, D\rangle$. It does not bound the nonlinear field D, the weighted sum's cross terms by itself, moving-patch rates, or the remaining high-strain and strain-remainder terms. The scale-critical coupled budget and arbitrary-data continuation remain open.

The full boundary sum retains spectral depletion. The cross terms can now be handled explicitly. For arbitrary constant real operators A_i, A_j, `PancakeWeightedBoundaryAssembly` proves

$$\int \langle A_i R_i, A_j E_j\rangle = 0.$$

Indeed the adjoint moves A_j to the retained factor, after which the negation-closed real Fourier orthogonality applies. Neither operator needs to be symmetric. Set

$$G = \sum_i \rho_i A_i w, \qquad \mathcal{R} = \sum_i A_i R_i, \qquad \mathcal{B} = \sum_i A_i E_i.$$

Then $G = \mathcal{R} + \mathcal{B}$ and

$$\int |G|^2 = \int |\mathcal{R}|^2 + \int |\mathcal{B}|^2.$$

Weighted Cauchy–Schwarz, checked in `PancakeWeightedBoundaryDepletion`, therefore gives

$$\boxed{\int |\mathcal{B}|^2 \le C \int \sum_i \rho_i |A_i w|^2.}$$

This is the squared norm of the *complete operator-weighted sum*. No cell count enters the bound, and the actual operator action has not been replaced by $\|A_i\|\,|w|$.

For symmetric S, write $K_0(S) = \lambda_{\max}(S) I - S$, $d_S(w) = \lambda_{\max}(S)|w|^2 - \langle w, Sw\rangle$ and $\sigma(S) = \lambda_{\max}(S) - \lambda_{\min}(S)$. The two proved spectral-coordinate identities yield

$$|K_0(S)w|^2 \le \sigma(S) d_S(w) \le 2\|S\| d_S(w).$$

`PancakeSpectralResidualDepletion` also proves $d_S(w) = 0 \iff K_0(S)w = 0$. These statements hold at eigenvalue collisions and refer to the entire top eigenspace; there is no inverse gap. With $A_i = K_0(S_i)$, the assembled boundary estimate thus becomes

$$\int |\mathcal{B}|^2 \le C \int \sum_i \rho_i \sigma(S_i) d_{S_i}(w).$$

Unfreezing must retain the same benefit. The collision-safe inequality $\|K_\delta(T) - K_\delta(S)\| \le 2\|T - S\|$ implies

$$|K_\delta(T)w|^2 \le 2|K_\delta(S)w|^2 + 8\|T - S\|^2 |w|^2.$$

`PancakeBoundaryFreezingError` consequently proves, for a continuous strain field $S(x)$ and constant sampled strains S_i,

$$Q(x) = \sum_i \rho_i(x) \|S_i - S(x)\|^2,$$
$$\int |\mathcal{B}|^2 \le 2C^2 \int |K_\delta(S(x))w(x)|^2 + 8C \int Q(x)|w(x)|^2,$$

where here $A_i = K_\delta(S_i)$. The unregularized case $\delta = 0$ is specialized in `PancakePhysicalBoundaryDepletion` to the actual filtered Fourier strain, its sampled values at the patch centers, and $w = \operatorname{Re}\operatorname{Rec}_M \operatorname{curl} u$. The exact three-term nonlinear pairing above then uses the single field $\mathcal{B}$ in its middle term, including all cross-cell contributions.

The variance is quantified in `PancakeBoundaryFreezingVariance`. If an exactly subordinate partition ψ_i has active-cell oscillation at most ε, $|\rho_i - \psi_i| \le \eta_i$, and the global oscillation from center i is at most L_i, then

$$Q(x) \le \varepsilon^2 + \sum_i \eta_i L_i^2.$$

For the constructed quadratic family with approximation tolerance $a > 0$, one may take $\eta_i = a(a+2)$ and $L_i = L$, obtaining $V = \varepsilon^2 + na(a+2)L^2$ with n cells. Provided $na(a+2) \le 1/2$, the same existentially constructed family has coverage between $1/2$ and $3/2$, off-cell leakage below a^2, and

$$\int |\mathcal{B}|^2 \le \frac{9}{2} \int |K_0(S)w|^2 + 12V \int |w|^2.$$

The partition and its active-cell/global oscillation bounds are explicit inputs to this last construction. Its Fourier approximation precision must pay the displayed aggregate tail; no uniform bandwidth or moving support cost is inferred. In particular, this aligns the boundary energy with the actual negative residual square in the material identity, but does not yet control $\int \langle \mathcal{B}, D \rangle$. Squaring D by Young's inequality would leave an unpaid nonlinear-output energy, not a closed arbitrary-data estimate.

Removing the projection artifact under a common smoothness envelope. The discarded field D is a finite-projection error. One can remove this error by a consistency limit under an explicit smoothness hypothesis, without claiming an initial-energy bound for $|D|^2$. For velocity coefficients u_k, put

$$m_s(u,k) = (1+|k|)^s |u_k|.$$

Suppose $g_k \ge 0$, $G = \sum_k g_k < \infty$, and $m_2(u,k) \le g_k$. `PancakeCurlOutputTail` checks the two physical curl factors, including the unit-torus derivative phase $2\pi i$. In particular, the ordered coefficient of $\operatorname{curl}(u \times \operatorname{curl} u)$ at an input pair (k,l) has norm at most

$$H_g(k,l) = (2\pi)^2 g_k g_l.$$

This majorant is summable over all lattice pairs. Define

$$\tau_M(g) = \sum_{k,l:\ k+l \notin M} H_g(k,l).$$

`PancakeSummableOutputTail` proves $\tau_M(g) \to 0$ as the finite retained sets exhaust the lattice: each fixed output eventually lies in M, and the entire pair series has a common summable bound. `PancakeExteriorReconstructionTail` proves the corresponding uniform reconstruction estimate, allowing arbitrary finite input-pair and output sets.

For mean-zero velocity, $u_0 = 0$, with $k \cdot u_k = 0$ on M, `PancakeZeroModeCurlBridge` identifies the actual nonviscous vorticity coefficients with this output curl even when $0 \in M$. `PancakeDiscardedNonlinearLimit` consequently gives

$$\|D_{M,O}(u)\|_{L^\infty} \le \tau_M(g).$$

The same finite core works for every coefficient field under the same g and every finite output set O; this is not restricted to a single fixed polynomial.

If $|\chi_k| \le B$ with $B \ge 0$, the actual physical fields satisfy

$$\|w_M\|_{L^\infty} \le 2\pi G, \qquad \|S_{\chi,M}\|_{L^\infty} \le 9(2\pi)BG, \qquad \|K_0(S_{\chi,M})\|_{L^\infty} \le 18(2\pi)BG.$$

The factor 9 is the explicit matrix-to-operator estimate used in `PancakeSmoothSpectralEnvelope`, not a frequency count. For normalized torus measure, `PancakeSmoothProjectionPairing` therefore proves

$$\left| \int \langle K_0(S_{\chi,M}) w_M, D_{M,O} \rangle \right| \le C(B,g)\tau_M(g), \qquad C(B,g) = 18(2\pi)^2 BG^2.$$

For a time-dependent coefficient path continuous in each retained coordinate, `PancakeSmoothProjectionTimeLimit` constructs joint continuity of the actual discarded nonlinearity and continuity of this spatial mean. If the same envelope and mean-zero/transversality hypotheses hold throughout $[0,T]$, with $T \ge 0$ and a fixed multiplier χ, then

$$\boxed{\int_0^T \left| \int \langle K_0(S_{\chi,M}) w_M, D_{M,O} \rangle \right| dt \le TC(B,g)\tau_M(g) \longrightarrow 0.}$$

The coefficient continuity used here is on $\mathbb{R}$; domination is required only on $[0,T]$. The convergence is uniform over paths with the same envelope and multiplier bound. Time integrability is proved, not inserted as an assumption.

This is a conditional consistency theorem, not a global regularity estimate. It neither constructs the common envelope from a solution, derives it from kinetic energy, constructs an arbitrary-data Galerkin family, nor passes the complete material identity to a continuum limit. The envelope may be used only where it has been established; its cost must not be relabelled an a priori global bound. Intrinsic high-strain, pressure, moving-weight and scale-summed misalignment estimates remain separate open obligations.

Joint input/output consistency against a constructed full series. An output-only limit does not justify restricting a continuum solution to a Galerkin system: high input frequencies can also contribute to retained outputs. The full comparison is now constructed explicitly. With $e_q(x) = \exp(2\pi i q \cdot x)$, set

$$C_{k,l}(u) = 2\pi i(k+l) \times (u_k \times \omega_l), \qquad \mathcal{N}_\infty(u,x) = \sum_{k,l} \mathrm{Re}\big(e_{k+l}(x) C_{k,l}(u)\big).$$

Here $\omega_l = 2\pi i\, l \times u_l$ is the physical curl coefficient. `PancakeInfinitePairReconstruction` constructs this type of real pair series, proves absolute summability under a summable coefficient majorant, and proves continuity by uniform convergence. Finite coefficient collection is identified exactly with the existing reconstruction.

For a mean-zero velocity the pair $C_{k,l}(u)$ is zero if $k = 0$, $l = 0$ or $k + l = 0$. Thus `PancakeZeroSafeInteractionTail` replaces H_g by

$$H_g^0(k,l) = \begin{cases} (2\pi)^2 g_k g_l, & k \ne 0,\ l \ne 0,\ k+l \ne 0, \\ 0, & \text{otherwise.} \end{cases}$$

Let $\mathcal{T}_M = \{(k,l) : k, l, k+l \in M\}$ and

$$\sigma_M(g) = \sum_{(k,l) \notin \mathcal{T}_M} H_g^0(k,l).$$

`PancakeInteractionTruncationTail` proves that this combined input/output tail tends to zero. Moreover, $\sigma_{M\setminus\{0\}}(g) = \sigma_M(g)$ exactly: excluding the zero frequency does not leave a spurious positive majorant contribution. This makes the exhaustion compatible with nonzero retained-mode sets.

Write $\mathcal{N}_M(u)$ for the real reconstruction on M of the actual finite nonviscous vorticity RHS, using M for both inputs. `PancakeFullNonlinearConsistency` proves, for $u_0 = 0$, transversality on M, and the preceding common second-moment envelope,

$$\|\mathcal{N}_\infty(u) - \mathcal{N}_M(u)\|_{L^\infty} \le \sigma_M(g) \longrightarrow 0.$$

Its finite-core theorem is uniform over all coefficient fields under g, including exhaustion through $M \setminus \{0\}$.

`PancakeParametricInfiniteCurl` proves joint continuity of the infinite series for continuously parameterized coefficients under a common envelope. The parameter space may be the compact interval $[0,T]$ itself. Accordingly `PancakeFullNonlinearTimeLimit` needs only coordinate continuity, the common envelope and the algebraic hypotheses on that interval, not on all of $\mathbb{R}$. For fixed $|\chi_k| \le B$, with $R_M = \mathcal{N}_\infty(u) - \mathcal{N}_M(u)$, it proves

$$\boxed{\int_0^T \int_{\mathbb{T}^3} |\langle K_0(S_{\chi,M})w_M, R_M\rangle|\, dx\, dt \le TC(B,g)\sigma_M(g) \longrightarrow 0.}$$

Absolute value is now inside both integrals. Spatial and temporal integrability are constructed, and the same conclusion holds with $M \setminus \{0\}$ throughout.

This closes the combined Fourier-series consistency estimate under the stated envelope. The spatial nonlinear identification is supplied next. Constructing a continuum solution and its envelope, passing the complete material inequality to the limit, and obtaining the intrinsic scale-critical dynamical budget remain separate obligations.

Identification with the actual real curl nonlinearity. The infinite interaction is now identified with a derivative-defined physical field. This step does not define curl by its Fourier symbol. For a frequency map $\kappa : I \to \mathbb{Z}^3$ and complex vector coefficients a_n, construct the series in the Banach space of continuous fields:

$$F_{\kappa,a}(x) = \sum_{n\in I} e_{\kappa(n)}(x)a_n, \qquad d_j a_n = 2\pi i\, \kappa(n)_j a_n.$$

`PancakeInfiniteFourierDerivative` differentiates its translation curve in that Banach space under absolute summability of a and $d_j a$. Evaluation gives the ordinary coordinate derivative on the torus. `PancakeIndexedFourierMoments` pays these hypotheses with

$$M_1^\kappa(a_n) = (1 + |\kappa(n)|)|a_n|, \qquad |d_j a_n|_\infty,\ |2\pi i\, \kappa(n) \times a_n|_\infty \le 2\pi M_1^\kappa(a_n).$$

Here the unsubscripted coefficient norm is Euclidean; its relation to the coordinate sup norm is proved explicitly. `PancakeInfiniteSpatialCurl` consequently proves

$$\nabla \times F_{\kappa,a} = F_{\kappa,\, 2\pi i\kappa\times a},$$

where the left side uses actual derivatives of coordinate translations.

For $\kappa(k,l) = k + l$, the product estimate is

$$(1 + |k + l|)|u_k \times v_l| \le (1 + |k|)|u_k|\,(1 + |l|)|v_l|.$$

Thus two summable first moments pay the first moment of the pair series. Continuous bilinearity and absolute double summation in `PancakeInfiniteFourierProduct` identify it exactly with $F_{\mathrm{id},u} \times F_{\mathrm{id},v}$. `PancakeInfiniteRealCurl` proves reality by conjugation and reindexing $k \mapsto -k$, and commutes real part with the actual derivatives.

Write $U = \operatorname{Re} F_{\mathrm{id},u}$. Assume the reality symmetry $u_{-k} = \overline{u_k}$ and a summable second moment $\sum_k (1+|k|)^2 |u_k| < \infty$. The theorem in `PancakeFullAnalyticNonlinearity` gives

$$\boxed{\mathcal{N}_\infty(u) = \nabla \times \big(U \times (\nabla \times U)\big).}$$

Both curls here are ordinary derivatives of real fields. The second moment pays the first moments of both u and its curl. No incompressibility or zero-mean hypothesis is needed for this identification itself. Under the additional mean-zero, transversality and common-envelope hypotheses used above, the retained nonlinear RHS therefore satisfies

$$\big\|\nabla \times \big(U \times (\nabla \times U)\big) - \mathcal{N}_M(u)\big\|_{L^\infty} \le \sigma_M(g) \longrightarrow 0.$$

This removes the spatial series-identification gap. This identification theorem alone does not construct a time-evolving Navier–Stokes solution or its common envelope. The higher-order local construction below supplies both on a positive interval, but does not bound that envelope globally. The intrinsic high-strain, pressure, moving-weight and scale-critical misalignment estimates remain open.

Constructing the common envelope from physical regularity. The envelope can now be obtained from defined spatial derivatives of a physical input field. For a scalar field $f : \mathbb{T}^3 \to \mathbb{C}$, set

$$D_j f(x) = \frac{d}{dh} f(x + h e_j)\Big|_{h=0}, \qquad J_{j,n} f = D_j^n f.$$

`PancakeContinuousCoordinateJet` uses these iterates themselves. Its order-six regularity predicate requires continuity of $J_{j,n} f$ for $0 \le n \le 6$ and coordinate differentiability of the iterates for $n < 6$. No independently chosen derivative fields are supplied.

Haar translation invariance gives

$$\widehat{f(\cdot + c)}(k) = e_k(c) \widehat{f}(k).$$

Differentiation under this integral, dominated by the norm of the continuous derivative, proves $\widehat{D_j f}(k) = 2\pi i k_j \widehat{f}(k)$ in `PancakePhysicalFourierDerivative`. Iteration yields

$$(2\pi |k_j|)^n |\widehat{f}(k)| \le \|J_{j,n} f\|_\infty.$$

If $B \ge 0$ bounds the sup norms of the three field components and all their pure sixth coordinate derivatives, `PancakePhysicalSixthDecay` proves the explicit vector estimate

$$(1+|k|)^6 |\widehat{f}(k)| \le 12288 B.$$

The proof selects a coordinate controlling the frequency norm, retains the physical 2π factor, and handles the low-frequency case with the field norm.

`PancakeLatticeDecayEnvelope` applies lattice summability in dimension three, with the zero frequency handled separately, to prove

$$w_k = (1+|k|)^{-4}, \qquad \sum_{k \in \mathbb{Z}^3} w_k < \infty.$$

Thus a uniform sixth-order coefficient bound A constructs $g_k = Aw_k$ and pays $m_2(\widehat{f}, k) \le g_k$. For a compact parameter space P, joint continuity on $P \times \mathbb{T}^3$ of the field and its defined sixth coordinate derivatives supplies the uniform B. The lower derivatives need the stated coordinate-jet regularity in each parameter slice. `PancakeCompactJetEnvelope` consequently constructs one nonnegative summable g for the whole physical family.

`PancakePhysicalFieldReconstruction` proves that the actual Fourier integrals reconstruct the original continuous field under this summability, and that a real input has conjugate-symmetric coefficients. Combining this identification with the envelope construction, `PancakePhysicalFamilyConsistency` proves: for a compact family f_p with the preceding regularity, reality, mean-zero coefficients and transversality, every $\varepsilon > 0$ admits one finite frequency core K such that, for every finite $M \supset K$,

$$\sup_{p\in P,\ x\in\mathbb{T}^3} \left|\nabla \times (f_p \times (\nabla \times f_p))(x) - \mathcal{N}_{M\setminus\{0\}}(\widehat{f_p}, x)\right| < \varepsilon.$$

The formal statement bounds each p, x uniformly; f_p is identified with its real Euclidean field. This theorem assumes physical regularity, not a Fourier envelope. It does not construct a Navier–Stokes evolution or prove that this regularity persists. A compact subinterval of an established smooth lifespan is an admissible parameter space; extending that lifespan still requires the intrinsic dynamical estimates.

Constructed local retained evolution from physical data. The finite coefficient evolution is now constructed, rather than supplied as a path satisfying an equation. Write

$$E_M = \mathbb{C}^{M\times\{1,2,3\}}, \qquad H_M = \{v \in E_M :\ k \cdot v_k = 0,\ v_{-k} = \overline{v_k}\},$$

where coefficients outside M are extended by zero. The subspace H_M is a real Hilbert space. Assume $M = -M$ and $0 \notin M$. `PancakeFiniteVelocitySmooth` proves that the previously defined physical velocity RHS, restricted to E_M, is smooth: its expression is finite and quadratic, and every inverse wave-number factor is constant with respect to the state.

`PancakeFiniteVelocityInvariant` proves directly that this vector field maps H_M into H_M. The Fourier Leray projection preserves conjugate symmetry, and its output is transverse. The transport convolution and viscous coefficient respect the same reality symmetry. An orthogonal projection gives a smooth map with codomain H_M; the checked inclusion identity proves that this projection changes no coefficient of the actual vector field on H_M.

Picard–Lindelöf on this complete finite-dimensional space yields, in `PancakeLocalVelocityExistence`, a time $T_M > 0$ and a coefficient curve u_M such that

$$\begin{aligned}
&(u_M)_k(0) = \widehat{u}_0(k) \quad (k \in M), \qquad (u_M)_k(t) = 0 \quad (k \notin M),\\
&\partial_t (u_M)_k = V_M(u_M)_k \quad (0 \le t \le T_M,\ k \in M),\\
&k \cdot (u_M)_k = 0, \qquad (u_M)_{-k} = \overline{(u_M)_k}.
\end{aligned}$$

A closed positive-time interval is chosen strictly inside the open existence interval, so the derivative statements hold at both endpoints. There is no input evolution or assumed invariant-subspace preservation.

The already checked energy and curl identities now apply to this constructed curve. `PancakeLocalVelocityEnergy` proves the actual finite vorticity equation and

$$U_M(t) + 2\nu \int_0^t \Omega_M(s)\, ds = U_M(0), \qquad 0 \le t \le T_M.$$

For $\nu \geq 0$, `PancakeLocalVelocityUnique` proves uniqueness on this interval: the energy ball is compact, the smooth finite vector field is Lipschitz on it, and the ODE uniqueness theorem applies.

Finally, `PancakeLocalL2Velocity` takes actual componentwise $L^2(\mathbb{T}^3)$ data, not a free Fourier coefficient sequence. Almost-everywhere reality gives conjugate symmetry of the Fourier integrals; incompressibility is stated as their Fourier divergence condition. Parseval yields, for every retained set and on its constructed lifespan,

$$U_M(t) \leq \|u_0\|_{L^2}^2, \qquad 2\nu \int_0^t \Omega_M(s)\, ds \leq \|u_0\|_{L^2}^2.$$

Thus the energy payment has no cutoff or mode-count loss. At this stage the quantifier order is $\forall M\, \exists T_M > 0$; the continuation construction below subsequently provides a common interval and then an infinite coefficient evolution. The nonzero truncations exclude the initial mean; their initial matching is asserted only on retained modes. Uniform higher-regularity estimates, passage to the continuum, and the intrinsic global alignment budget remain necessary.

Cutoff-independent local second-order energy control. Set

$$w(k) = (1+|k|)^2, \qquad E_M(u) = \sum_{k\in M} w(k)^2 |u_k|^2, \qquad D_M(u) = \sum_{k\in M} w(k)^2 |k|^2 |u_k|^2,$$

and let $L = \sum_{k\in\mathbb{Z}^3} (1+|k|)^{-4} < \infty$ be the already constructed lattice sum. All coefficient norms here are Euclidean. `PancakeFiniteConvolutionL2` proves a finite convolution estimate without a mode-count factor. If a_M is the zero extension of a sequence on M, translation injectivity and Cauchy–Schwarz give

$$\sum_{k,l\in M} a_M(k+l) b_k c_l \leq \|a\|_{\ell^2(M)} \|b\|_{\ell^2(M)} \sum_{l\in M} c_l \quad (c_l \geq 0),$$

and the analogous bound with the two inputs exchanged. `PancakeSobolevWeights` proves

$$w(k+l) \leq 2\big(w(k)+w(l)\big), \qquad \sum_{k\in M} a_k \leq \sqrt{L} \left(\sum_{k\in M} w(k)^2 a_k^2 \right)^{1/2}.$$

Applying these two input placements to $a_k = |u_k|$ gives the weighted convection majorant $4\sqrt{L}\, E_M \sqrt{D_M}$.

`PancakeSobolevVelocityRate` identifies this majorant with the existing physical velocity equation, not a replacement convolution. The Leray projection does no work against the transverse output coefficient; the nonzero transverse source coefficients justify the existing curl-to-velocity recovery identity. Retaining the physical 2π factor and the exact viscosity term yields

$$E_M' + 2\nu(2\pi)^2 D_M \leq 8(2\pi)\sqrt{L}\, E_M \sqrt{D_M}.$$

The derivative is obtained from the actual coefficient ODE. `PancakeSobolevAbsorption` then proves, for $\nu > 0$,

$$\boxed{E_M' + \nu(2\pi)^2 D_M \leq C_\nu E_M^2, \qquad C_\nu = \frac{16L}{\nu}.}$$

There is no cutoff radius or cardinality in C_ν.

For any initial bound $E_M(0) \leq B_0$, $B_0 \geq 0$, define

$$\tau_\nu(B_0) = \frac{1}{2(C_\nu+1)(B_0+1)} > 0.$$

`PancakeQuadraticGrowthWindow` uses monotonicity of $(E+1)^{-1} + C_\nu t$ to handle zero initial energy without division by it. `PancakeUniformLocalSobolev` applies this scalar comparison to the constructed local coefficient solution:

$$E_M(t) \le 2(B_0+1), \qquad 0 \le t \le T_M, \quad t \le \tau_\nu(B_0).$$

A nonnegative summable initial envelope $m_2(u_0,k) \le g_k$ pays $E_M(0) \le (\sum_k g_k)^2$ for every M. `PancakePhysicalInitialWindow` constructs such an envelope from the defined order-six continuous coordinate derivatives of the original physical initial field. It combines that construction with actual local Galerkin existence, assuming reality and the Fourier divergence condition. No regularity or Fourier envelope of a future trajectory is an input.

This cutoff-independent local *a priori* estimate bounds each solution on the intersection of its existing lifespan with the common window. The following continuation construction extends that conclusion to actual solutions on the entire common window. The quadratic growth inequality is not a global regularity bound.

Constructed continuation and a common Galerkin interval. `PancakeDissipativeFiniteODE` proves a finite-dimensional continuation theorem directly. Let F be a smooth vector field on a finite-dimensional real Hilbert space, with $\langle x, F(x)\rangle \le 0$ for every x. Choose a smooth scalar bump χ that is one on $|x| \le |x_0|+1$, nonnegative everywhere, and zero outside a larger ball. The field $G = \chi F$ is smooth with compact support, hence globally bounded and Lipschitz. Picard–Lindelöf on $[-1, T+1]$, with a spatial ball enlarged to pay its global speed bound, constructs a trajectory for every prescribed finite $T \ge 0$. There is no small-time restriction on the Lipschitz constant in this construction.

Along the constructed curve,

$$\frac{d}{dt}|a(t)|^2 = 2\chi(a(t))\langle a(t), F(a(t))\rangle \le 0 \qquad (0 \le t \le T).$$

Thus $|a(t)| \le |x_0|$, so $\chi(a(t)) = 1$ throughout the required interval. The output therefore solves $a' = F(a)$, not merely the auxiliary cutoff equation. The larger initial time interval supplies two-sided derivatives at both endpoints of $[0,T]$.

`PancakeFiniteVelocityContinuation` applies this construction to the already defined physical invariant subspace. Its actual vector field satisfies

$$2\langle v, F_M(v)\rangle = -2\nu \sum_{k\in M} |\widehat{\operatorname{curl} u}(k)|^2 \le 0 \qquad (\nu \ge 0).$$

The equality uses the exact retained kinetic cancellation. The codomain projection is proved equal to the original velocity RHS, so neither that projection nor the auxiliary bump changes the dynamics. A physical retained solution now exists on every prescribed finite horizon. Uniqueness proves agreement with any previously constructed solution on their common interval; no separate extension hypothesis is supplied.

Combining this with the preceding second-order estimate, `PancakeCommonGalerkinWindow` closes the lifespan quantifier gap:

$$\boxed{\exists\, \tau > 0,\ B \ge 0 \quad \forall\, M \quad \exists\, u_M \text{ on } [0,\tau] \quad \forall\, t \in [0,\tau],\ E_M(u_M(t)) \le B.}$$

Here M is any symmetric finite set excluding zero. The initial coefficients are the Fourier integrals of the physical field with the defined order-six continuous coordinate derivatives, reality and the

Fourier divergence condition. The interval and bound precede the choice of M. Separately, actual L^2 initial data give solutions on any finite horizon with the same Parseval-paid kinetic and integrated enstrophy bounds, without a uniform high-regularity claim for those data.

`PancakeSobolevTimeBudget` also proves continuity, and hence integrability, of the finite viscous quantity directly from the coefficient equation. The absorbed differential estimate yields

$$E_M(t) + \nu(2\pi)^2 \int_0^t D_M(s)\, ds \le E_M(0) + C_\nu B^2 t.$$

Consequently the common physical family has the cutoff-independent time budget

$$\nu(2\pi)^2 \int_0^t D_M(s)\, ds \le H, \qquad H = B + C_\nu B^2 \tau, \qquad 0 \le t \le \tau.$$

This constructs the common finite family and its local bounds. Convergence to a continuum evolution, propagation of the regularity needed by the full nonlinear consistency theorem, and the intrinsic global alignment budget remain to be proved. In particular, finite horizon existence at every cutoff is not continuum global regularity; the bound $E' \le C_\nu E^2$ remains a local comparison.

A constructed, time-uniform coefficient limit. The fixed-output estimate can use kinetic energy alone. Incompressibility gives

$$k \cdot u_k = 0, \quad k + l = q \quad \Longrightarrow \quad l \cdot u_k = q \cdot u_k.$$

The already checked convolution-fiber bound, together with this identity, yields in `PancakeFixedOutputVelocityBound`

$$\left| \sum_{\substack{k,l \in M \\ k+l=q}} (u_k \cdot l) u_l \right| \le |q|\, U_M(u), \qquad U_M(u) = \sum_{k \in M} |u_k|^2.$$

The Euclidean norm of the actual Leray symbol is proved contractive, including its defined zero-mode branch. Retaining viscosity and the unit-torus derivative factor gives

$$|V_M(u)_q| \le (2\pi)|q|\, U_M(u) + |\nu|(2\pi)^2 |q|^2 |u_q|.$$

There is no input-frequency or mode-count factor.

For a constructed solution with $U_M(t) \le U$, `PancakeCoefficientEquicontinuity` proves

$$|(u_M)_q(t)| \le \sqrt{U}, \qquad |\partial_t (u_M)_{q,i}(t)| \le A_q := (2\pi)|q|U + |\nu|(2\pi)^2 |q|^2 \sqrt{U}.$$

Every fixed component is therefore A_q-Lipschitz on the common interval. Excluded coefficients are identically zero. Dependence on the fixed output q is explicit; a uniform Lipschitz constant over all output frequencies is not asserted.

`PancakeCountablePathCompactness` proves a reusable extraction theorem. For each coordinate, the bounded equicontinuous paths have compact closure in the uniform norm by Arzelà–Ascoli. The countable product of these compact closures is compact and first countable. Sequential compactness therefore supplies one strictly increasing subsequence for every coordinate simultaneously, rather than a different subsequence for each coordinate.

`PancakeGalerkinCoefficientLimit` takes the explicit nested sets

$$M_n = \{k \in \mathbb{Z}^3 : \ |k_i| \le n \text{ for all } i\} \setminus \{0\}$$

and proves their symmetry, exclusion of zero, monotonicity, and eventual inclusion of every nonzero mode. The common physical existence theorem supplies u_{M_n}, and $U_{M_n} \le E_{M_n} \le B$ supplies the kinetic bound. A single extraction $\phi : \mathbb{N} \to \mathbb{N}$ and continuous coefficient paths $a_{q,i}$ are constructed, with

$$\sup_{0\le t\le T} |(u_{M_{\phi(n)}})_{q,i}(t) - a_{q,i}(t)| \longrightarrow 0 \qquad \text{for every fixed } q, i.$$

The original physical initial field is an input; these finite trajectories and the subsequential coefficient field are outputs.

`PancakeCoefficientLimitProperties` proves that this same coefficient limit retains conjugate symmetry, Fourier transversality, and zero mean. Every nonzero initial coefficient is preserved. Full initial matching is proved under the explicit mean-zero initial condition. For every finite set F, the finite-support energy comparison and passage to the coefficient limit give

$$\sum_{q\in F} (1+|q|)^4 |a_q(t)|^2 \le B.$$

Consequently the infinite series is proved summable, with

$$\boxed{\sum_{q\in\mathbb{Z}^3} (1+|q|)^4 |a_q(t)|^2 \le B \qquad (0 \le t \le T).}$$

This is a constructed infinite coefficient field with physical constraints and a genuine Sobolev energy bound. The next step proves nonlinear convergence and the coefficient evolution equation. The local H^2 bound also does not by itself provide the common summable pointwise second-moment envelope used by the earlier higher-regularity consistency theorem.

Passing the actual nonlinear equation to the coefficient limit. The common second-order bound supplies a more direct route than a second-moment pointwise envelope. Define

$$d(k) = (1+|k|)^{-4}, \qquad f_q(u,k) = (q\cdot u_k) u_{q-k}.$$

The singleton energy inequality gives $|u_k|^2 \le B d(k)$. For each fixed output q, Cauchy–Schwarz and $2ab \le a^2 + b^2$ give

$$|f_q(u,k)| \le \frac{|q|B}{2}(d(k) + d(q-k)) =: G_{B,q}(k), \qquad \sum_k G_{B,q}(k) < \infty.$$

The same bound holds for every approximation and every time in $[0,T]$. `PancakeSobolevFiberEnvelope` proves the componentwise version of this bound and its summability. Its constant depends on the fixed output: this is not a frequency-uniform critical estimate.

`PancakeConvectionPathLimit` constructs the series in the complete normed space $C_b([0,T];\mathbb{C})$. Each summand converges in that space, because both coefficient paths converge uniformly in time. Dominated convergence for series, using $G_{B,q}$, therefore yields

$$\sup_{0\le t\le T} \left| \sum_k f_q(u_{M_{\phi(n)}}(t), k) - \sum_k f_q(a(t),k) \right| \longrightarrow 0.$$

This step does not interchange an unjustified pointwise limit with a quadratic infinite sum.

`PancakeConvectionFiberIdentity` identifies this series exactly with the finite convection tensor on each supported approximation. When $k + l = q$, source transversality gives $l \cdot u_k = q \cdot u_k$; finite support eliminates all terms outside the retained set. Thus the derivative is on q before taking the infinite limit. `PancakeVelocityPathLimit` carries this convergence through the existing Leray projection and viscous multiplier, identifying the resulting paths with the actual finite velocity right-hand sides.

For $q \neq 0$, the mode belongs to every sufficiently large member of the subsequence. Uniform convergence of the derivatives and pointwise convergence of the coefficients then give, in `PancakeCoefficientLimitEquation`,

$$\boxed{\partial_t a_q(t) = -2\pi i\, P_q \sum_{k\in\mathbb{Z}^3} (q \cdot a_k(t)) a_{q-k}(t) - \nu(2\pi)^2 |q|^2 a_q(t), \qquad 0 < t < T.}$$

The zero mode is treated separately: both its coefficient and its right-hand side vanish. A constant extension beyond $[0, T]$ supplies a real-domain representative for the derivative theorem; no equation is claimed outside the open local interval.

Finally, `PancakeLocalInfiniteVelocity` proves the existence theorem

`exists_physical_localInfiniteVelocitySolution`.

For positive viscosity and mean-zero, real physical initial data with the stated sixth coordinate jets and Fourier incompressibility, it constructs $T > 0$, $B \geq 0$, and a continuous coefficient evolution with the exact initial Fourier coefficients, reality, zero mean, transversality, the bounded summable second-order energy, and the displayed nonlinear coefficient equation. The future evolution and nonlinear convergence are outputs, not hypotheses.

This closes the local infinite coefficient-equation construction. The next step supplies its continuous spatial realization. Higher regularity for the material strain identities and the intrinsic all-time alignment/pressure/weight estimates remain to be proved. In particular, the local inequality $E' \leq C_\nu E^2$ does not provide the global continuation estimate.

Joint spatial reconstruction and actual Fourier recovery. The second-order bound also suffices to reconstruct a continuous spatial field without the stronger pointwise second-moment envelope. For a finite set $F \subset \mathbb{Z}^3$, put

$$R_B(F) = \sqrt{B}\sqrt{L - \sum_{k\in F} (1 + |k|)^{-4}}\,.$$

The expression under the square root is nonnegative and tends to zero as F increases. For every finite M disjoint from F, weighted Cauchy–Schwarz gives

$$\sum_{k\in M} |a_k(t)| \leq \left(\sum_{k\in M} (1 + |k|)^{-4}\right)^{1/2} \left(\sum_{k\in M} (1 + |k|)^4 |a_k(t)|^2\right)^{1/2} \leq R_B(F).$$

`PancakeSobolevAbsoluteTail` proves this estimate uniformly over every field satisfying the same finite-partial-energy bound.

`PancakeSobolevSpaceTime` uses it in the Banach space of bounded continuous functions on $[0, T] \times \mathbb{T}^3$. The finite Fourier tails are uniformly small, so the series converges in that space. Its sum U is jointly continuous and, componentwise,

$$U_i(t, x) = \sum_k e^{2\pi i k\cdot x} a_{k,i}(t), \qquad \sup_{t,x} \left| U_i(t, x) - \sum_{k\in F} e^{2\pi i k\cdot x} a_{k,i}(t) \right| \leq R_B(F).$$

The argument proves summability of the time–space function series from small finite tails; it does not assume summability of the individual time-supremum coefficient bounds.

Absolute coefficient summability at each time permits termwise Fourier integration. Character orthogonality, formalized in `PancakeSpatialCoefficientRecovery`, gives $\widehat{U(t)}(q) = a_q(t)$ for every q. `PancakeLocalSpatialVelocity` then proves reality of U, exact matching $U(0) = u_0$ for the original physical input, zero mean and incompressibility in Fourier form, and the nonlinear evolution equation for the actual Fourier integrals of U.

Finally, `PancakeGalerkinSpatialLimit` combines two uniform tails with the finitely many converging coefficient paths between them. The same constructed Galerkin subsequence therefore satisfies

$$\sup_{(t,x)\in[0,T]\times\mathbb{T}^3} |(U_{M_{\phi(n)}})_i(t,x) - U_i(t,x)| \longrightarrow 0 \qquad (i = 1,2,3).$$

Thus the local coefficient construction now has a jointly continuous real spatial realization and a strong spatial approximation theorem. Joint continuity alone is not a smooth global Navier–Stokes theorem; the next construction supplies higher-order local control and its physical curl.

Higher-order local control and a constructed nonlinear envelope. The local argument now applies to each fixed integer order $m \geq 2$. Write

$$w_m(k) = (1+|k|)^m, \qquad E_m(M,u) = \sum_{k\in M} w_m(k)^2|u_k|^2, \qquad D_m(M,u) = \sum_{k\in M} w_m(k)^2|k|^2|u_k|^2.$$

The inverse-square weight sum is at most $L = \sum_k (1+|k|)^{-4}$, independently of m and the finite mode set. The elementary splitting inequality

$$w_m(k+l) \leq 2^m\big(w_m(k) + w_m(l)\big)$$

and the finite convolution estimate give the weighted triad bound $2^{m+1}\sqrt{L}\,E_m\sqrt{D_m}$ in `PancakeHigherSobolevWeights`. `PancakeHigherSobolevVelocityRate` retains the exact Leray cancellation and the physical viscous symbol, hence

$$E_m' + 2\nu(2\pi)^2 D_m \leq 2^{m+2}(2\pi)\sqrt{L}\,E_m\sqrt{D_m}.$$

Absorbing half the viscous term gives

$$E_m' + \nu(2\pi)^2 D_m \leq C_{m,\nu}E_m^2, \qquad C_{m,\nu} = \frac{4(2^m)^2 L}{\nu}.$$

This coarse order-dependent constant is not claimed sharp. It has no frequency-cutoff or mode-count dependence.

The initial energy is constructed from actual physical derivatives. `PancakeHigherInitialEnergy` generalizes the coordinate Fourier decay argument to order N. Compactness bounds the finitely many original components and their pure order-N coordinate derivatives by nonnegative constants B_0, B_N, and gives

$$(1+|k|)^N|\widehat{f}(k)| \leq A_N, \qquad A_N = 3\cdot 4^N(B_0 + B_N).$$

For $N = m+2$ this yields $E_m(M,\widehat{f}) \leq A_{m+2}^2 L$ for every finite M. Given this initial allowance $\mathcal{B}_0$, the common time and bound are

$$T_m = \frac{1}{2(C_{m,\nu}+1)(\mathcal{B}_0+1)} > 0, \qquad E_m(M, u_M(t)) \leq 2(\mathcal{B}_0+1) \quad (0 \leq t \leq T_m).$$

`PancakeHigherSobolevWindow` proves the estimate and `PancakeHigherGalerkinWindow` constructs the finite solutions on this interval. No future trajectory is an input.

Finite partial energies are continuous functions of finitely many coefficients. Their bounds therefore pass to the extracted coefficient limit. `PancakeHigherLocalVelocity` constructs a `LocalInfiniteVelocitySolution` whose coefficients are exactly that curve, and proves its higher-order bound. Thus the nonlinear equation and the energy estimate hold for the same solution. The assertion is for each fixed m; a common interval for all orders simultaneously is not asserted by this comparison argument.

A singleton energy bound gives $(1+|k|)^m|a_k(t)| \le \sqrt{\mathcal{B}}$. Taking $m = p+4$ constructs the time-independent summable envelope

$$(1+|k|)^p|a_k(t)| \le g(k) := \sqrt{\mathcal{B}}(1+|k|)^{-4}, \qquad 0 \le t \le T_m.$$

In particular, order-eight initial coordinate jets supply the common second-moment envelope along an actual sixth-order-controlled evolution. The module `PancakeLocalPhysicalNonlinearity` applies the existing derivative theorems to this solution: its coordinate derivatives and curl are the differentiated Fourier series, and the full nonlinear interaction is the derivative-defined real field $\nabla \times (U \times (\nabla \times U))$.

The same module also discharges the common-envelope hypothesis in the local, time-integrated nonlinear alignment consistency theorem. For each filter bound $C \ge 0$ and $\varepsilon > 0$, it constructs a finite core such that every larger cutoff, after removing zero, satisfies

$$\int_0^{T_m} \int_{\mathbb{T}^3} |\langle G_{\chi,M}(x)\omega_M(x), \nabla \times (U \times (\nabla \times U))(x) - \mathcal{N}_M(a)(x)\rangle| \, dx\, dt < \varepsilon \qquad (\sup_k |\chi(k)| \le C).$$

All fields in this display are evaluated at time t, and $G_{\chi,M}$ is the previously constructed filtered strain-gap weight. The physical solution and its envelope are now outputs of this theorem, not supplied evolutions or smoothness bounds. This closes a local nonlinear-consistency hypothesis gap. It does not bound the full misalignment-strain accumulation through an arbitrary lifespan, close the pressure and moving-weight budgets, or prove global BKM continuation. The following step supplies the local pointwise velocity equation; the higher material-strain identities still require their bridges.

From coefficient evolution to the pointwise projected PDE. The second-moment envelope now pays the full velocity right-hand side, summed over every output frequency. Define

$$(g * g)(q) = \sum_k g(k)g(q-k), \qquad H_{\nu,g}(q) = 3(2\pi)(g*g)(q) + |\nu|(2\pi)^2 g(q).$$

In `PancakeInfiniteVelocityEnvelope`, the map $(q,k) \mapsto (k, q-k)$ reindexes a summable product envelope, proving $g * g \in \ell^1$. The first moments of the two inputs absorb the output derivative in convection, and the second moment absorbs viscosity. Thus, for the infinite velocity RHS $R_\nu(a)$,

$$|R_\nu(a(t))_q| \le H_{\nu,g}(q), \qquad \sum_q H_{\nu,g}(q) < \infty \quad (0 \le t \le T).$$

The factor three is the proved comparison from componentwise bounds to the Euclidean coefficient norm, not a mode-count loss.

`PancakeInfiniteFourierTime` realizes each coefficient-to-mode map as a continuous linear map into $C(\mathbb{T}^3;\mathbb{C}^3)$. The summable derivative envelope permits time differentiation of the series on $(0,T)$ in this Banach space:

$$\frac{d}{dt}U(t) = F_{\mathrm{id},R_\nu(a(t))}.$$

The right-hand side is continuous in the uniform spatial norm on $[0,T]$, and jointly continuous in time and space. No derivative equation is claimed for the artificial extension beyond the local interval.

The spatial operators are ordinary derivatives, not prescribed symbols. `PancakeInfiniteSpatialLaplacian` proves, for every pair of coordinate directions j,l,

$$\partial_l\partial_j F_{\mathrm{id},a} = F_{\mathrm{id},(2\pi i)^2 k_l k_j a_k}, \qquad \Delta F_{\mathrm{id},a} = F_{\mathrm{id},(2\pi i)^2|k|^2 a_k}.$$

Consequently the reconstructed viscous term is exactly $\nu\Delta U$.

The nonlinear identification uses actual Fourier integrals. `PancakePhysicalConvectionCoefficients` first proves the product formula for a continuous scalar field f,

$$\widehat{U_i f}(q) = \sum_k a_{k,i}\widehat{f}(q-k).$$

Applying it to the coordinate derivatives of U, and using $k\cdot a_k = 0$, yields

$$\widehat{(U\cdot\nabla)U}_i(q) = 2\pi i\sum_k((q-k)\cdot a_k)a_{q-k,i} = 2\pi i\sum_k (q\cdot a_k)a_{q-k,i}.$$

All product integrations and finite/infinite sum exchanges have their summability proofs. The multiplier in the coefficient evolution therefore acts on the Fourier coefficients of the physical advection field.

Finally, `PancakeLocalProjectedEquation` defines the spatial Leray operator from those actual Fourier integrals: $\mathbb{P}f = F_{\mathrm{id},\,P_q\widehat{f}(q)}$. Its application to advection has an explicitly summable coefficient series. The constructed local field satisfies, pointwise,

$$\boxed{\partial_t U = -\mathbb{P}\big((U\cdot\nabla)U\big) + \nu\Delta U, \qquad \nabla\cdot U = 0, \qquad 0<t<T.}$$

The theorem `exists_physical_local_projected_equation` constructs this field from the order-eight physical initial jets, positive viscosity, real conjugacy, mean-zero and Fourier transversality hypotheses. It also proves $U(0)=f$ and reality of the evolving field. The divergence in the display is computed from actual coordinate derivatives.

This is a local Navier–Stokes equation in pressure-projected form. The following construction restores the scalar pressure for this same velocity. Neither result establishes the route's all-time strain or misalignment bounds.

Scalar pressure, ordinary Hessian, and the unprojected equation. Write

$$C(a)_q = \sum_k (q\cdot a_k)a_{q-k}, \qquad p_q = -\frac{q\cdot C(a)_q}{|q|^2}\quad (q\neq 0), \qquad p_0 = 0.$$

In `PancakeInfinitePressureCoefficients`, this definition satisfies the pressure Poisson coefficient identity at every frequency, including zero:

$$|q|^2 p_q = -q\cdot C(a)_q, \qquad |q|\,|p_q| \le |C(a)_q|.$$

A nonzero integer wavevector has Euclidean norm at least one. Thus $|p_q|\le|C(a)_q|$ as well, and the existing summable convection envelope pays both pressure and its first derivatives. Conjugate symmetry of the velocity gives $C(a)_{-q} = -\overline{C(a)_q}$ and $p_{-q} = \overline{p_q}$; the reconstructed pressure is real.

`PancakeInfiniteSpatialPressure` defines the continuous scalar field $p = F_{\mathrm{id},(p_q)}$. Termwise differentiation is justified by absolute summability, so its ordinary spatial gradient is

$$\partial_j p = F_{\mathrm{id},(2\pi i)q_j p_q}.$$

Actual Fourier integrals recover every p_q, and the zero mode gives $\int_{\mathbb{T}^3} p\,dx = 0$. This is a normalization of the constructed pressure, not an assumed auxiliary field. The same module reconstructs the actual advection from its Fourier coefficients.

The coefficient identity

$$R_\nu(a)_q = -(2\pi i)C(a)_q - (2\pi i)q\,p_q + \nu(2\pi i)^2|q|^2 a_q$$

therefore lifts, in `PancakeLocalPressureEquation`, to

$$\boxed{\partial_t U = -(U\cdot\nabla)U - \nabla p + \nu\Delta U, \qquad \nabla\cdot U = 0, \qquad 0 < t < T.}$$

The theorem `exists_physical_local_pressure_equation` constructs this same local velocity and its real zero-mean pressure from order-eight physical initial jets and positive viscosity. It retains initial matching, velocity reality, and ordinary spatial divergence zero.

The second pressure derivatives require an output derivative of convection. The second input moments already pay it:

$$\begin{aligned} |q|\,|(q\cdot a_k)a_{q-k,i}| &\le g(k)g(q-k),\\ |q|\,|C(a)_q| &\le 3(g*g)(q),\\ |(2\pi i)^2 q_j q_l p_q| &\le (2\pi)^2|q|\,|C(a)_q|. \end{aligned}$$

Here $|q| \le (1+|k|)(1+|q-k|)$ supplies each of the two output factors. There is no additional cell-count or output-frequency factor. `PancakeInfinitePressureHessian` uses these bounds to construct ordinary continuous mixed second derivatives of p and prove their symmetry. `PancakeInfinitePressurePoisson` identifies the trace of that actual Hessian with the divergence of actual advection:

$$\Delta p = -\nabla\cdot\big((U\cdot\nabla)U\big).$$

The finite coordinate sums and infinite mode sums are interchanged only after their summability is proved.

Finally, `PancakeLocalPressureContinuity` uses the same common local envelope and the already constructed convection paths. Pressure and its first and second derivative series are continuous in time in the uniform spatial norm; p, ∇p, and $\nabla^2 p$ are jointly continuous on $[0,T]\times\mathbb{T}^3$.

These are local pressure geometry and momentum identities, not an all-time pressure budget. The following step differentiates this same evolving velocity gradient and preserves the signed strain source. No global bound follows merely from the local envelope or the construction of the Hessian.

The actual local gradient and strain equations. A common third absolute moment now pays the differentiated velocity equation. The construction uses `exists_physical_localMomentEnvelope` with $p = 3$, so physical initial coordinate jets of order nine provide one interval $[0,T]$, one evolving velocity, and a summable envelope

$$(1+|q|)^3|a_q(t)| \le g(q) \qquad (0\le t\le T).$$

`PancakeHigherDerivativeMoments` proves monotonicity of these moments and the one-derivative cost

$$\mathrm{moment}_m(\partial_j a)_q \le 2\pi\ \mathrm{moment}_{m+1}(a)_q.$$

In `PancakeVelocityGradientEnvelope`, the same convolution envelope used above bounds

$$|q|\,|R_\nu(a)_q| \le H_{\nu,g}(q), \qquad |(\partial_j R_\nu(a))_q| \le 2\pi H_{\nu,g}(q).$$

The nonlinear output derivative is paid by the two input moments already used for the pressure Hessian. Differentiated viscosity costs $|q|^3|a_q|$, paid by the third moment.

`PancakeLocalVelocityGradientTime` applies the continuous-field series differentiation theorem to the derivative coefficients:

$$\frac{d}{dt}F_{\mathrm{id},\,\partial_j a(t)} = F_{\mathrm{id},\,\partial_j R_\nu(a(t))} \quad \text{in } C(\mathbb{T}^3;\mathbb{C}^3).$$

At each spatial point this is the time derivative of the ordinary $\partial_j U$, not merely of a formal multiplier. The identification uses equality in a neighborhood inside $(0,T)$, and makes no claim for the artificially extended coefficient curve outside its local interval.

The spatial product rule, proved in `PancakeSpatialGradientDifferentiation`, separates

$$\partial_j\big((U\cdot\nabla)U\big) = (U\cdot\nabla)(\partial_j U) + \sum_m (\partial_j U_m)\,\partial_m U.$$

The pressure term differentiates to its already constructed Hessian. `PancakeSpatialViscousDifferentiation` proves $\partial_j\Delta U = \Delta\partial_j U$ from summable ordinary third derivatives. Uniqueness of these spatial derivatives, applied to the actual momentum equation, gives the differentiated RHS.

Let $A_{ij} = \partial_j U_i$. In `PancakeLocalGradientEquation`, the product term is exactly $(A^2)_{ij}$, and the constructed field obeys

$$\partial_t A + (U\cdot\nabla)A = -A^2 - \nabla^2 p + \nu\Delta A.$$

Here matrix transport and the matrix Laplacian are defined by ordinary coordinate derivatives of their columns. The existence theorem `exists_physical_local_gradient_equation` supplies this identity directly from the physical initial data, on the same positive local interval.

Set $S = (A + A^{\mathsf{T}})/2$ and $W = (A - A^{\mathsf{T}})/2$. `PancakeMatrixDifferentialSymmetry` proves that symmetrization commutes with the ordinary matrix transport and Laplacian under their explicit differentiability hypotheses. `PancakeSpatialGradientRegularity` discharges those hypotheses for this A using the same third velocity moment. The algebraic identity

$$\mathrm{sym}(A^2) = S^2 + W^2$$

cancels the mixed strain–spin terms. Therefore `PancakeLocalStrainEquation` proves

$$\boxed{\partial_t S + (U\cdot\nabla)S = -S^2 - W^2 - \nabla^2 p + \nu\Delta S, \qquad 0 < t < T.}$$

Its theorem `exists_physical_local_strain_equation` constructs the local solution from order-nine physical initial jets. Neither the strain equation nor its differentiability is an independent future-evolution hypothesis.

Finally, `PancakeLocalStrainReality` proves that the physical derivative multiplier preserves coefficient conjugacy. Thus A, S, and W are real; S is symmetric and trace-free, and W is antisymmetric. These are geometric properties of the actual spatial derivatives, not prescribed matrix data.

This completes the local Eulerian strain identity. The following construction evaluates it along actual material trajectories; neither identity alone supplies the global critical estimate.

Constructed material flow and its strain equation. Write $G = \sum_q g(q)$ for the common local envelope above. `PancakeInfiniteVelocityLipschitz` proves, on the real periodic lift,

$$\|U(t,x)\| \le G, \qquad \|U(t,x) - U(t,y)\| \le 6\pi G\,\|x-y\|.$$

These estimates use the coordinate sup norm; the coefficient Euclidean norm in the absolute moment dominates that norm explicitly. The character estimate is obtained by differentiating along a straight real segment and using

$$|q\cdot v| \le 3|q|\,\|v\|, \qquad |\chi_q(x) - \chi_q(y)| \le 6\pi|q|\,\|x-y\|.$$

Summability of the first absolute moment justifies the infinite sum. No chosen chart radius or number of localization cells enters the bound.

`PancakeLocalMaterialTrajectories` defines the real lift of the constructed velocity and proves its time continuity, boundedness, and spatial Lipschitz estimate. The Picard–Lindelof hypotheses hold on the entire existing interval $[0,T]$: for an initial point x, a ball of radius GT contains the possible displacement. This constructs X_x with

$$X_x(0) = x, \qquad \dot X_x(t) = \operatorname{Re} U(t,[X_x(t)]).$$

The equation holds with one-sided derivatives at the endpoints and ordinary derivatives in $(0,T)$. The spatial field is real, so taking real parts changes no physical velocity. `PancakeLocalMaterialFlow` constructs trajectories simultaneously for every initial point and proves uniqueness on $[0,T]$. With $K = 6\pi G$, its Gronwall estimate is

$$\|X_x(t) - X_y(t)\| \le e^{Kt}\|x-y\|, \qquad 0 \le t \le T.$$

This is a local flow regularity estimate, not a scale-critical bound: G is the local high-regularity envelope.

The material derivative is proved directly for the infinite series. `PancakeInfiniteDirectionalTransport` defines

$$D_v a_q = (2\pi i)(q\cdot v)a_q = \sum_j v_j\,\partial_j a_q$$

and identifies its reconstruction with ordinary spatial transport. `PancakeInfiniteMaterialSeries` proves

$$\frac{d}{dt}F_{a(t)}([X(t)]) = F_{\dot a(t)}([X(t)]) + F_{D_{\dot X(t)}a(t)}([X(t)]).$$

For a common first-moment bound g, derivative-coefficient bound h, and speed bound M, the term derivatives are dominated by the summable envelope $h(q) + 6\pi M g(q)$. Both coefficient evolution and character motion are present in the product rule.

`PancakeLocalMaterialGradient` applies this result to the actual gradient coefficients. The time derivative and moving-character term cancel Eulerian advection exactly:

$$\frac{d}{dt}A(t,[X_x(t)]) = (-A^2 - \nabla^2 p + \nu\Delta A)(t,[X_x(t)]).$$

The common third velocity moment pays the differentiated equation and the spatial derivatives. No material gradient equation is assumed. `PancakeLocalMaterialStrain` then symmetrizes, using the checked ordinary Laplacian commutation:

$$\boxed{\frac{d}{dt}S(t,[X_x(t)]) = (-S^2 - W^2 - \nabla^2 p + \nu\Delta S)(t,[X_x(t)]), \qquad 0 < t < T.}$$

The theorem `PancakePhysicalMaterialStrain.exists_physical_material_strain` constructs one velocity and one such flow from the order-nine physical initial jets, with initial matching and the strain equation along every trajectory.

Finally, `PancakeMaterialFlowQuotient` proves that initial lifts representing the same torus point give the same torus trajectory. The difference of the initial lifts is a period; translating one real trajectory by it preserves the ODE, and uniqueness identifies the two. Thus the real flow factors through a torus-valued particle map.

This is a deterministic local material construction. Spatial Jacobians, measure preservation, and a stochastic flow have not been constructed by these modules. The next dynamical connection must identify the actual vorticity-direction evolution along this flow and attach the collision-safe spectral-defect estimate. Pressure, moving-weight, and misalignment accumulation over an arbitrary lifespan still require the global critical bound; the finite local envelope is not such a bound.

The eigenvalue-gap denominator also appears in spatial frame freezing. In `PancakeEigenframePerturbation`, if $Ae = \lambda e$, $Bf = \mu f$, B is self-adjoint, and e, f are unit vectors, then

$$(\lambda - \mu)\langle f, e\rangle = \langle f, (A - B)e\rangle, \qquad |\lambda - \mu|\,|\langle f, e\rangle| \le \|A - B\|.$$

Consequently, at every positive threshold, either the eigenvalues are in the near-degenerate band or the frame overlap is bounded by the strain oscillation divided by that threshold. This is the exact gap/near-degeneracy alternative needed by the variable-frame remainder; it is not yet the scale-summed bound needed for closure.

Two further modules turn this into an intrinsic local dynamics statement. `PancakeStrainSpectralFrame` constructs, for every symmetric three-dimensional strain operator, an ordered orthonormal eigenframe and proves the three-coordinate Parseval and Rayleigh identities. Thus the spectral coordinates used below are constructed objects, not assumed data. For a unit vorticity direction ξ and a moving unit top eigenvector e, `PancakeAlignmentDynamics` defines

$$E(\xi, e) = 1 - \langle \xi, e\rangle^2$$

and proves the exact simple-gap identity

$$\dot{E} = -2\langle \xi, e\rangle^2(\lambda_0 - \langle \xi, S\xi\rangle) - 2\langle \xi, e\rangle(\langle d, e\rangle + \langle \xi, \theta\rangle),$$

where d is the non-strain direction forcing and $\theta = \dot{e}$. The Rayleigh defect is bounded below by the top spectral gap times E, and the module integrates this damping to obtain the residence estimate

$$\int_0^T (\lambda_0 - \lambda_1) E\, dt \le E(0) + 2\int_0^T |\langle d, e\rangle + \langle \xi, \theta\rangle|\, dt$$

under its stated regularity and simple-gap hypotheses. It also proves the complementary upper bound by the full spectral width.

These identities expose rather than hide the hard cases. Substituting the usual eigenvector perturbation estimate $|\theta| \lesssim |\dot{S}|/(\lambda_0 - \lambda_1)$ into the residence inequality and applying a naive Young inequality produces still worse inverse-gap costs; it is not a closure. Moreover, a large trace-free strain may be isotropic on a strongly expanding two-plane. Such a partially degenerate plane supplies damping toward the plane but does not select a coherent line inside it. Controlling those intervals requires a projector/cluster formulation or a genuine time-integrated cancellation, and remains part of the dynamical pin.

G Formal scope of the earlier derivations

The H1 obstruction remains a theorem about the explicit normalized Fourier shear model in `BenH1Break`. Its Sobolev ratio is derived analytically and checked independently. It is not a complete Lean construction of the volume-preserving diffeomorphism group or its Sobolev charts. The derivative loss nevertheless identifies the failure in the manuscript's local adjoint bound; the repaired history route does not rely on that bound.

The cotangent identities and conditional pushdown retain their stated hypotheses. In particular, the averaged transport equation is supplied as an input; a Brownian expectation theorem is not obtained merely by constructing pathwise flows. The prescribed-strain examples are model results. The coherent Fourier estimates and projector identities retain their finite-sector, regularity, gap and localization hypotheses.

The scale-local pancake route is still conditional on its dynamical strain and misalignment estimates, including the residual forcing, collision sectors and all-scale aggregation. Those are obligations for that architecture, not an assertion that all routes must use the same pointwise budget. Actual continuation theorems are now available separately on both domains; what they require is the arbitrary-data finite-horizon bound described in Section 1.

The current parent-reserve and coupled-return calculations in Section 3 are analytic PDE estimates with exact computational certificates. The observer and polynomial obstruction in Section 4 have the explicitly stated Lean boundary. No arbitrary-data global-regularity theorem follows from these conditional interfaces or from the favorable finite-order corrections.

Provenance

The original route is Ben Goertzel's *Global Regularity of 3D Navier–Stokes via SG–Cole–Hopf Program*, v7, with the subsequent pancake-closure proposal. The SG audit, conditional cotangent repair, microlocal development and later history calculations are work of the MeTTapedia formalization effort. The reconstruction uses the Constantin–Iyer stochastic-Lagrangian approach; the continuation criteria retain the Beale–Kato–Majda and Leray analytic foundations.

The integration reuses OpenAI/NavierStokesAndEuler and the Alpöge–Buckmaster `fluid_lean` libraries with their attribution and licenses. OpenAI's imported singularity construction is forced. The d-calculus tolerance is reused unchanged from MeTTapedia revision `3b432626d`; the history application reuses the existing GSLT signed-boundary and non-factorization theorems. The current source map and certification boundaries are given in Section 6.

Background references: J. T. Beale, T. Kato, A. Majda, *Comm. Math. Phys.* **94** (1984), 61–66; P. Constantin, G. Iyer, *Comm. Pure Appl. Math.* **61** (2008), 330–345; D. Ebin, J. Marsden, *Ann. of Math.* **92** (1970), 102–163; H. Bahouri, J.-Y. Chemin, R. Danchin, *Fourier Analysis and Nonlinear Partial Differential Equations*, Springer, 2011; P. Constantin, C. Fefferman, *Indiana Univ. Math. J.* **42** (1993), 775–789; J. Tom, M. Carbone, A. D. Bragg, *J. Fluid Mech.* **910** (2021), A24.